NorthCUA Technical Report

Turbocharging Computer-Use Agents through Experience at Scale

NorthCUA Team

ABSTRACT

Scaling computer-use agents from individual actions to complex software work requires experience that connects decisions to complete outcomes. We present NorthCUA, an integrated system for turbocharging computer-use agents through experience at scale. Its Experience Flywheel supplies diverse executable tasks to the NorthCUA Harness, which sustains long interactions, records decisions, and checks task outcomes. The Flywheel uses these records to curate verified trajectories. For unsolved tasks, recursive self-improvement evolves reusable skills and preserves verified milestones so later attempts build on earlier progress. Staged Learning combines this experience with grounding data from the Flywheel to develop GUI competence through mid-training and long-horizon supervised fine-tuning. The policy then learns from fresh attempts in the same Harness through joint desktop–web reinforcement learning, closing the loop between experience production and policy improvement. The resulting Nano, Mini, and Pro family achieves state-of-the-art results among open-source models across GUI grounding, desktop execution, and Web interaction under the reported evaluation settings (Figure 1). For more responsive Web workflows, NorthCUA-Ultrafast pairs direct action selection with open-ended generation on pre-RL Nano, offering a measured quality–latency tradeoff. Together, these advances establish NorthCUA as a high-performing model family and a reusable foundation for agents that carry complex software work from intent to completion.

Page beipo.top/research/northcua/

Models huggingface.co/beipo-ai/NorthCUA-Pro

Code github.com/china-qijizhifeng/NorthCUA-Harness

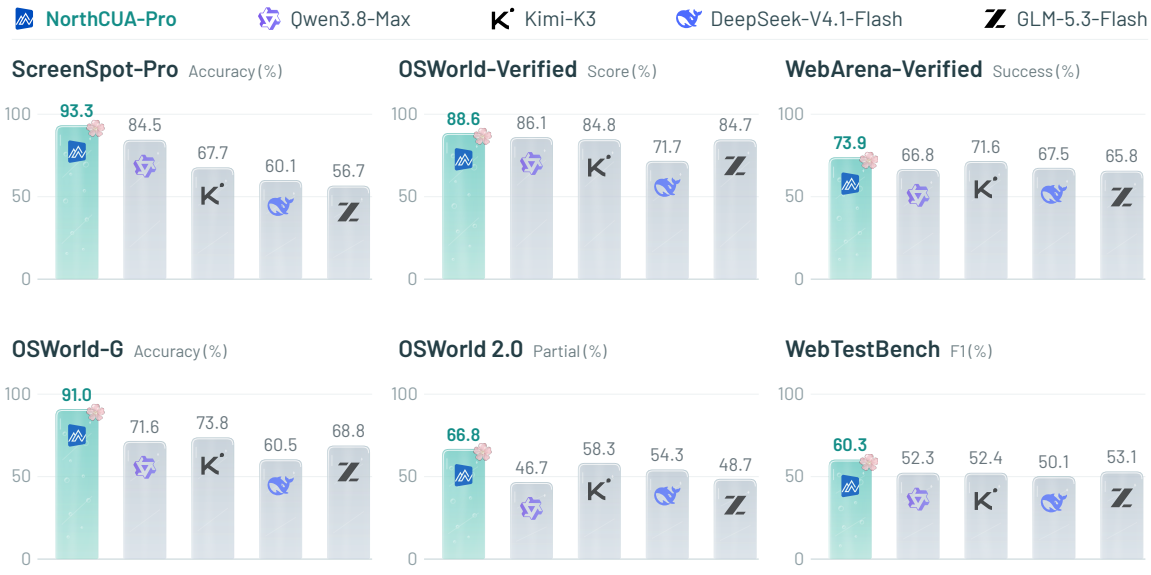


Figure 1. **NorthCUA-Pro performance compared with leading models.**

1 Introduction

Graphical user interfaces (GUIs) remain the primary way people use many everyday and professional applications. For AI systems, reliable interaction with these interfaces is a critical last mile between understanding a user's request and completing the work in those applications (Zhang et al., 2025a,c). Computer-use agents (CUAs) bridge this gap by interpreting the screen and translating natural-language goals into mouse and keyboard actions (Zhang et al., 2025b, Wang et al., 2025b). Advances in visual grounding and action generation have expanded the operations these agents can perform (Qin et al., 2025, Wang et al., 2025c, Zhou et al., 2025). As tasks extend into longer workflows, however, earlier actions shape the software state on which subsequent decisions depend. Reliable completion therefore requires agents to preserve intermediate progress and the user's constraints, adapt to changing interfaces, and recover from errors while keeping the overall goal in view (Zhang et al., 2026, Lu et al., 2026).

Training agents to coordinate such workflows therefore requires experience that develops precise interface skills and connects successive decisions to complete task outcomes. Obtaining this experience at scale begins with constructing varied tasks, each of which needs a request grounded in available software and content, a usable initial environment, and a reliable way to judge success (Ge et al., 2026, Zhou et al., 2026b). Once these tasks are available, collecting experience from them requires an agent framework that sustains long interactions and records decisions and outcomes across desktop and web environments. The resulting demonstrations can teach the policy how to complete workflows, but independent execution may lead it into states absent from those demonstrations. Continued improvement therefore requires fresh interaction and outcome feedback from the policy's own attempts (Wang et al., 2025c, Jung et al., 2026, Lu et al., 2026). Meeting these linked needs calls for co-designing how tasks are constructed, how agents execute them, and how the resulting experience is used for learning.

We present NorthCUA, an integrated system for *turbocharging computer-use agents through experience at scale*. Our central idea is to co-design the production and use of experience, making software interaction a continuing source of learning. We connect executable tasks to policy updates through three components (Figure 2), focusing on GUI-based software operation. The **Experience Flywheel** constructs executable tasks for the **NorthCUA Harness** to run and assess. The recorded trajectories and outcome evidence guide curation, supplying **Staged Learning** with demonstrations that complement the Flywheel's separately constructed grounding examples. This mixture develops interface understanding and workflow completion. Shared execution and outcome interfaces also let the learner collect fresh attempts from the policy itself, extending the same system from demonstration collection to continued policy improvement.

NorthCUA Harness: full-lifecycle infrastructure. We design the NorthCUA Harness as *full-lifecycle infrastructure* that executes software tasks and turns the resulting attempts into inspectable experience for learning. This requires keeping the agent's decisions connected to its workspace and outcomes, from environment preparation through execution and evaluation to record preservation. Within the shared runtime, visual context and task memory help agents build on progress across long workflows (Zhang et al., 2026, 2025d). An evaluator uses the execution record and live workspace to check task completion and execution quality, supplementing task-specific verifiers where their coverage is incomplete (Zhuge et al., 2025). To scale this process, the Batch Rollout Service coordinates parallel workers and overlaps lifecycle stages, supporting efficient, high-concurrency experience collection (Section 3).

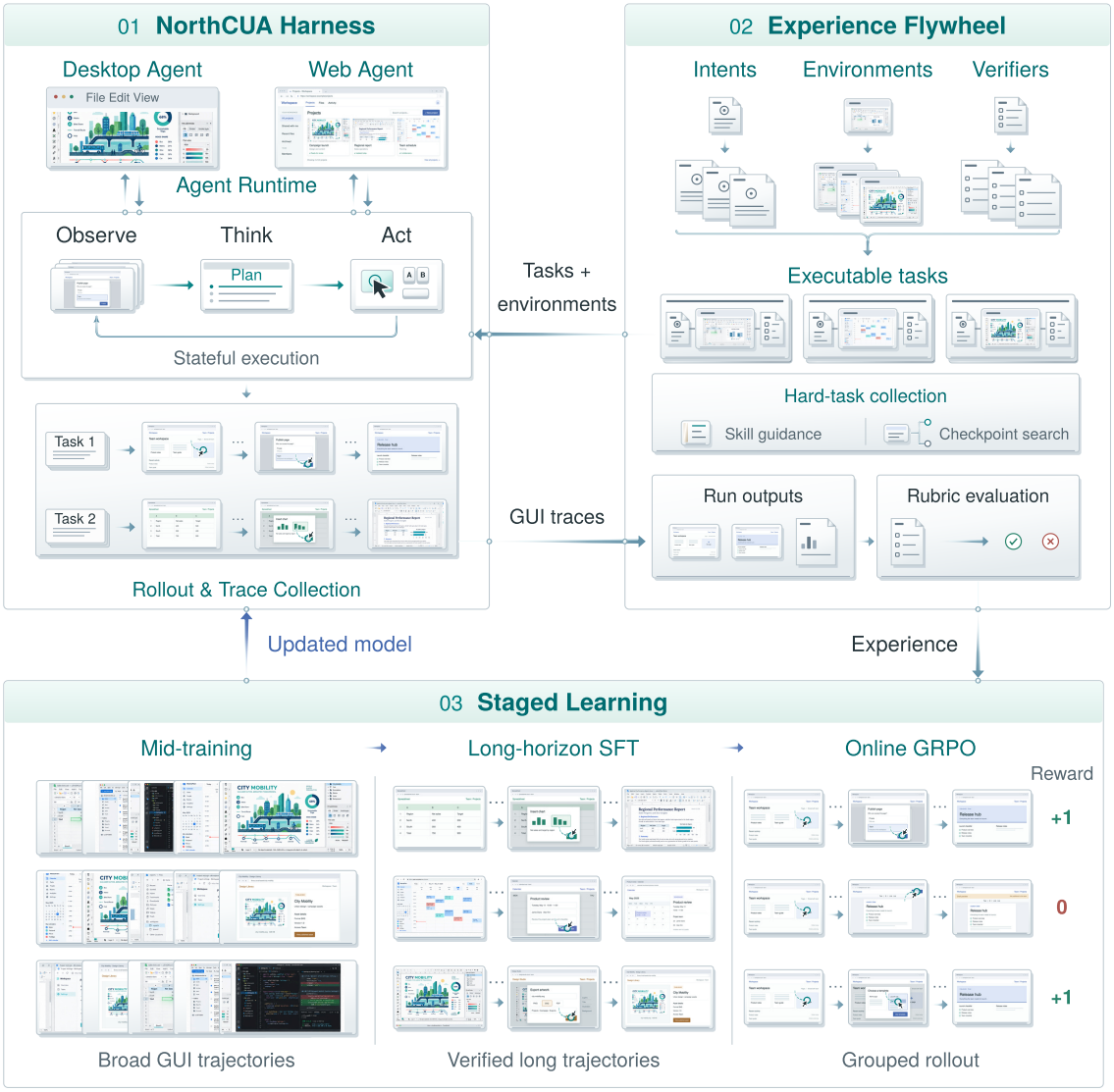


Figure 2. The NorthCUA experience loop. 01 The harness executes desktop and web tasks, records their trajectories, and retains workspaces through required live inspection. **02** The Experience Flywheel expands intents, environments, and verifiers into executable tasks, then curates assessed trajectories. The hard-task panel denotes skill-guided and checkpoint-assisted collection through the harness (Section 4.3). **03** Staged Learning converts complementary experience into an updated policy for subsequent execution. GUI states, cursor cues, and rewards are schematic. Trajectory rows omit intermediate actions, and GRPO retains valid failures alongside successes.

Experience Flywheel: scaling grounding and agentic experience. With a runtime for collecting and checking software experience, the next challenge is to supply enough varied, useful work for the agent to learn. The Experience Flywheel addresses two levels of supervision: single-step grounding examples for locating controls and assessed agent trajectories for completing tasks. Grounding pairs screenshots with queries and target locations. Visual and structural annotations provide broad coverage (Lu et al., 2024, Li et al., 2025a), while application documentation helps explain a control's function when appearance alone is insufficient. At the trajectory level, three complementary construction routes supply tasks to the harness. *Scaling Intents* derives requests from available software and content, while *Scaling Environments* builds the workspaces

needed by sourced requests or reconstructs reference websites. For tasks with executable checks, *Scaling Verifiers* creates further tasks by adapting requests, initial states, and verifiers together (Ge et al., 2026, Lu et al., 2026, Lv et al., 2026). When a valid task remains unsolved in ordinary collection, we use recursive self-improvement (RSI) for hard-task collection. Skill-guided exploration retains tested procedures for later attempts, while milestone-guided search preserves software states and agent context at verified milestones to explore alternative ways of completing the remaining work. Task outcomes and execution quality determine which trajectories become demonstrations for learning (Section 4).

Staged Learning: from experience to capability. A model's ability to coordinate a long workflow builds on familiarity with the interfaces and operations it uses. Staged Learning follows this progression, starting with *Broad GUI mid-training* on grounding examples and desktop and web trajectories judged to complete their tasks (Wang et al., 2025c, Jung et al., 2026). The broad trajectory corpus spans over four million GUI interaction steps across dozens of applications. Building on this operational foundation, *Verified long-horizon supervised fine-tuning* (*SFT*) concentrates on carefully selected, complete workflows whose outcomes pass executable task verifiers and whose intermediate results support later decisions. Independent execution can still lead the policy into states absent from its demonstrations. *Joint desktop–web agentic reinforcement learning* therefore lets one policy practice in both domains and learn from fresh successes and failures. Task verifiers score these attempts, and group-relative policy optimization (GRPO, Shao et al., 2024) uses their outcomes to update the policy (Section 5).

Because collection and learning share task, execution, and assessment interfaces, the updated policy can return to the same Harness for further interaction (Figure 2). Its new attempts provide additional experience under the same outcome checks, keeping policy improvement connected to the software work that motivates it. This reusable loop gives NorthCUA a foundation for continued learning across desktop and web environments.

This experience loop yields NorthCUA-Nano (9B), NorthCUA-Mini (35B), and NorthCUA-Pro (397B), built on the Qwen3.5 model family (Qwen Team, 2026a). Evaluation on six benchmarks spans GUI grounding, desktop execution, and Web interaction, covering precise control, long-task completion, and defect detection. Across the reported settings, NorthCUA achieves state-of-the-art results among open-source models in all three domains (Figure 1). At 397B parameters, Pro exceeds the reported grounding, OSWorld-Verified, and WebArena-Verified scores of trillion-parameter Qwen3.8-Max and Kimi-K3 (Qwen Team, 2026c, Kimi Team, 2026). The smaller tiers also sustain strong task execution, with the 35B Mini surpassing Kimi-K3 on OSWorld-Verified and Qwen3.8-Max on WebArena-Verified. Section 6 gives the benchmark results and evaluation protocols.

Longer workflows also make decision latency consequential. We therefore explore NorthCUA-Ultrafast, a Jev-inspired extension of pre-RL Nano that pairs direct selection for finite actions with generation for open-ended outputs (Almeida, 2026). Evaluation quantifies the resulting quality–latency tradeoff for Web execution (Section 6.8). We release all three models, the full NorthCUA Harness code, and this technical report to support further research and deployment. Together, they provide a shared foundation for extending computer use to broader software coverage, longer workflows, and faster execution.

2 Scope and System Overview

2.1 Task Scope and Characteristics

We study two capability targets in desktop and web environments (Figure 3): locating a requested control in a screenshot and completing a task through software interaction. We evaluate these targets separately.

Single-step UI grounding. Given a screenshot and a natural-language query, the model predicts the coordinates (x, y) of the target UI element in the original image (Cheng et al., 2024, Li et al., 2025b). The model completes the task with one prediction over a fixed screenshot. The query can describe the element's appearance or function, such as locating the zoom tool in Figure 3(a).

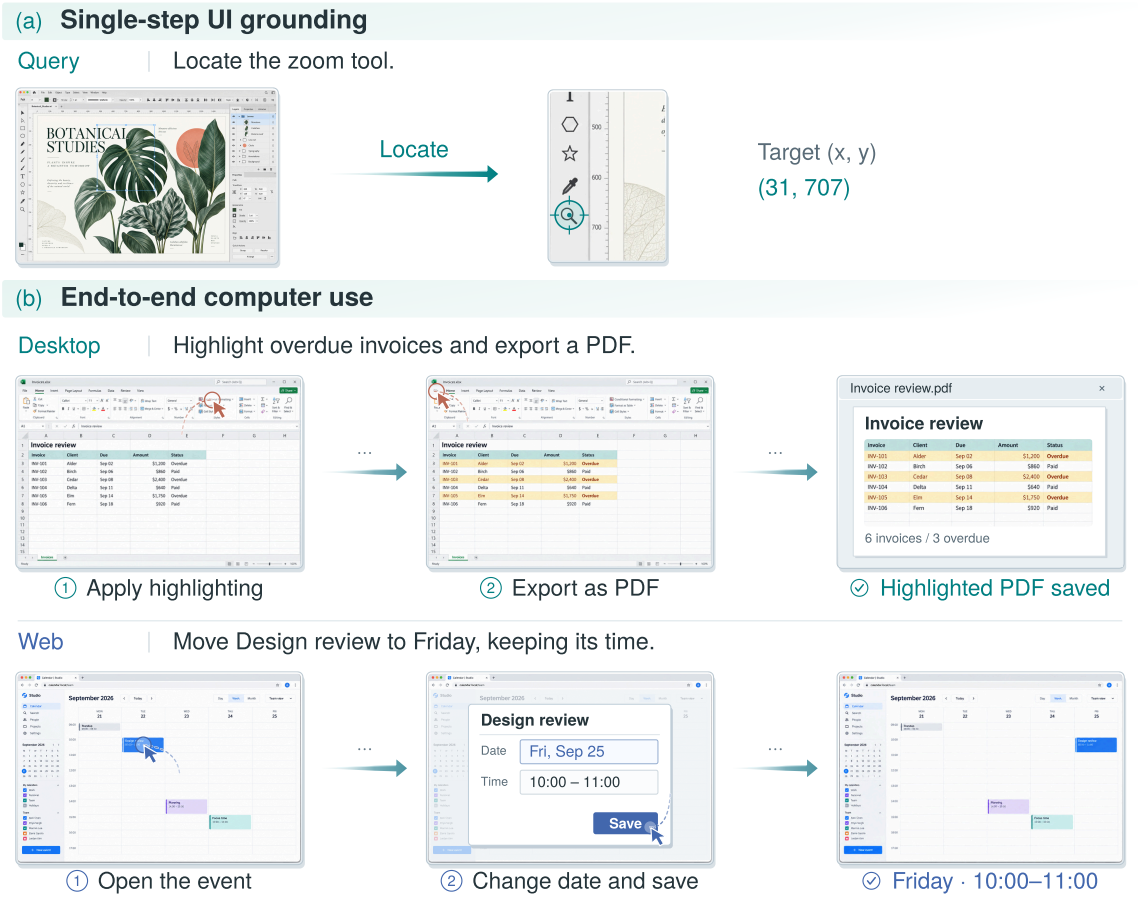


Figure 3. Task scope and interaction. (a) Single-step grounding predicts a target location in a fixed screenshot. The enlarged output view marks the zoom tool, with absolute pixel coordinates in the original image. (b) Desktop and web agents change software state through successive actions, with completion assessed from the resulting artifacts or state. Web observations also include DOM information. Generated GUI scenes, action overlays, and outcomes are illustrative. The displayed transitions omit intermediate interactions.

End-to-end computer use. An end-to-end task combines a user request, an initial software environment, and conditions for judging completion. The environment provides the applications, files, and website records needed for the request. The agent observes the current state, takes an action, and uses the resulting feedback to decide what to do next. Later decisions can use information and results from earlier steps. The request and completion conditions stay fixed throughout the attempt, so the final files or software state can be checked against the original goal (Zhou et al., 2024, Xie et al., 2024). Figure 3(b) illustrates the two interaction settings:

- **Desktop agents** observe desktop screenshots and operate applications through mouse and keyboard actions, including clicking, typing, scrolling, and dragging. Tasks may span multiple applications and files.
- **Web agents** observe webpage screenshots and document object model (DOM) information, and interact through browser tools for navigation, element-based or coordinate-based actions, text input, and tab management.

Both settings follow the same observation, action, and feedback loop, using their respective desktop or browser interfaces. The acting policy uses these interfaces to perform the task. Preparation, annotation, and evaluation can use additional programmatic tools with separate permissions.

2.2 System Organization and Experience Lifecycle

The experience loop in Figure 2 connects task construction to execution and learning. Its handoffs specify how an executable task becomes an assessed record and then a training sample.

The Experience Flywheel packages each executable task as a CaseSpec, a task specification containing the request, input materials, environment configuration, and setup instructions, together with requirements for assessing the result. For the invoice task in Figure 3(b), this means providing the workbook and spreadsheet application and specifying that the exported PDF should highlight overdue invoices.

Using this specification, the NorthCUA Harness prepares the environment, uploads input files, and runs setup and launch commands before starting the agent on the request. Observations, actions, and feedback are recorded throughout the attempt so that its outcome can be assessed in the context of the work that produced it. Task-specific checks, an evaluator, or both inspect the result against the declared requirements. For the invoice task, this means checking the saved PDF and its highlighted contents. These outcome and execution-quality assessments give the Experience Flywheel the evidence needed to select demonstrations. Valid tasks that remain unsolved can instead enter the hard-task collection branch, where skill-guided exploration or milestone-guided search produces further attempts for the same assessment process (Section 4.3).

Staged Learning turns this collected experience into capability in three steps. Broad GUI mid-training combines diverse task trajectories with the Flywheel's separate screenshot–query–target examples, building a foundation in interface understanding and software operation. Grounding supervision is used only in this stage. Verified long-horizon SFT then focuses on high-quality, complete workflows with checked outcomes, teaching the policy to carry information and intermediate results across dependent steps. This prepares the model to learn further from its own interaction.

For joint desktop–web agentic reinforcement learning, we first select tasks on which the SFT policy's trial attempts have different outcomes. This focuses initial collection on tasks with useful

outcome differences for comparing executions. The Harness collects fresh groups of attempts, executable verifiers score their outcomes, and GRPO uses comparisons within each task group to update one shared policy (Section 5.3). The updated policy returns to the Harness for subsequent attempts under the same task and outcome interfaces.

The following sections explain these components in detail. Section 3 presents the execution and assessment framework, Section 4 describes task construction and experience curation, and Section 5 develops the staged learning recipe.

3 NorthCUA Harness: Full-Lifecycle Infrastructure for Computer Use

The NorthCUA Harness provides *full-lifecycle infrastructure* for executing desktop and web tasks and turning each attempt into inspectable experience for learning. During execution, a persistent workspace and retained task context let the agent build on earlier progress. We retain its execution record and resulting artifacts for outcome assessment, keeping the original environment available when live checks need to inspect the state the agent produced. Resources are released after the required inspection and record preservation are complete. By carrying each attempt through assessment and preservation, this lifecycle keeps decisions, outputs, and judgments connected to the same task attempt (Section 2).

Figure 4 organizes this lifecycle at two levels. Within one task attempt, the unified agent runtime (**01**) supplies the shared interaction loop and platform tool interfaces. Long-horizon support (**02**) operates within this execution, keeping the next decision informed while retaining the complete trajectory. After acting ends, a tool-using evaluator (**03**) checks the declared requirements using that trajectory and the live workspace. The Batch Rollout Service coordinates the parallel task lifecycles shown below. Each isolated environment proceeds through preparation, execution, and assessment. Centralized control and recording manage these attempts, keeping each workspace available through inspection and output capture. This distinction separates the mechanisms that make one attempt inspectable from the service that turns them into sustained experience production. Online RL reuses the execution and outcome interfaces under training-side lifecycle management (Section 5.3), extending the same foundation from standalone collection to policy learning.

3.1 Unified Execution Across Desktop and Web

Desktop and web agents follow the same basic loop: observe the software, choose an action, execute it, and use the feedback to continue. What differs is how each platform exposes its state and controls. Building on shared-runtime approaches to computer use (Zhang et al., 2026, Yan et al., 2025), we design *Agent Kit* to separate this common loop from platform-specific observation and action tools. Long-horizon support and evaluation reuse this same execution foundation.

3.1.1 Shared Agent Runtime

We use a shared runtime to organize interaction as a ReAct-style loop with visual feedback (Yao et al., 2023). It assembles the request, observation, and relevant history into the next model input, invokes the policy, and executes the resulting tool calls. Returned observations and action results then inform the next decision. Common runtime services manage progress, budgets,

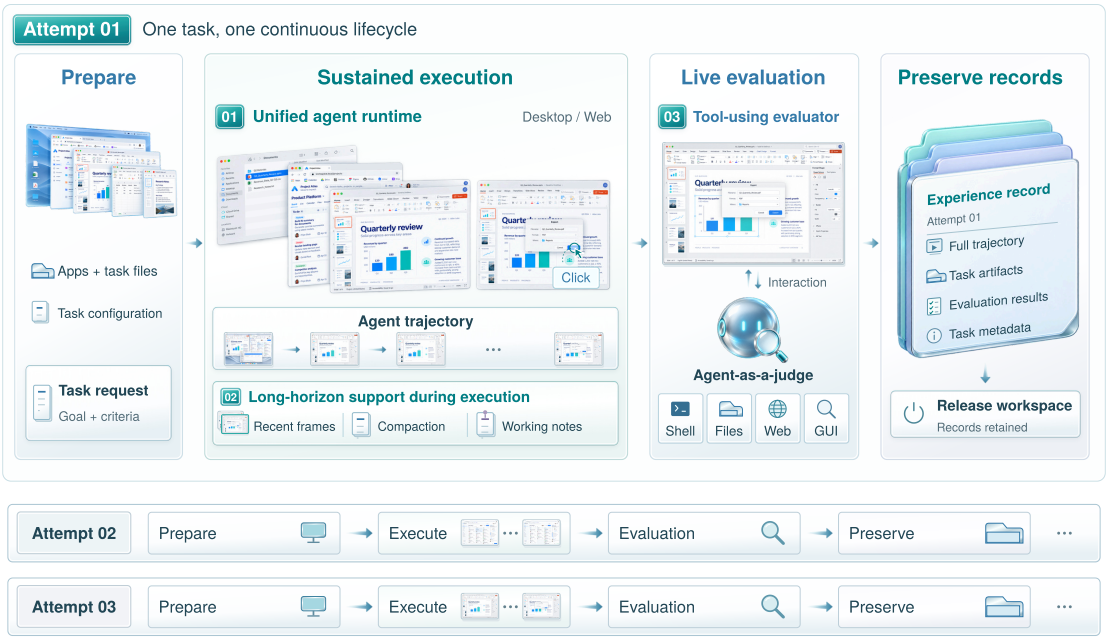


Figure 4. NorthCUA Harness lifecycle. Within the expanded attempt, the shared runtime and long-horizon support sustain GUI interaction while retaining the full trajectory. The evaluator inspects recorded behavior and the live environment. Trajectory, task artifacts, evaluation results, and task metadata are saved in an experience record before the workspace is released. The lower lanes show parallel instances of this lifecycle, coordinated by the batch collection service. GUI scenes, agent imagery, and trajectories are illustrative.

and termination, while environment-specific implementations supply the observations and operations. To build an agent for another platform, developers supply its perception functions, action tools, and observation processing hooks. The shared runtime then carries these components through the complete decision and recording loop, without rebuilding the agent flow. Section 3.1.3 details the desktop and web instances.

CUA execution also involves several timescales. Application and browser sessions persist across actions, a user request can require many decisions, and each decision produces feedback needed by the next. We represent these lifetimes through three nested scopes:

- **Session** maintains the agent's execution state and associated software resources, such as a browser session, across a sequence of user requests.
- **Round** tracks the processing of one user request, including its multi-step execution, budget, and termination status.
- **Step** records one policy decision and the resulting tool execution and feedback. Each step can contain multiple tool calls.

These runtime scopes organize interaction, while the task attempt supplies the collection-level identity used to associate actions, artifacts, and assessment. Each recorded action retains its request and session context, supporting task-level inspection and collection recovery. Session resources are managed within the broader environment lifecycle coordinated by the collection or training system. Appendix H.1 gives the execution-scope details.

3.1.2 Tool Integration and Dispatch

Desktop input, browser interaction, and workspace inspection expose different operations, whose implementations may run beside the agent or near a remotely hosted application. To make these capabilities available to the shared decision loop, we use a common tool interface that allows their implementations to evolve independently.

Tool definitions describe operations and their arguments, while dispatch connects each call to its implementation and returns the result to the requesting step. The Agent Kit integrates local functions and tools exposed through Model Context Protocol (MCP), whose standard tool descriptions and invocation messages support capability discovery and execution (Model Context Protocol, 2025). Both paths feed the same interaction history and recording services. Platform-specific controls can therefore be added through tool bindings while reusing task progression and feedback handling.

To reduce the repeated inference overhead of long GUI workflows, we allow the model to issue multiple action calls in a single response. The runtime executes these calls serially in the specified order, allowing actions that can be planned together to share one model invocation.

For computer use, tool feedback also needs to describe the state after an operation. Post-action observation hooks attach fresh screenshots or page observations to state-changing tool results, giving the next decision a current view of the interface. Section 3.2.1 develops this action-aware feedback design. The same tool separation supports centralized agents operating remote devices, whose collection lifecycle is described in Section 3.4.

3.1.3 Desktop and Web Instantiations

We build desktop and web agents on this runtime by supplying the observations, tools, and execution context appropriate to each platform. Table 1 compares the two instances along these interfaces:

- **Desktop agent.** The agent uses desktop screenshots to observe applications and mouse and keyboard tools to click, type, scroll, and drag. Its platform adapter manages the active application, foreground window, and input focus so that an action reaches the intended control. This supports workflows across application windows and files, with screenshots, action results, and output artifacts retained for later assessment. Appendix H.2.1 lists its actions and parameter schemas. Appendix J.2 follows a recorded animation task in which these same desktop actions open a terminal, enter scripts and return to the presentation UI for playback checks.
- **Web agent.** The agent observes page screenshots together with DOM-derived information. Browser tools support navigation, element-reference or coordinate-based actions, form entry, and tab operations. Its adapter maintains the active tab, page state, and element references as navigation changes the interface. The resulting record includes page observations, action results, and browser artifacts. Appendix H.2.2 specifies the corresponding browser tools and argument constraints.

The adapters handle these platform differences, while both agents reuse model invocation, request progression, context management, and trajectory recording.

The same separation provides a path to additional platforms. Mobile extensions, for example, would supply screen observations, touch operations, and application/session handling around

Table 1. Desktop and web interfaces used in this report. Platform adapters provide observations and action execution, while the Agent Kit supplies the shared runtime and trajectory-recording services.

Interface	Desktop agent	Web agent
Interaction surface	Applications, windows, and system interfaces	Pages, tabs, and browser state
Policy observations	Desktop screenshots	Page screenshots and DOM-derived information
Environment-facing actions	Mouse-and-keyboard GUI operations	Navigation, element-reference or coordinate actions, keyboard input, forms, and tab operations
Execution context	Active application, foreground window, and input focus	Active tab, page state, and element references
Retained evidence	Screenshots, action results, and application artifacts	Screenshots, page snapshots, action results, and browser artifacts

the shared runtime. Across the desktop and web instantiations, this reusable execution foundation also exposes a common challenge: carrying earlier constraints and intermediate results into decisions made much later in a workflow.

3.2 Sustaining Long-Horizon Interaction

Long-horizon execution is a central challenge in practical computer use. Real software work can require many fine-grained GUI operations across applications, files, and intermediate artifacts, with later actions depending on earlier discoveries and decisions (Yuan et al., 2026). Editing a document may still require a value read from a spreadsheet and an output requirement encountered much earlier. Meanwhile, repeated visual observations and tool feedback enlarge the history supplied to each decision (Lu et al., 2026, Lv et al., 2026). Sustaining these workflows requires controlling context growth while keeping current state, earlier progress, and task-relevant knowledge available for continued action.

We separate the *decision context* presented to the model from the *complete execution record* retained for inspection and learning. To keep later decisions informed within a limited context budget, we design three complementary mechanisms: *action-aware visual context* for recent states and operations, *task-preserving compaction* for earlier progress and remaining work, and *structured working memory* for facts and decisions the agent chooses to retain. Figure 5 follows one invoice workflow across these mechanisms. The visual window shows the steps leading to export, the summary keeps export marked as pending, and an agent-written note carries the overdue total into the report. Each view supplies information needed for continued action without replacing the full execution record.

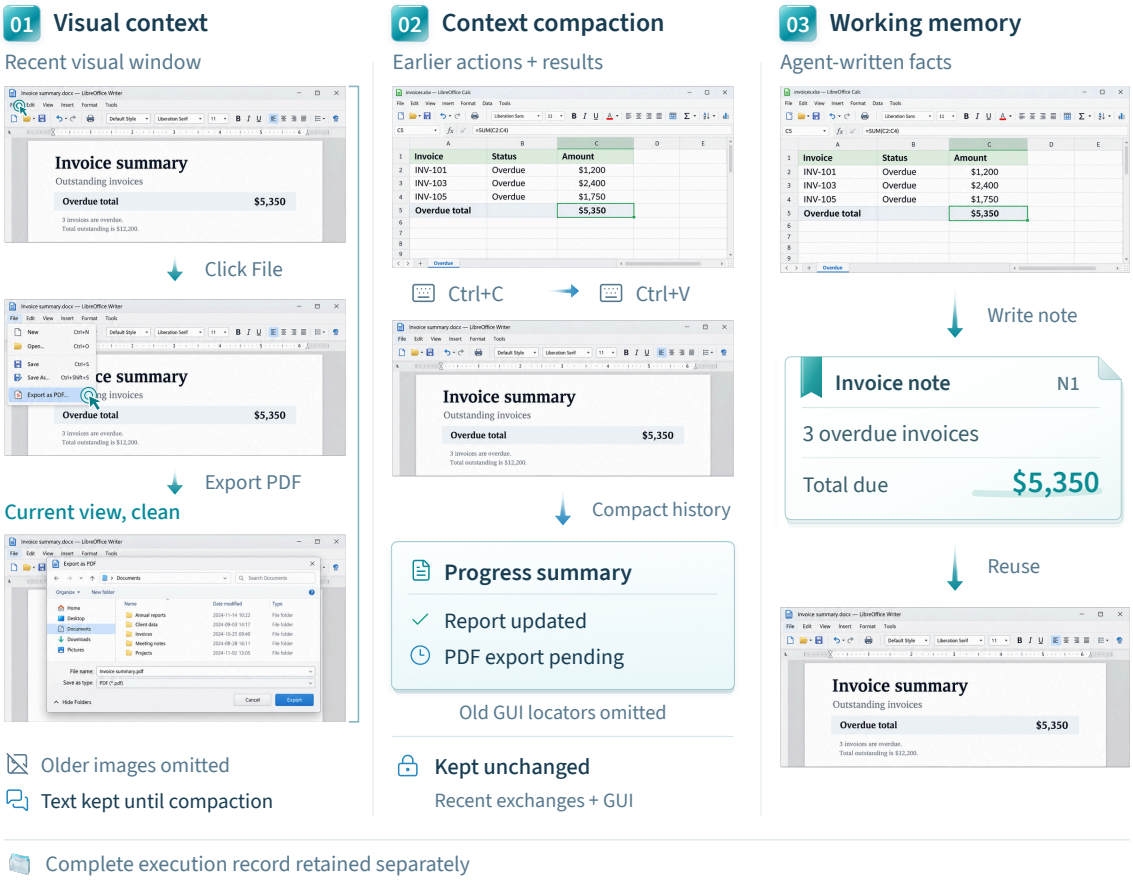


Figure 5. Context for long-horizon execution. The complete trajectory is retained separately from the next-action context. The visual window contains historical pre-action overlays and the latest clean tool-result image. Textual action and result history can outlast image eviction until compaction. Compaction summarizes earlier actions and results, omits old GUI locators, and preserves recent complete exchanges and the current GUI. An active agent-written note carries the overdue total from workbook to report. PDF export remains unconfirmed. GUI scenes and the displayed visual-window size are illustrative, with intermediate interactions omitted.

3.2.1 Action-Aware Visual Context

Repeated GUI interaction produces an image-heavy history whose size grows with the workflow (Lu et al., 2026). Its visual content serves two related purposes: the current screen guides the next action, and recent states help the agent interpret what it just did. The resulting screen alone can leave the location of a preceding click or the path of a drag unclear, particularly after a window or page changes. Visual context therefore needs to preserve both current state and the spatial meaning of recent operations within a limited image budget.

We use a sliding window to provide a **bounded visual history** of up to k recent frames, including the latest clean screenshot. Textual action and tool-result history survives image eviction, retaining a longer account of the interaction until the text-history policy calls for compaction. To keep the newest frame current, post-action observation hooks supply **fresh post-action feedback** alongside state-changing tool results. The next policy invocation receives this screenshot directly, avoiding a dedicated model turn solely to request the resulting screen.

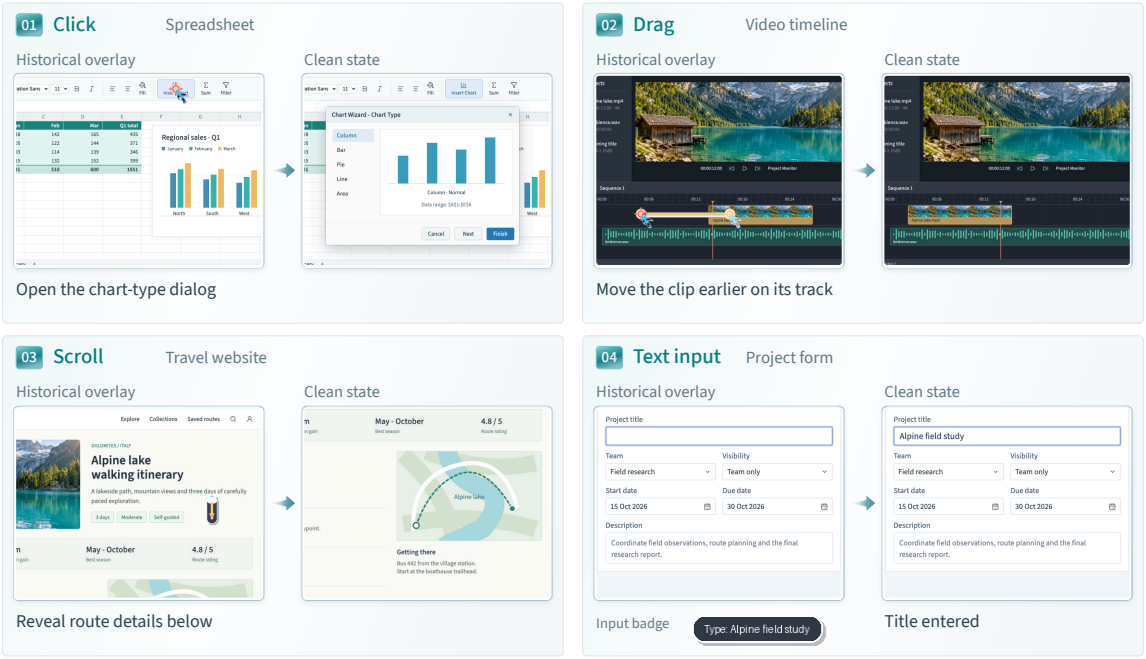


Figure 6. Action-aware visual feedback. Matched crops show the historical pre-action view with its NorthCUA overlay (left) and the clean resulting state (right). **01** A click opens the chart dialog. **02** A drag moves a video clip earlier on its track. **03** Scrolling reveals route details below. **04** A text badge records the payload entered into the focused field. Screenshots come from interactive GUI mockups with generated scenery. Overlays use the unmodified runtime renderer. The input badge is enlarged below its historical view.

To help the agent connect its previous operation with the change now visible on screen, we design a complementary action view in the historical frames. Visual prompting work shows how spatial marks can make image regions easier to refer to (Yang et al., 2023). Our **historical action overlays** mark the corresponding *pre-action* image, using different cues for different operations. As shown in Figure 6, a click marker locates the control that opens a dialog, a drag path links the clip's old and new positions, and a scroll cue identifies the direction in which the page moved. Text and key badges record operations that are less clear from a spatial marker alone. Figure 6 pairs each annotated pre-action view with its clean resulting state to make the operation and observed change easy to compare. In the runtime's decision context, historical overlays identify recent operations, while the latest clean screenshot provides the current interface for the next decision. Replacing an earlier clean frame with an overlay changes the image history. The training-segment rule in Section 5.3.3 preserves the input seen before that change.

This bounded visual history preserves both the spatial meaning of recent actions and the current interface state. The textual record continues to accumulate across these visual windows, creating the next context-management challenge.

3.2.2 Task-Preserving Context Compaction

Bounding images leaves another source of context growth: assistant messages, reasoning, tool feedback, and intermediate task information accumulate throughout the workflow. Earlier constraints and discoveries may still be needed when this text approaches the model's usable input budget. Context-management methods consolidate long histories to retain useful information

within finite prompts (Ye et al., 2026). We adapt this principle to computer use by designing task-preserving compaction around three priorities. When older history needs to be summarized, we preserve the immediate action–feedback context, retain task progress, and avoid carrying stale GUI locators into later decisions:

- **Protect current interactions.** The task request, latest GUI observation, and latest few steps remain protected. Assistant actions stay grouped with their tool results, and open exchanges remain intact until their responses arrive. This preserves the immediate action–feedback context, including the current GUI observation whether it appears in a user message or a tool result.
- **Summarize task progress.** The summary organizes goals, constraints, supported facts, decisions, completed and failed work, current state, open items, and artifact references around task continuation. Desktop and web profiles instruct the summarizer to retain semantic software state and observed effects while condensing routine navigation. For example, an extracted spreadsheet value, its source file, and a pending export requirement can therefore remain available after the navigation history is compressed. The instructions also distinguish issued actions, observed effects, and unresolved completion claims: an export command and an observed saved file represent different evidence about progress.
- **Handle transient GUI locators.** Coordinates and element references can become stale after a page update or application switch. The summary-input projection filters these volatile fields, and the summary instructions exclude old locators from current executable state. The retained current observation remains available for grounding subsequent actions.

Later compaction updates the existing summary with newly eligible history, with source references available for inspection. This gives continued work a compact account of its dependencies while keeping the full execution record available for outcome assessment. Alongside this budget-triggered consolidation, the agent can actively preserve important information as it is discovered.

3.2.3 Structured Working Memory

During a long task, an agent encounters information with clear future value: a figure read from another file, a requirement for the final deliverable, an unsuccessful approach, or a plan for remaining work. Assistant messages can mention these details in the chronological history, where their later visibility depends on history retention and summarization. Giving selected information an explicit working-memory representation lets the agent decide what to retain and revise during execution, following the broader principle of agent-controlled memory management (Packer et al., 2024).

To let the agent preserve useful information when it discovers it, we design a dedicated text-only `note` tool. The agent can call this tool proactively to record a spreadsheet value before switching to a document, or to preserve a correction while revising an unsuccessful approach. The right-hand panel of Figure 5 shows this transfer: note N1 records the overdue total from the workbook for later use in the report. Notes are stored independently of the history summary and can be presented in subsequent decision contexts. This complements compaction: a summary consolidates accumulated work under budget pressure, while a note explicitly preserves information the agent expects to use again.

Notes serve three purposes: (i) retaining reusable knowledge and choices through *facts*, *constraints*, and *decisions*, (ii) recording lessons and risks through *corrections* and *precautions*, and (iii) tracking progress and unfinished work through *checkpoints*. This organization encourages concise entries with a clear role in later decisions.

As work progresses, replacement notes can supersede outdated entries. The memory service maintains them within the authorized task and session, then supplies a budgeted view of active entries to prompt construction. A corrected value or revised plan can therefore replace the corresponding obsolete item in working context, keeping remembered information aligned with the agent's evolving understanding of the task.

Together, these mechanisms keep the current screen, accumulated progress, and reusable knowledge available for continued action. The complete execution record connects these decisions to the resulting workspace, providing the evidence for outcome evaluation and subsequent training. Appendix H.1 gives the runtime settings, and Appendix E.1 and Section 5.3.3 describe their training interfaces.

Appendix J.1 follows a photo-delivery task in which a payment constraint remains available from reconciliation through the final email.

3.3 Agentic Outcome Evaluation

Selecting GUI trajectories for learning requires accurate outcome judgments across a growing variety of tasks. Dedicated verifiers become difficult to write and validate when correctness depends on content, presentation, or application-specific state (Rosset et al., 2026). We therefore use an adaptive evaluator that interprets the declared task requirements and gathers the evidence needed to assess the work. Figure 7 shows how recorded work and active inspection together support an evidence-linked assessment.

We build the evaluator as an *agent-as-a-judge* (Zhuge et al., 2025), reusing Agent Kit tools. After the acting phase ends, it selectively retrieves saved records and artifacts and can inspect the original workspace using authorized shell, GUI, and browser tools. It adapts its inspection to each requirement and seeks further evidence when the available observations leave the outcome unclear. This combines reusable inspection capabilities with task-specific judgment, extending assessment to requirements that lack dedicated executable checks. Task-grounded rubrics guide this process, and multidimensional reports connect the findings to experience selection and failure analysis. In Figure 7's export example, an issued action does not establish that the saved PDF contains the requested information. The evaluator checks file presence and opens the result to compare its contents with the source, gathering the evidence behind its final report. We require this report to associate each criterion-level judgment with its evidence, uncertainty, and evaluator operations.

3.3.1 Artifact and Live-State Inspection

The evaluator receives the task request, assessment criteria, and a compact view of the execution records. Requirements may concern an earlier action, a final file, or a state that is visible only inside an application. To reach these different forms of evidence, it uses two complementary tool groups:

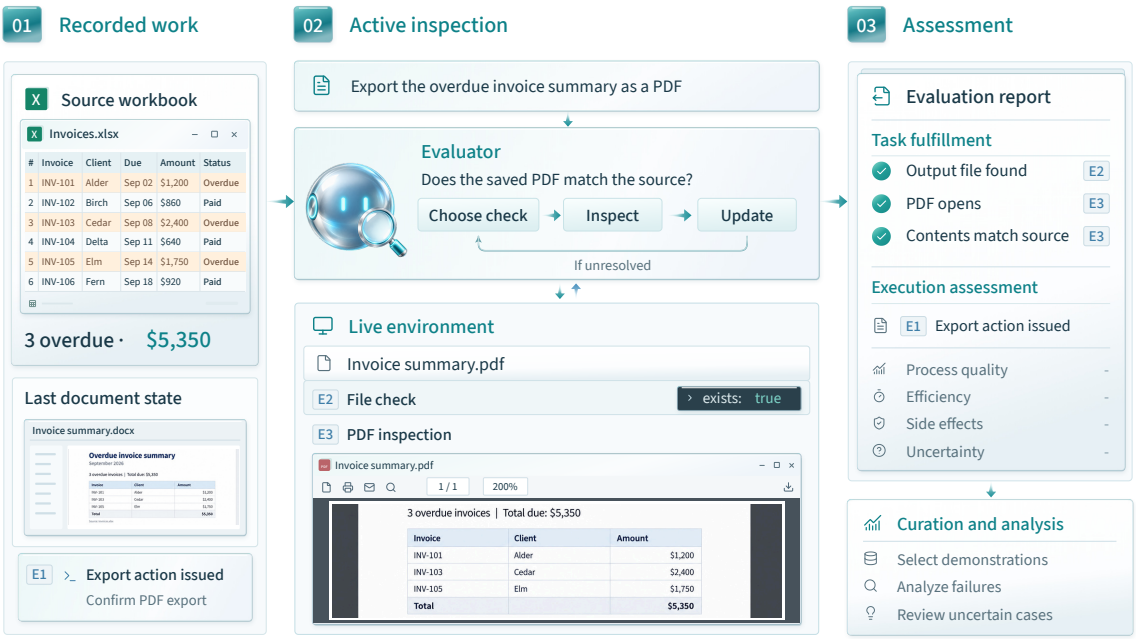


Figure 7. Agentic outcome evaluation. Open questions guide selective retrieval and live inspection in the task environment. E1 records an issued export, E2 establishes file presence, and E3 exposes PDF contents for comparison with the source workbook. Unresolved questions prompt further checks and can remain unresolved in the final report. Evidence-linked findings distinguish task fulfillment from execution assessment. GUI views, tool results, actions, and findings are illustrative. Generated document views are cropped, intermediate actor actions are omitted, and selected quality fields are shown without ratings.

- **Recorded evidence.** Retrieval tools expose selected steps, screenshots, tool results, and retained artifacts. They let the evaluator inspect relevant parts of a long trajectory as questions arise, while the full record remains available outside its current context.
- **Live task state.** Live tools inspect the original device or sandbox through fresh screenshots, read-only shell commands, and GUI or browser operations. Web probes can reuse the actor's browser session, including its tabs and page state. This access makes environment-resident results available for checking while the workspace is still alive.

Appendix H.2.3 lists the evaluator's retrieval tools, authorized live interfaces, and report schema.

The evaluator chooses between these sources according to the requirement under review, continuing its investigation when an observation leaves an important question open. This environment-aware approach is also studied in Shi et al. (2026). For a document export, for example, a recorded action can show that export was attempted, a file check can establish whether an output exists, and an application inspection can reveal its rendered contents. Comparing these contents with the task inputs supports judgments about their correctness. Each observation answers a different part of the acceptance question. Selective retrieval and live access thus connect the history of an operation to its actual result.

To preserve attribution, the assessment protocol pairs pre-inspection terminal evidence with a separate record of evaluator operations. Each judgment is linked to the submitted work and the observations used to assess it, making the investigation traceable to the original attempt.

Table 2. Evaluator reporting dimensions and their uses in trajectory curation and failure analysis.

Dimension	Assessment	Analysis and selection use
Task completion	Criterion fulfillment, overall verdict, and completion rating	Identify supported completion examples and unresolved requirements
Process quality	Suitability of actions, tool use, and avoidable mistakes	Distinguish reliable demonstrations from fragile paths to the same result
Efficiency	Repetition, detours, unnecessary steps, and recovery behavior	Compare successful paths and apply cohort-specific efficiency criteria
Safety and side effects	Sensitive operations, unsolicited changes, and observed risks	Review unsafe or undesired behavior separately from completion
Failure diagnosis	Failure mechanism, controllability, and supported divergence location	Analyze capability gaps and distinguish them from environment blockers
Judgment uncertainty	Confidence, missing information, and uncertain verdicts	Route unresolved cases for further checking

The rubric then connects this evidence to task fulfillment and execution quality. Appendix H.1 details the inspection and attribution protocol.

3.3.2 Task-Grounded, Multidimensional Rubrics

The value of a GUI trajectory depends on both its result and the behavior it demonstrates. Successful attempts can contain unnecessary detours or unintended changes, while an attempt blocked late by the environment can contain useful completed subtasks. Data curation therefore needs answers to two complementary questions: (i) *which task requirements were fulfilled*, to establish what the trajectory accomplished, and (ii) *how the agent performed the work*, to assess the behavior available for learning. We therefore design the rubric to combine task-grounded acceptance criteria with multidimensional execution assessment (Rosset et al., 2026).

For **task fulfillment**, the request and declared acceptance conditions define the intended outcome. The rubric expresses these obligations as specific, individually checkable questions. Export tasks may require the right contents, destination, and format. Explicitly required procedures also form part of acceptance. Request-derived criteria are generated before scoring the trajectory, with the original task requirements governing their interpretation. Assessing these obligations individually identifies complete, partial, and unresolved fulfillment, including which requested subtasks have supporting evidence.

For **execution assessment**, the report records process quality, efficiency, side effects, and failure diagnosis alongside judgment uncertainty. These dimensions explain the path taken to the result: repeated ineffective actions may reduce its value as a demonstration, while a successful recovery can provide useful behavior to study. Failure diagnosis distinguishes agent errors from environment blockers, and uncertainty identifies judgments that need further checking. Findings are linked to the relevant requirements and recorded steps so that their scope can be inspected. Table 2 summarizes the dimensions and their uses.

Together, the rubric and execution report describe both the result achieved and the behavior that produced it. Their links to specific requirements and recorded steps make the judgments useful for selecting complete demonstrations, reviewing local segments, and analyzing recurring errors.

3.3.3 From Evaluation to Experience Selection

We use the report's separate rubric scores to select data for different learning goals. Task fulfillment identifies completed work, while process quality, efficiency, and side-effect assessments let us adjust selection thresholds for each data cohort. For example, a cohort of concise successful demonstrations can apply stricter efficiency criteria, while a cohort meant to teach recovery can retain useful corrections and exploration within successful attempts.

At the **trajectory level**, these criteria select complete demonstrations. At the **segment level**, requirement-linked evidence can guide review of completed subtasks or recovery sequences, with the original decision context and local outcome kept alongside the behavior. Figure 7's report separates completion from execution quality so that these dimensions can be used together in selection. Section 5.1 applies the reports to trajectory curation, and Section 5.2 retains complete episodes for long-horizon SFT.

3.4 Reliable Batch Rollout at Scale

The preceding mechanisms turn one task attempt into assessed experience. Producing a large training corpus requires many such attempts to run in parallel, each with its own software state. Their preparation, GUI execution, evaluation, and recording take different amounts of time and consume different resources (Xie et al., 2024, Yuan et al., 2026). Simply launching more agents can therefore create a backlog at another stage, while an interruption after execution can leave useful work without a complete record.

To support efficient large-scale collection, we design a **Batch Rollout Service** that coordinates the entire task lifecycle. Following the environment-lifecycle perspective also explored in Peng et al. (2026), we combine centralized agent control with isolated workspaces, prepare upcoming tasks alongside ongoing execution, and manage load and recovery through record delivery. Figure 8 shows the two relationships behind this design. Panel (a) connects one controller to independent device workers. Panel (b) shows how their lifecycles overlap: the same-time marker crosses evaluation, execution, and preparation in three different attempts. The service coordinates this concurrent work so that completed attempts produce usable experience records.

3.4.1 Remote Execution with Centralized Agent Control

As collection grows, prompts, model settings, and agent logic continue to change. Placing that logic inside every task environment would require updating those environments whenever the agent changes. We therefore **separate centralized agent control from device execution**, as shown in Figure 8(a):

- **Centralized controller.** One service manages multiple agent runners and their device workers. It maintains prompts and agent configuration, assembles context, invokes the model, and dispatches actions.
- **Device worker.** Each worker runs inside one task's isolated sandbox. It executes tool calls and captures observations from that environment. The worker starts with its sandbox and ends when the sandbox is released.

The controller maintains a long-lived WebSocket connection to each worker, sending actions and receiving observations over the same connection. Related distributed device orchestration

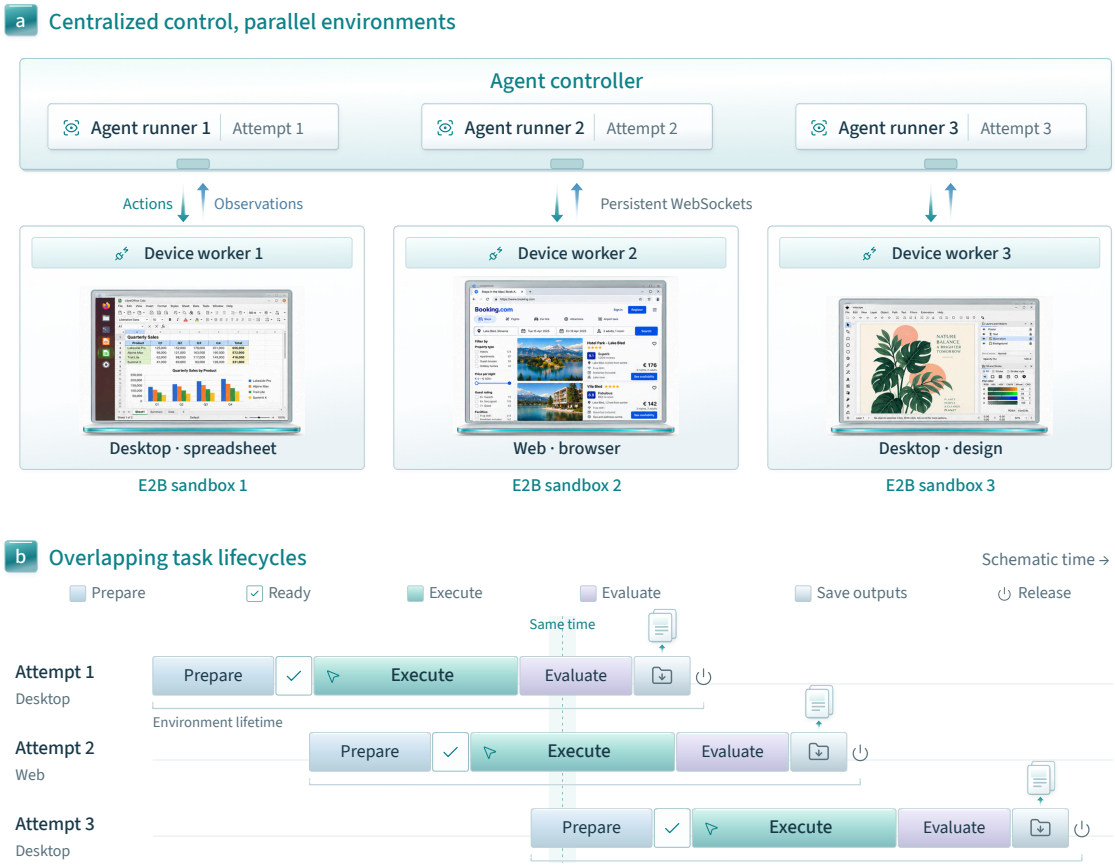


Figure 8. Batch rollout at scale. (a) A centralized controller manages agent runners connected to device workers in independent E2B sandboxes over long-lived WebSocket connections. Each worker shares its sandbox's lifecycle. (b) Staggered lifecycles overlap preparation, execution, and evaluation. Ready intervals mark prepared environments awaiting dispatch. Sandboxes remain available through live checks and output capture. Record icons mark captured outputs that persist after release. The vertical time slice marks concurrent stages. Timings and concurrency are schematic, and GUI scenes are generated illustrations.

appears in Zhang et al. (2025d). Here, each runner communicates with the worker assigned to its task, keeping the concurrent attempts separate.

This division lets us update compatible prompt and model configurations centrally for later batches while reusing the same device-worker implementation and software templates. The device layer remains focused on observation and tool execution, so routine agent changes do not require changing that layer. We save each batch's configuration for resumed collection and retain its trajectories on the controller side, where they survive sandbox release. With control and recording shared, scaling execution then depends on preparing enough independent task environments.

3.4.2 Isolated Environments with Pipelined Preparation

A new attempt needs the files and software state specified by its task. Reusing a workspace changed by an earlier attempt could alter those starting conditions (Xie et al., 2024). We therefore give each attempt its own initialized sandbox in our fresh-start collection mode. For Linux desktop and browser tasks, we choose E2B to create and manage these isolated environments

(E2B, 2026b), with its desktop extension providing the graphical workspace (E2B, 2026a). Custom templates supply the operating system and shared applications, while per-task setup uploads files, launches applications, and prepares the required state. This lets tasks reuse a software template without inheriting another attempt's changes.

Creating a fresh environment takes time, so preparing tasks only when an agent is ready would leave collection waiting on setup. We address this with a **task-bound preparation queue**: upcoming environments are prepared while earlier attempts are executing or being evaluated. Figure 8(b) makes this overlap visible in the offset Prepare blocks. Each ready workspace is assigned once to its matching task, provided enough time remains for execution and cleanup. It stays available through the required outcome checks and output capture, then is released. Preparation can thus run ahead without sacrificing fresh starting conditions or losing the evidence needed for assessment.

3.4.3 Completion-Oriented Scheduling and Recovery

Preparation, execution, and evaluation progress at different rates. Independent admission can leave prepared environments waiting for execution or completed attempts waiting for evaluation. We coordinate admission with two controls. **Stage-specific concurrency limits** bound the number of attempts admitted to each stage. Because resource demand can vary within these limits, **controller-pressure feedback** slows new admissions when the shared controller remains busy. The first control limits local demand, while the second responds to load across the concurrent attempts.

When a connection is lost or a downstream process fails, we first check whether the affected attempt is still running or has completed. Completed execution records can support retries of record-based evaluation or artifact processing, preserving the work already done. If the task itself needs to be rerun, we start it in a fresh environment.

Before delivery, we check that the required trajectory, observations, and assessments are complete and linked to the same attempt. Appendix H.3 gives the configuration and recovery details.

Together, these designs carry the same runtime from individual task execution to sustained delivery of complete, inspectable experience. The Experience Flywheel in Section 4 builds on this foundation to construct tasks and curate the resulting records for model training.

4 Experience Flywheel: Scaling Grounding and Agentic Experience

The NorthCUA Harness provides the execution infrastructure, but training a computer-use model also requires a large and varied supply of experience. The model needs examples of which control a request refers to, as well as trajectories that show how successive decisions achieve a complete goal. We design the Experience Flywheel to construct and curate these two levels of supervision. **Single-step grounding** pairs a screenshot and a query with the location of one visible control. **End-to-end task construction** supplies user goals, prepared workspaces, and outcome checks from which the harness can collect multi-step trajectories.

These levels require different construction methods. For grounding, we start from visible controls and generate language that identifies them. For complete tasks, we connect available software and content, sourced requests, or verified seed tasks to the environments and assessments

they need. This gives three complementary routes for scaling intents, environments, and verifiers. Executing the resulting tasks supplies experience for learning, although successful trajectories can remain scarce for the hardest workflows. We address that remaining gap through recursive exploration that retains useful skills and verified progress across attempts. Together, these methods supply localization examples and curated trajectories for the staged learning in Section 5.

4.1 Single-Step UI Grounding

Single-step grounding takes a screenshot and a natural-language query, then predicts the location of the requested control (Cheng et al., 2024). Each training example contains a screenshot, a query that identifies a visible control such as a button or toolbar icon, and the control's coordinate in the original image. The main construction challenge is to write queries that identify the same control across different visual states and ways of describing an operation, especially in specialized applications (Lu et al., 2024, Li et al., 2025a).

We construct these examples through two complementary routes:

1. **Broad visual and structural coverage.** We use screenshot evidence and available interface metadata to describe visible controls across diverse applications and states (Figure 9).
2. **Documentation-grounded functional localization.** We retrieve application knowledge when a control's appearance leaves its function unclear, then use the matched function to generate queries (Figure 10).

The first route expands interface and language coverage. The second fills gaps that visual evidence alone leaves unresolved. Both preserve the target's original location and produce the same screenshot–query–coordinate format, so their candidates can pass through shared quality and coverage checks.

4.1.1 Broad Visual and Structural Coverage

To obtain broad grounding coverage, we first need screenshots of varied interfaces, then a way to turn their visible controls into language targets. We use the three-step procedure in Figure 9. Panel (a) collects interface states, including states encountered during agent work. Panel (b) associates one visible control with its appearance and original location. This target record gives the query generator in (c) a fixed referent: it can vary how the control is described while keeping the location label unchanged.

To collect these states, we combine public UI datasets, web-collected GUI screenshots, and screenshots from agent trajectories, including real Linux desktop interaction. Trajectories contribute intermediate states such as open dialogs, expanded menus, and partially completed forms. We then use OmniParser, a screen-parsing system that detects interactive regions and describes icons, to obtain candidate control boxes and descriptions (Lu et al., 2024). Accessibility annotations, when available for the same state, add control names and roles. These sources provide complementary evidence for identifying a target. Visual parsing supports screenshots collected without accessibility metadata, while names and roles help connect visible controls to the language used in requests.

For each target, we consolidate this evidence into one control record that keeps the crop and original image position together with the available name, role, color, appearance, and surrounding

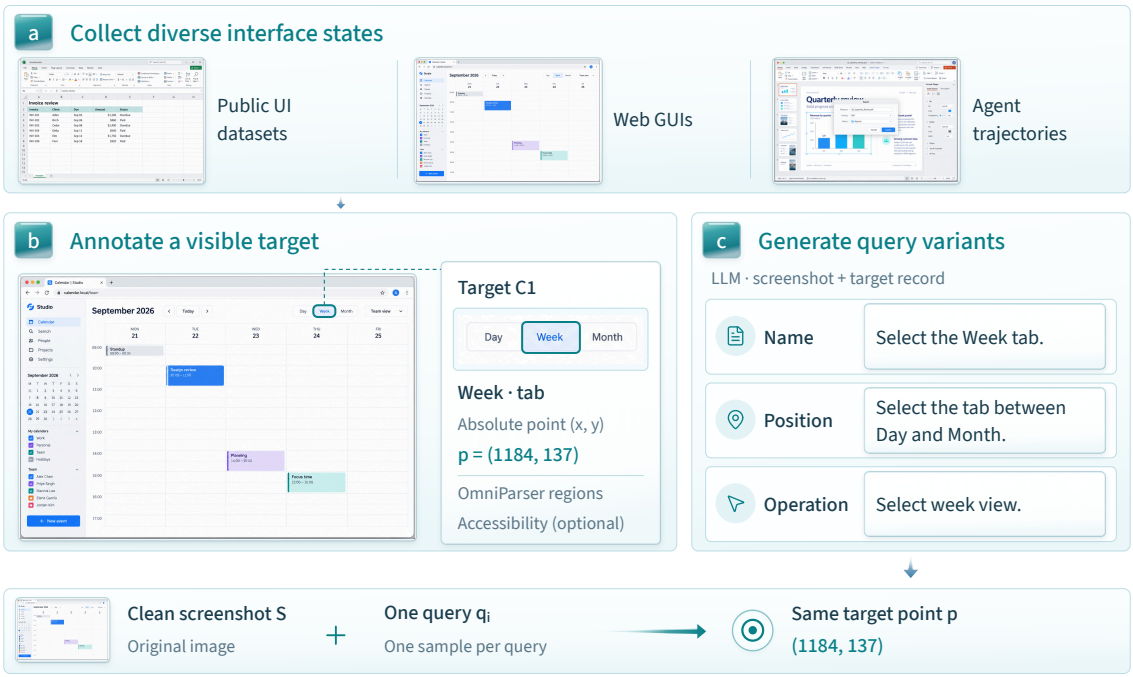


Figure 9. Constructing GUI grounding examples. (a) Screenshots cover public UI datasets, web-collected interfaces, and states encountered in agent trajectories. (b) Visual annotations and available accessibility metadata describe one target in its original screenshot. (c) Name-, position-, and operation-based queries each form a separate candidate example with the same clean screenshot and absolute target point. The enlarged Day–Week–Month control group supplies the local evidence for these descriptions. GUI assets are illustrative. Target outlines explain construction, while training uses the unmarked 1536×1024 source image.

context (Zhang et al., 2026). For the calendar example in Figure 9(b), the record identifies the blue *Week* tab between *Day* and *Month*, together with a point inside its clickable region. The full screenshot preserves the neighboring controls needed to make a description specific.

With this record in place, we use an LLM to generate several queries from the record and full screenshot. The descriptions refer to the control's name, appearance, position, or an operation supported by the available evidence. In the example, *Select the Week tab* names the control directly, while *Select week view* expresses its purpose. Each query becomes a separate candidate example paired with the clean original screenshot and the same coordinate. This variation broadens the language supervision without changing which control the model learns to locate (Chen et al., 2026a).

This route supplies broad coverage of interfaces and visual states. When a control's operation depends on application-specific knowledge, the documentation route provides the missing functional description.

4.1.2 Documentation-Grounded Functional Localization

Some controls belong to specialized applications whose functions are hard to infer from a screenshot alone. Parsing may locate a toolbar icon while a general-purpose LLM remains unsure what the icon does. This distinction matters when several visually similar controls perform different

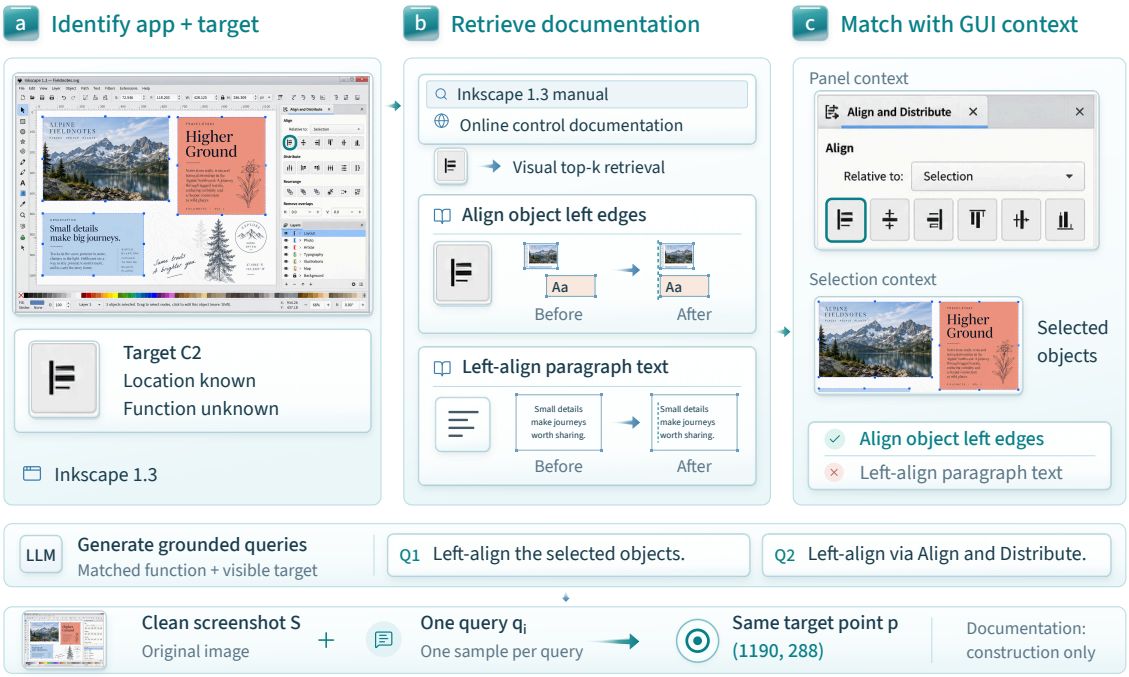


Figure 10. Documentation-guided control disambiguation. (a) The screenshot supplies the application, version, and located target. (b) Online manual search retrieves control images and descriptions for visual top-k candidates. One aligns the left edges of separate objects, while the other aligns paragraph text within a fixed text object. (c) Panel and selection context determine the function match, which guides query generation. Each query retains the original screenshot and target point. The GUI, documentation cards, and before/after operation diagrams are illustrative. Documentation supports sample construction.

operations, as in a layout editor with alignment controls for different parts of a document (Wang et al., 2025c).

We design a data-construction agent to supply the missing application knowledge before query generation. Figure 10 illustrates three steps. The agent identifies the application and target, then searches online documentation for candidate functions. It uses the surrounding GUI to select the function that fits the visible context.

In the alignment example, the same-looking alignment symbol could refer to moving objects or formatting text. Matching its documented function resolves what a query can reliably ask the agent to do with the already located control.

Identify the application and target. This route begins with the same OmniParser-based screen parsing, which provides candidate control boxes and crops while preserving their locations in the original screenshot (Lu et al., 2024). Given the screenshot and a target crop, the construction agent identifies the application and any visible version information, such as Inkscape 1.3 in panel (a). These details constrain the documentation search to the software in which the control appears, while the crop provides the visual target to match.

Retrieve documented functions. Using that application identity, the agent searches online user guides and manuals and extracts illustrated controls with their function descriptions. We keep each illustration with its function description and page context, so that a visually similar icon retains the information needed to interpret its operation. Image-embedding similarity then nar-

rows these references to top-k candidates (Radford et al., 2021). Panel (b) shows both parts of this step: the online manual search and the candidate operations retrieved for the target icon.

Match with GUI context and generate queries. Visual similarity supplies candidates but can leave the operation ambiguous. The agent therefore compares their descriptions with the full screenshot, neighboring controls, and active tool group. This contextual check matters because similar icons can denote different operations.

In Figure 10, object alignment and paragraph alignment provide plausible visual candidates. Object alignment moves selected canvas objects to align their left edges, whereas paragraph alignment left-aligns the text within a text object. The visible *Align and Distribute* panel identifies the relevant tool group, and the selected canvas objects provide complementary evidence for the object-alignment interpretation. The accepted match therefore links a specific function description to the control already located in the screenshot.

Once the function is resolved, we give the accepted description, target crop, and full screenshot to an LLM to generate query variants. One can request the documented operation, such as aligning selected objects along their left edges. Another can identify the same control through its tool group and appearance. All variants inherit the original target coordinate. Thus, application knowledge enriches the query while preserving the common grounding format. The documentation and retrieval candidates remain construction-side evidence, and the resulting samples enter the shared quality checks below.

4.1.3 Quality Control and Coverage

Both routes produce the same candidate format: a clean screenshot, a query, and a target coordinate. We use two complementary checks before adding a sample to the corpus:

- **Correctness check.** Geometry checks confirm that the point lies inside the annotated control. Multimodal review then checks that the control supports the described operation, the query refers to it, and the description is unambiguous among nearby controls. Documentation-derived examples also carry their accepted function match into this review.
- **Coverage check.** Deduplication removes repeated screenshots and near-identical descriptions. Selection retains useful variation across applications, UI states, control types, and query styles, so that different ways of referring to a control remain represented.

These checks address different reasons a larger candidate pool may fail to provide useful supervision: incorrect target labels and repetitive coverage. Geometry or identity failures return to construction when they can be repaired, while unresolved cases are excluded. Selection then retains accepted examples across interfaces and language styles. Appendix B records the annotation, matching, and curation settings.

The accepted examples enter mid-training as direct localization supervision. Screenshots acquired during multi-step collection can also enrich this pool with states encountered in actual software work. The next subsection extends the same construction logic from identifying one control to carrying out a complete user goal.

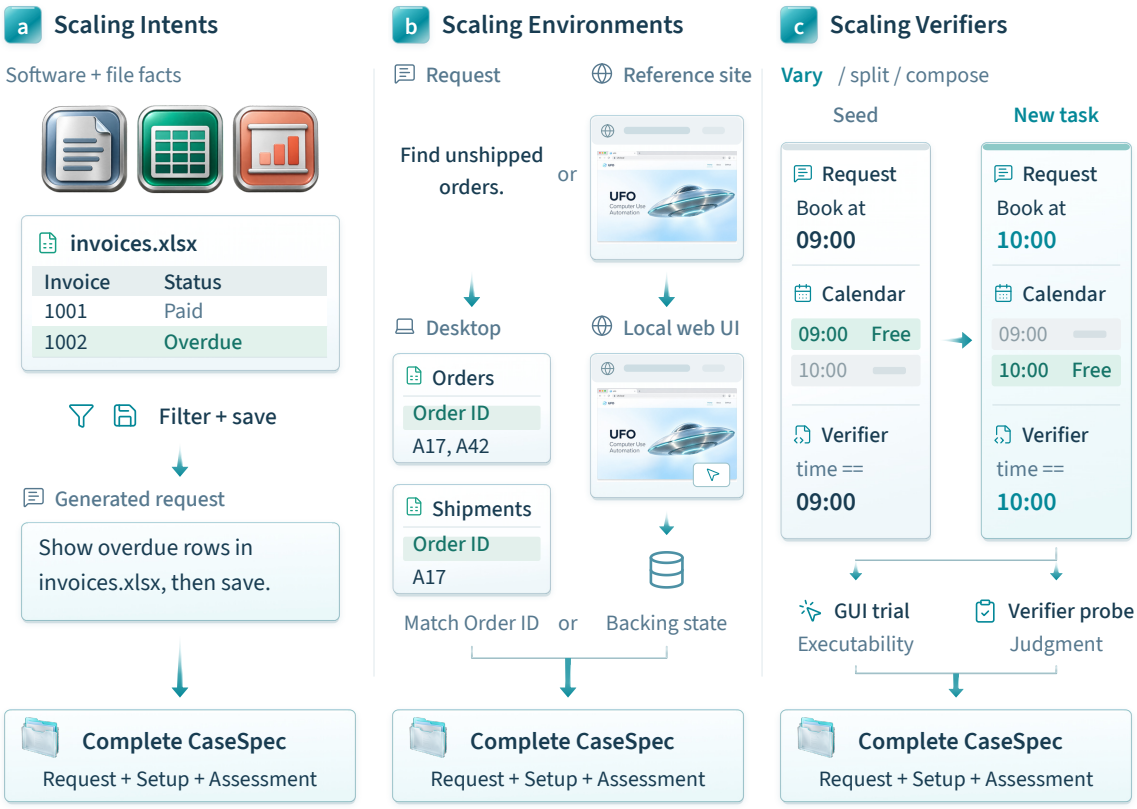


Figure 11. Three task-construction routes. Scaling Intents binds functions to file or website content before writing requests. Scaling Environments prepares files and applications for sourced requests or reconstructs website interfaces and backing state from references. Scaling Verifiers jointly adapts requests, setup, and executable checks, then tests tasks and verifiers. Each alternative route produces a complete task binding a request, environment setup, and outcome assessment. GUI trials test completion from the initial state, while result probes test acceptance of correct outcomes and rejection of incorrect ones. GUI assets and task packages are illustrative.

4.2 Scaling Intents, Environments, and Verifiers

Grounding examples teach the model to locate a requested control. Learning to plan and act across a complete workflow also requires interaction trajectories (Wang et al., 2025b, Jung et al., 2026, Wang et al., 2026a). To collect them, we must turn each user goal into an executable task by providing the workspace it needs and a reliable assessment of its outcome.

Figure 11 compares three construction routes by what they start with and what they generate:

- **Scaling Intents** starts from applications, files, or website content and generates requests supported by their functions and facts.
- **Scaling Environments** starts from sourced requests or reference websites and builds the files, applications, or local website needed to execute tasks in that setting.
- **Scaling Verifiers** starts from tasks whose outcomes can be checked by reliable scripts. It generates related tasks by adapting the request, setup, and executable verifier together.

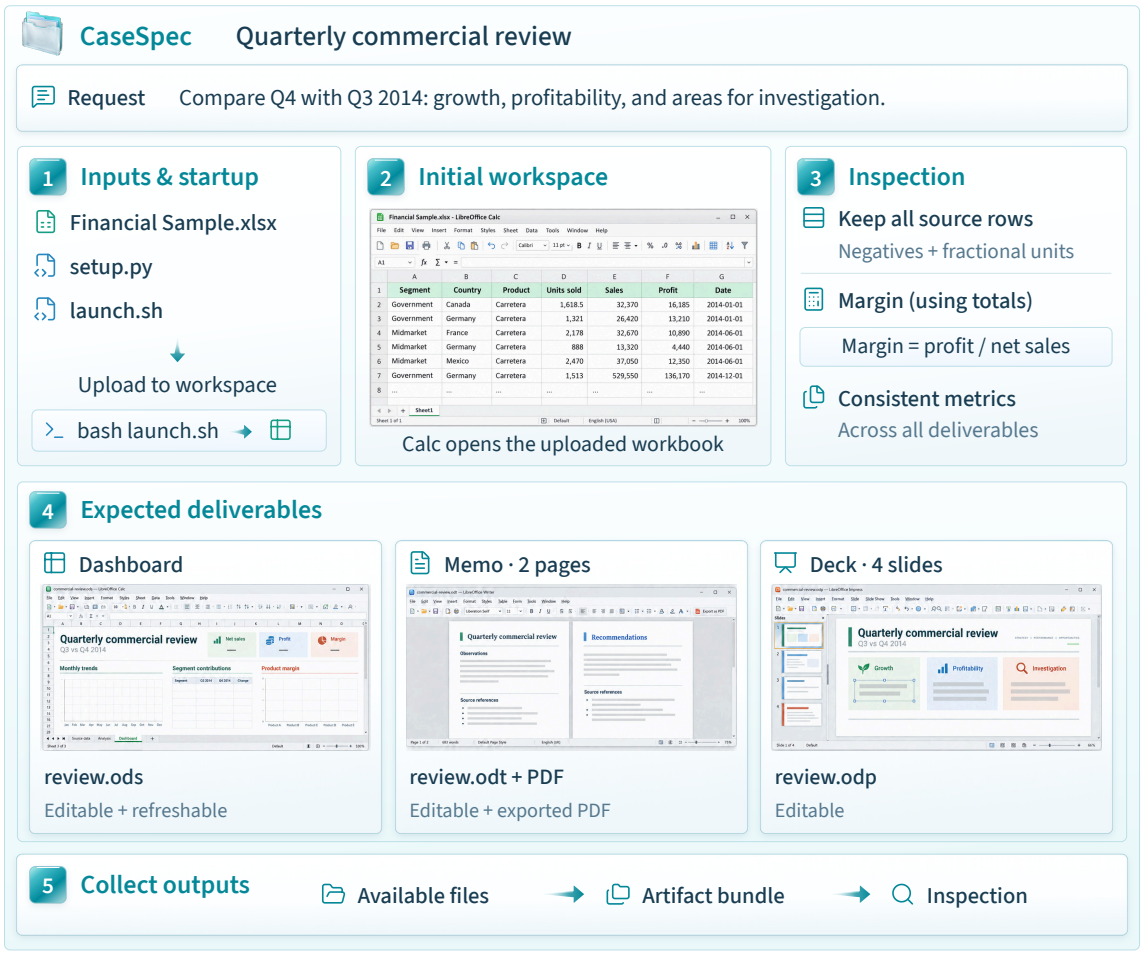


Figure 12. A concrete executable task contract. The saved commercial-review CaseSpec binds a request, input files, startup, and output collection. The initial-workspace illustration redraws representative source records in Calc. Requested deliverables comprise a refreshable dashboard, an editable two-page memo with PDF export, and an editable four-slide review deck. Inspection requirements summarize the query, while post-run collection gathers available files for assessment. GUI views, requested output layouts, abbreviated filenames, and the launch command are illustrative.

Each column ends with a complete task: a request, workspace initialization, and outcome assessment. The routes thus expand task supply from different resources while retaining the same requirements for execution and checking. Depending on the task, assessment uses an executable verifier or evaluator inspection of the declared acceptance conditions.

We package the output of each route in the same CaseSpec contract introduced in Section 2. Figure 12 shows one saved commercial-review task: its request asks for a quarterly comparison, its workbook and startup instructions establish the initial Calc workspace, and its requirements specify the deliverables and how they should agree.

The dashboard, memo, and review deck in Figure 12 are outputs the agent is asked to create. The post-run hook gathers available files for inspection against those requirements. This example shows what all three routes need to establish, even though they start from different resources.

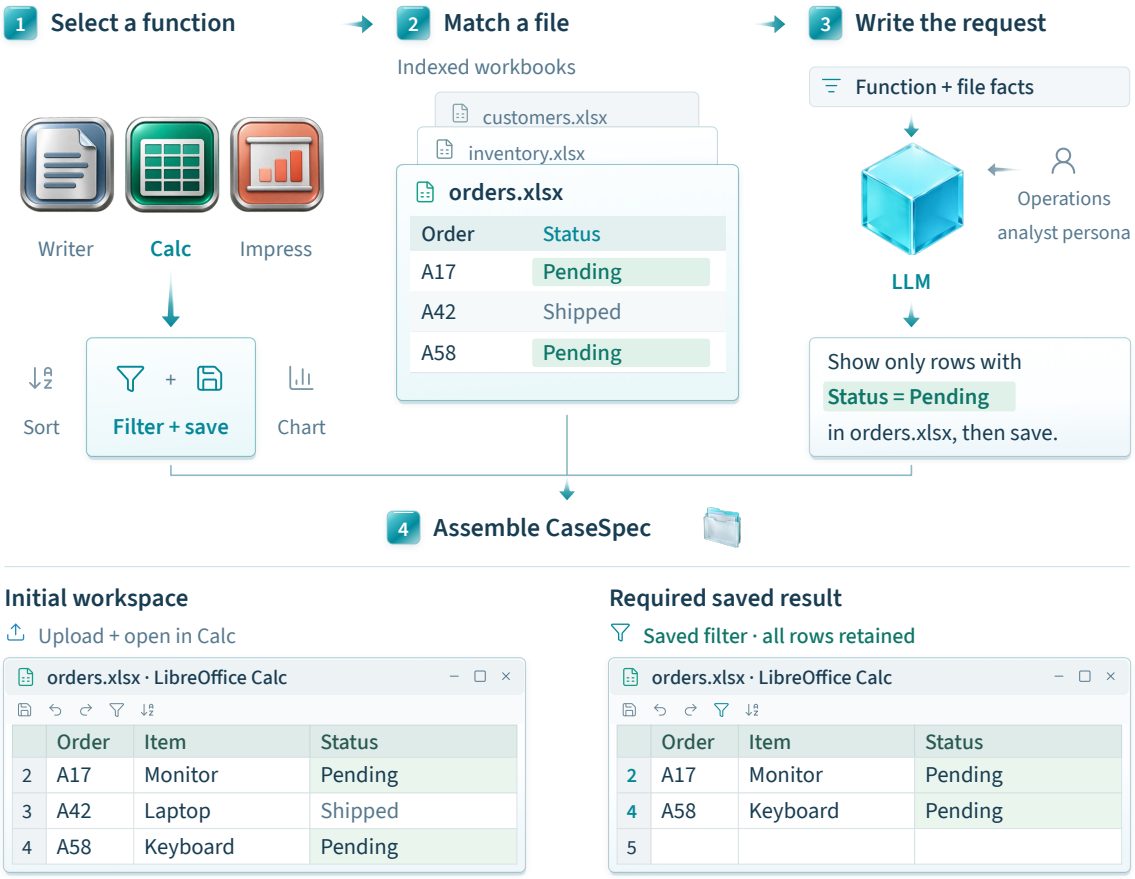


Figure 13. Constructing a desktop base task. The selected Calc function narrows compatible workbooks. The `orders.xlsx` summary grounds the `Status = Pending` condition, while the persona guides request wording. CaseSpec binds this request and file to startup instructions and the required saved filter. The required view retains A17 and A58 and hides A42 without deleting source rows. Software and GUI assets, records, and setup are illustrative. The right-hand view specifies a requirement, not an observed rollout.

4.2.1 Scaling Intents

Applications and working materials are readily available from software templates, public file collections, and websites. However, these resources do not come with a varied set of natural user requests grounded in their contents and capabilities. We design Scaling Intents to generate such requests from the operations the software supports and the content already present. This grounding gives the requested work concrete subjects and feasible actions, while variation in goals and phrasing broadens the collection. We first describe desktop base tasks, then extend the construction to dependent long-horizon workflows and web content.

Desktop base tasks. For a desktop base task, we need to connect a supported software operation to material on which it makes sense. Figure 13 illustrates the three steps used to generate the request. We first choose an application and one of its supported functions. That choice establishes the requirements for selecting a compatible source file. The selected file's content then supplies the facts from which an LLM writes a grounded request. Once the request is defined, we prepare its starting workspace and assessment to form the complete task shown in the lower panel.

Select a function. We begin with E2B templates that provide installed applications and their dependencies for office work, media, file management, browsing, and system settings. Each application has a profile of supported operations and readable file formats. We choose the target application and an eligible operation according to the desired coverage of software and functions. Selecting Calc and filtering, for example, determines both what the eventual request will ask for and what kind of input it needs.

Match a file. To supply these inputs, we collect tens of thousands of files from public and open resources and supplement them with agent-generated materials (Ge et al., 2026). We index their formats and extract bounded content summaries, including spreadsheet fields and sample values or document text excerpts. With the function selected, we use its format and material requirements to find usable candidates. A spreadsheet-filtering task needs a readable workbook whose fields support a meaningful selection condition. Sampling favors files used less often within the construction run, and the selected files' summaries become inputs to request generation. For settings tasks, the relevant material is the application preference state instead of a source file.

Write the request. We give an LLM the selected function and file summaries, together with a persona describing the user's role and motivation (Jung et al., 2026). The function fixes the operation, and the summaries identify the concrete subjects, fields, and values to which it can be applied. For example, a workbook summary that identifies a `Status` column containing `Pending` supports a request such as *"Filter the open spreadsheet to show only the pending records and save it."* The filtering condition comes from the workbook, and the persona guides how the user expresses the need.

Assemble CaseSpec. The request now describes feasible work, but the executing agent still needs the corresponding files and application state. To establish this starting point, we specify where the harness should place the selected files in the sandbox, which software template and dependencies to use, and which application or document to open before the first action. In the example, opening the workbook in Calc makes "the open spreadsheet" an unambiguous reference. Tasks involving several files instead identify them by name or workspace-relative path.

The same requested change determines the completion criteria. The workbook should be saved with the pending-record filter applied and all source rows retained. Figure 13 uses matched views of the initial workbook and the required saved filter, which hides nonmatching rows without deleting them. The request, workbook, startup instructions, and required result form one CaseSpec. The same selection therefore supplies the content of the request, the file the agent operates on, and the result to inspect.

Desktop long-horizon tasks. Base-task generation connects a function to suitable content, but this local match does not establish how several operations should depend on one another. Longer work, especially across applications, needs a coherent goal whose later stages use information or results produced earlier (Yuan et al., 2026, Ge et al., 2026).

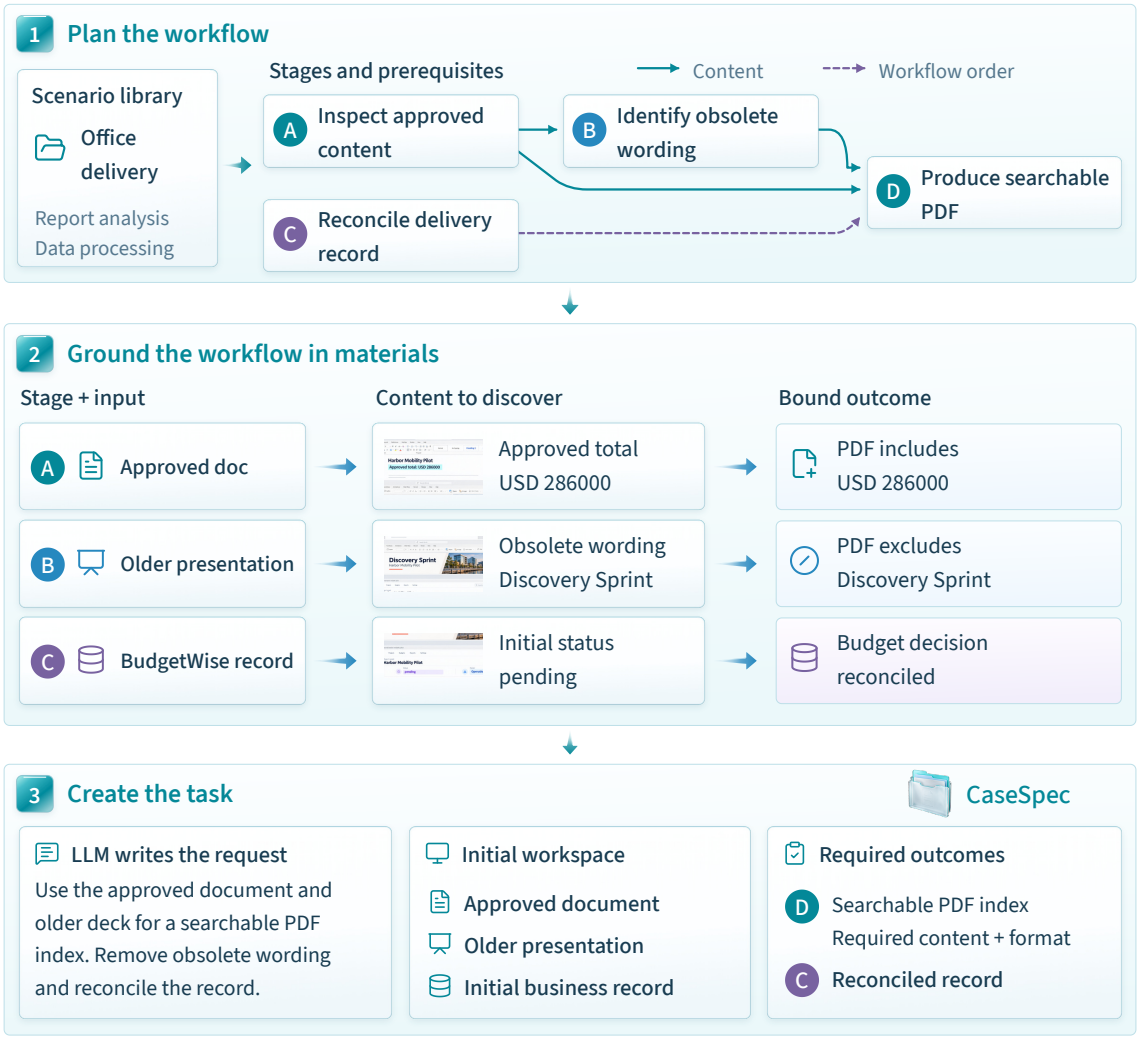


Figure 14. Constructing long-horizon tasks from blueprints. (1) Blueprint selection assembles work stages and prerequisites. Solid edges carry content dependencies. The dashed C-to-D edge sets workflow order. A node can require several GUI actions. (2) Materials bind the approved amount and obsolete wording to inclusion and exclusion conditions for D's PDF. C binds an initial record to the required budget-decision update. (3) The grounded plan guides LLM request writing and supplies the workspace and outcome requirements in one CaseSpec. Its two outputs are the PDF and updated record. Discovery facts remain in the prepared files. The structure follows a saved plan. GUI details, the shortened request, and prospective outcomes are illustrative.

We therefore plan the workflow before choosing its concrete materials and writing the request. The procedure in Figure 14 has three steps. First, we assemble work stages into a dependency graph so that they contribute to a shared goal. We then prepare the files and records that make each stage executable and its required outcome specific. Finally, we use this grounded plan to write the request and package the starting workspace and outcome requirements. The resulting task asks for a coherent workflow whose operations are connected by a shared purpose and explicit prerequisites.

Plan the workflow. We encode recurring workflows as *scenario blueprints*, each defining a final deliverable, candidate stages, their applications and materials, and their prerequisites. For the office scenario, a source review, a delivery-record update, and a final document all contribute to preparing the same delivery package.

For each instance, target task length, application coverage, and desired interaction challenges guide the choice of a blueprint and a compatible stage combination. We favor less-used scenarios and components, then use seeded sampling among eligible combinations to vary the work. A combination is retained when its prerequisites are present and every selected stage contributes, through its dependencies, to the overall workflow goal. Operation variants further specify the requested result, such as requiring the PDF's text to remain searchable.

We represent the selected workflow as a directed acyclic graph (DAG), a planning abstraction also used for cross-device orchestration in UFO3 (Zhang et al., 2025d). Here the graph specifies the task-construction plan. Each node is a work stage, such as inspecting a document for approved content, and may require several GUI actions. Edges can express use of earlier information or artifacts, or a required execution order. We retain the blueprint's dependencies and, when constructing longer workflows, add precedence between selected stages that use shared facts.

Figure 14(1) shows an office-delivery example of this planning process. The shared goal is to produce a current delivery package and reconcile its business record. We organize that work into four stages. Source inspection (A) establishes the approved content, which guides the search for obsolete wording in an older deck (B). Record reconciliation (C) can begin independently. The final PDF stage (D) uses the findings from A and B. The blueprint also places the record update before PDF production, shown by the dashed C-to-D edge. This edge specifies workflow order, while the updated record remains a separate deliverable with its own check.

Before preparing the materials, we check the assembled plan for cycles, missing prerequisites, and inconsistent production or use of files. The checked plan associates each stage with its application, required inputs, and expected output, so material preparation can follow the dependencies of the workflow.

Ground the workflow in materials. The DAG specifies what each stage should accomplish, but its nodes still need concrete information to operate on. We ground each node by selecting or generating the files and service records required by that subtask. Existing-file selection follows the same principles as base-task selection, matching application support, file format, and content. Long-horizon construction also considers whether the selected files belong to a shared project or directory and whether their contents support the relationships between stages. File summaries supply the names, values, and other details used to specify stage goals and acceptance conditions.

When the workflow calls for generated materials, we instead choose scenario facts and place them in the files or records the agent will inspect. The office sample follows this path in Figure 14(2). The approved document supplies a value that the final PDF should include, while the older deck contains obsolete wording that the PDF should exclude. The business record supplies the initial state for a reconciliation update. Each material therefore has a role in both the work stage and its outcome condition. The example requires two results: the PDF and the updated record.

We check that the required information is available through the selected applications and that generated files contain their assigned content. Linking each requirement to its source and its

later use keeps the workflow grounded across stages. The output is a task plan with concrete input files, specified initial record values, stage goals, and corresponding outcome criteria.

Create the executable task. With the work and its materials fixed, we use an LLM to write the user request. It receives the user-visible goal, references to the available materials, and required constraints, together with a selected persona and expression style. These choices vary how users describe their work while preserving the constructed task. Facts designated for discovery remain in the workspace for the executing agent to find. The request can ask for a reconciled delivery record and a searchable PDF index without revealing the values or obsolete terms the agent needs to discover. Query checks compare the wording with the task plan, return inconsistencies for revision, and filter semantically duplicate requests.

Figure 14(3) brings the request-writing, workspace, and assessment branches of the same grounded plan together. As in base-task construction, initialization instructions place the files and open the required applications. Here they also prepare the business record used by the workflow. Outcome assessment checks the requested PDF, record changes, and source preservation. Together with the request, these components form the CaseSpec executed by the harness. The DAG organizes task construction, while the executing agent plans its own GUI actions.

The shared task plan keeps the goal, workspace, and assessment aligned as the collection expands to longer, multi-application workflows.

Web tasks. An existing website supplies records and pages that can be read or changed through interface controls. We generate requests from this material in three steps. Examining the content and controls first establishes what work the site supports. We then select a goal involving that content and ask an LLM to express it as a user request. The chosen goal determines the initial records to prepare and the outcome to assess, completing the web task.

We first identify the information visible on the relevant pages and the controls used to access or modify it. On an illustrative task-management site, these include task names, due dates, and status values, together with controls for filtering the task list, opening a task, and changing its status. This combination supports requests to find tasks with particular properties and apply a corresponding update.

Using this account of the site, we select conditions that match records in the starting data and an operation available through the interface. We then use an LLM to express the grounded goal as a user request, for example: *"Mark tasks due before 1 September 2026 that are still In progress as Needs follow-up."* The due dates and current statuses determine which tasks qualify, and the status control provides the requested operation. Other requests use different items, conditions, and available actions to cover information retrieval, comparison, and form filling.

The generated request determines which initial state to prepare and which result to assess. We record its entry page, startup instructions to open that page in the browser, and any setup needed to make the selected records available. In the example, the starting records establish which task identifiers satisfy the date and status conditions. Outcome assessment checks the updated status of those same tasks after execution. Keeping their identities tied to the starting data lets the check follow the requested items even after their status changes.

Across desktop and web, these constructions turn available software and content into user requests with defined starting states and outcomes to assess. The resulting CaseSpec binds the goal and requirements to the selected resources and initialization instructions. The next route

starts with an existing request or reference website and builds the workspace needed to support it.

4.2.2 Scaling Environments

Scaling Intents uses the workspaces already available to generate new requests. Sourced requests offer a complementary resource: they describe useful work, but often omit the files, dependencies, or application state needed to execute it. Reference websites and software issues likewise describe interactions without necessarily providing an isolated environment for collection. We design Scaling Environments to build these missing execution conditions. For desktop requests, this means preparing the files and applications assumed by the user. For Web tasks, we collect open-source frontends and reconstruct workflows from public URLs to create local environments that we can initialize and reuse. We also develop repository-based frontend testing as a distinct web-agent application, where a runnable project lets browser interaction test a candidate code repair.

Desktop environments. Forums contain user-reported software problems, while tutorials and help resources describe practical workflows and their intended results (Wang et al., 2025b). These sources provide goals grounded in software use, but their descriptions often assume attachments or workspace state that the collector does not have. We collect tens of thousands of such requests and task seeds from forums, wikiHow, and help resources. The construction problem is then to supply the inputs and startup conditions each seed assumes while preserving the work it asks the agent to do.

We therefore design an environment-building agent, EnvAgent, to interpret each request and prepare its input files and startup plan. Figure 15 illustrates three construction steps. The agent first retrieves and reads candidate files to establish which requirements are already supported. Any missing inputs then guide file generation, using the retrieved material to keep related records consistent. With the inputs available, the agent specifies the request and startup plan and checks that they agree. A failed check returns the affected input or instruction for revision before the package becomes a CaseSpec. The order-comparison example below shows how the retrieved and generated materials contribute to the same task.

Prepare the input materials. EnvAgent uses the request and selected software template to identify the applications and inputs the task needs. Comparing orders and shipments, for example, requires two tables that refer to the same orders, with a shared identifier through which their records can be matched. A request restricted to a particular month also requires the corresponding date information. We reuse the indexed file formats and content summaries from Scaling Intents to locate candidates, using project and date information to find related tables. EnvAgent then reads their records to confirm that the identifiers refer to the same orders and that the files contain both shipped and outstanding orders for the comparison. If the retrieved materials meet the input requirements, the agent proceeds to request and startup specification. Otherwise, the unmet requirements guide generation.

Figure 15(2) shows an example of this step. Given a suitable order table but no matching shipment table, EnvAgent uses file tools and reusable construction skills to create a shipment table with matching identifiers and business context. The resulting input set gives the requested comparison concrete files to operate on.

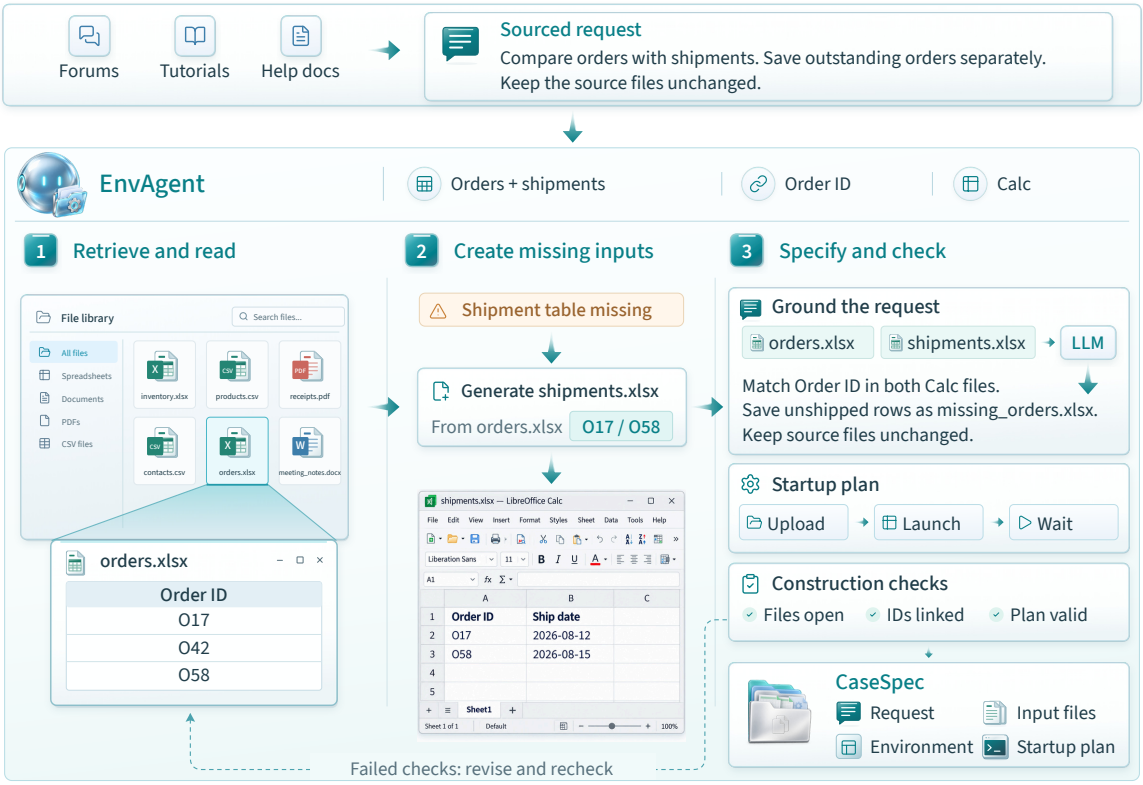


Figure 15. Request-driven desktop environment construction. EnvAgent identifies the required resources, retrieves an order table, and uses its identifiers to construct a missing shipment table. An LLM grounds the sourced request in the prepared filenames and their shared Order ID field. EnvAgent specifies startup and checks the materials and preparation plan, revising inputs or instructions when checks fail before producing a CaseSpec. The GUI agent subsequently performs the requested comparison. GUI views and records are illustrative.

Specify and check the task environment. The files establish which names and fields the request can refer to. We use the original request and these file details to guide an LLM's replacement of general references with concrete ones, preserving the goal and constraints. In the example, the request now names the two workbooks and their shared `Order ID` field. It asks for unmatched orders in a separate file and preserves any original source-protection or time restriction. The GUI agent still needs to identify the outstanding orders and create the result file.

The refined request also determines how execution should begin. EnvAgent specifies that both inputs be placed in a project directory, Calc be launched, and the order table be opened. The startup plan waits for the application window to be ready, with the shipment table available alongside the order table. These instructions give the harness the sequence needed to present the files and application assumed by the request.

Before handoff, EnvAgent checks the prepared materials and startup plan. File tools establish whether the inputs exist and have valid formats, while the agent reviews whether their contents support the requested comparison, whether the template provides the required applications, and whether startup actions place the files before opening them. Findings such as a missing identifier field or an incomplete startup action lead to revisions of the affected material or instruction. After a file changes, EnvAgent submits a new inspection report before finalization, so the report describes the materials that will actually be delivered.

The checked request, files, environment configuration, and startup plan form the CaseSpec from which the harness creates the E2B instance and establishes the starting application state. With the input files in place, the GUI agent performs the comparison and saves `missing_orders.xlsx`, whose contents are then checked against the required orders alongside preservation of the source files. Construction checks thus establish the starting conditions, while outcome assessment evaluates the work performed from those conditions.

Web environments. We collect Web frontends from open-source websites as starting points for local task environments. To extend this pool, we also reconstruct selected workflows from public reference URLs (Chae et al., 2026). These references provide realistic pages and interactions, while local control lets us initialize records, repeat attempts, and construct multiple tasks within the same site.

Our URL-to-environment construction builds on WebAnything (Xie et al., 2026) and follows the four steps in Figure 16. We first explore the reference to identify the required pages and behavior. Those observations guide a local interface connected to backing state. Once it runs, browser tests check the interactions and return discrepancies for repair. We then restore the intended starting data and package the checked site for task collection. These steps turn an external example into an interactive environment with controlled initial conditions.

The builder begins at the reference URL and explores the workflows selected for reconstruction, retrieving relevant pages and accessible frontend assets. For record editing, this covers the list page, detail view, editing controls, and navigation between them. An existing request can further focus the exploration. The resulting observations establish which pages the local site needs and what each interaction should do.

We use these observations and accessible frontend assets to build the local pages. Their interactions also specify the backend behavior to reproduce. When the reference's backing data or implementation is unavailable, the builder creates a local schema, initial records, and update logic for the observed workflow. In the record-editing example, the list and detail views refer to the same record, and saving an edit updates that record for subsequent visits. Linking the reconstructed controls to these state changes makes the workflow available for actual browser interaction. If construction raises questions about a page or control, the builder returns to the reference site for targeted inspection before continuing the implementation.

Once the local workflow is runnable, a browser testing agent checks its visual correspondence with the reference and the effects of its controls. For the illustrated check, it opens a record, edits and saves it, and refreshes the page to inspect the stored value. An unreachable detail page leads to a navigation repair, while a lost update leads to a correction of the backend state change. After revision, the affected interactions are tested again. Figure 16(3) shows a persistence test on record `R17`. The agent changes *Delivery plan* to *Delivery plan v2*, saves, and reloads. Seeing the edited title after reload provides the evidence for this particular check.

These tests modify the records that future tasks will use. Before collection, we therefore restore the declared initial records and entry-page state, keeping construction-test edits separate from task setup. We package this starting state with the checked site for isolated E2B execution. Its content and tested functionality also support Scaling Intents, allowing the same environment to supply multiple user tasks.

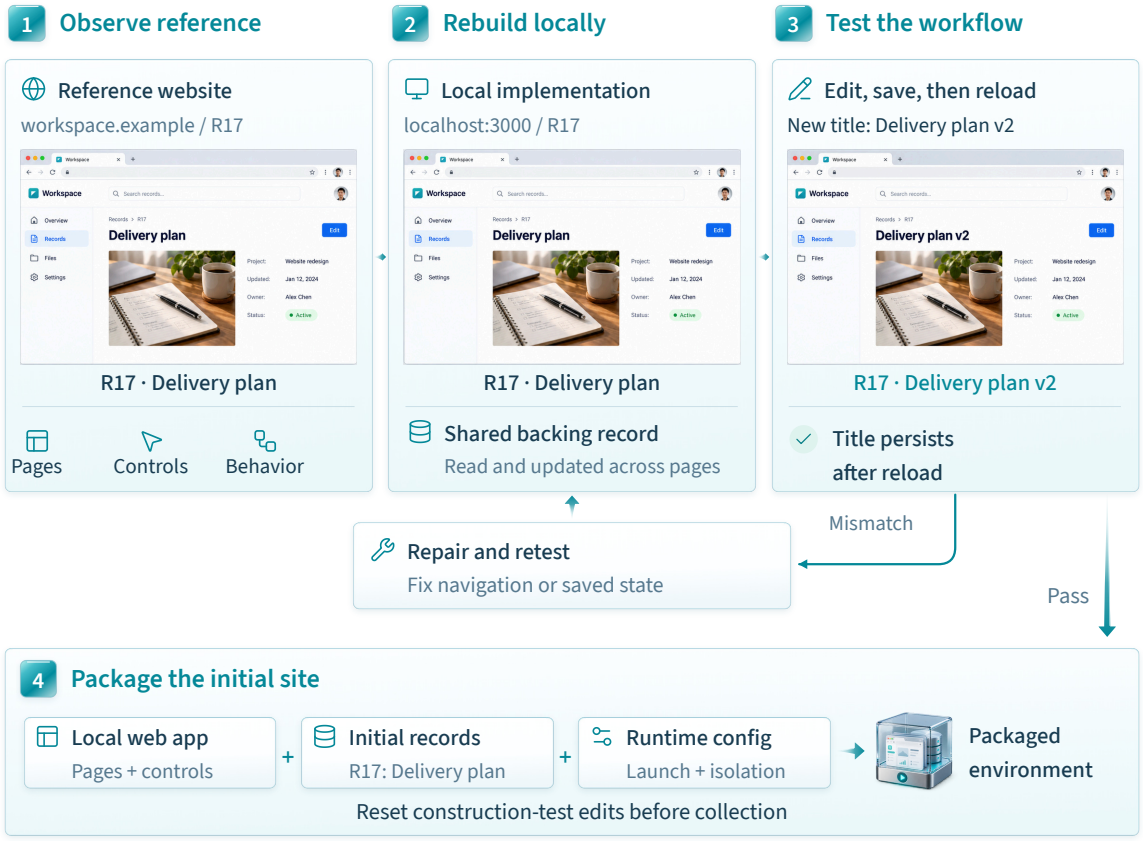


Figure 16. Reconstructing interactive websites from references. The aligned GUI views follow record R17 from the reference site to the local implementation and a reload test. Observed pages and interactions guide the interface and shared backing record. The test edits and saves the title, then reloads the record to check persistence. Discrepancies return to implementation for repair and retesting. The checked site is packaged with its declared initial records and runtime configuration, separately from construction-test edits. GUI states and URLs are illustrative.

Repository-based frontend testing. Frontend testing is an important application of web agents because code that builds successfully can still behave incorrectly when a user interacts with it (Yang et al., 2025). We therefore treat issue-driven testing as a distinct setting for environment construction and experience collection. A repository supplies the application, and its issue report specifies behavior that browser interaction can test.

Figure 17 shows how browser testing guides successive code revisions. We first prepare a runnable project and turn the issue into a fixed test expectation. A coding agent then revises and launches the application, making a candidate repair available to the web agent. The web agent replays the reported interactions and compares their effects with the expectation. Finally, its action trace, screenshots, and verdict return to the coding agent in a test report. This evidence directs the next repair while keeping every iteration focused on the same issue.

Environment preparation begins with GitHub frontend repositories whose issues describe browser-reproducible behavior. The builder installs dependencies, starts the project, and prepares the pages and data involved in the issue. This makes the reported problem available for direct interaction and supplies the workspace in which repairs will be tested.

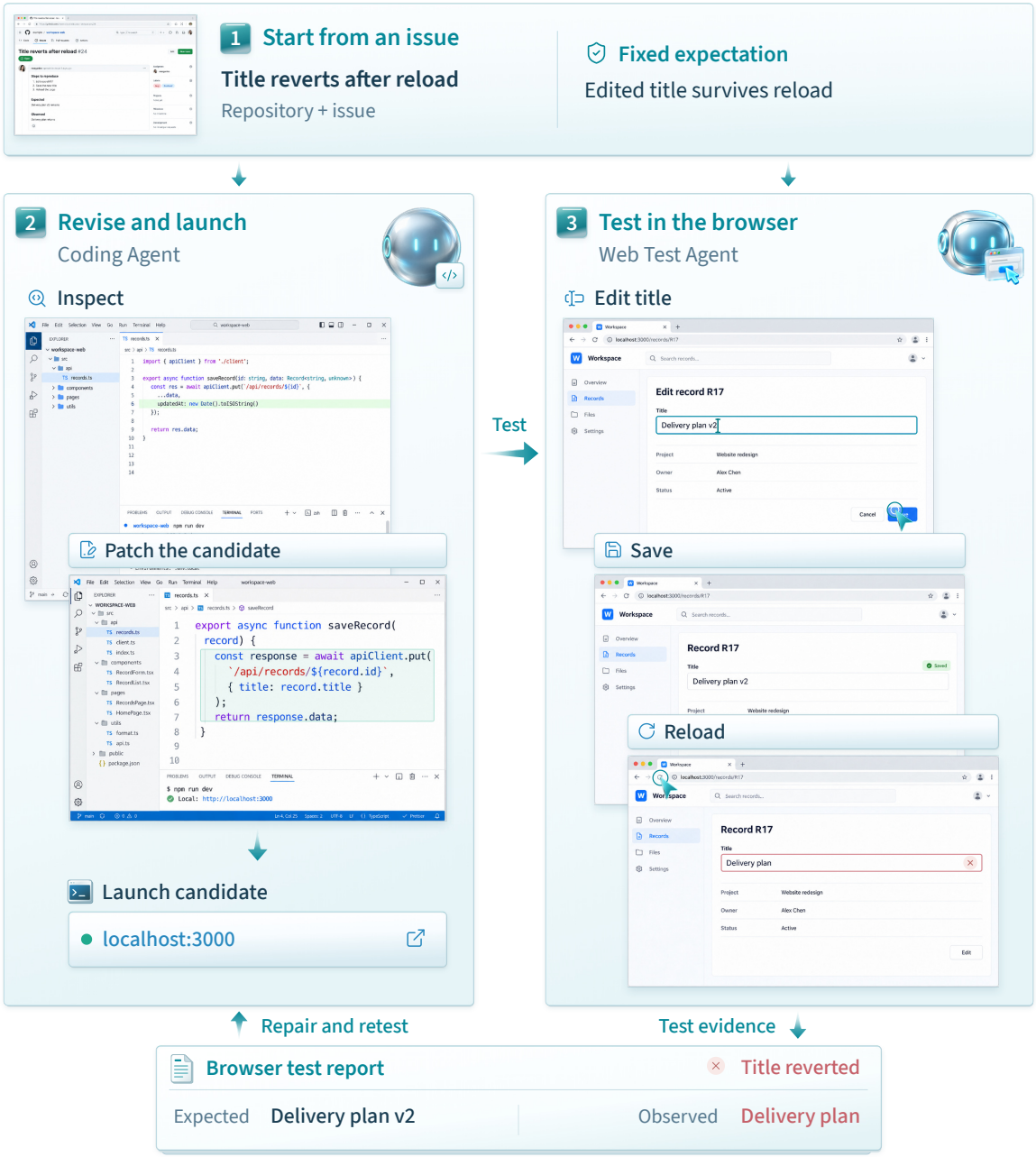


Figure 17. Browser interaction guides iterative code repair. (1) The issue fixes the expected behavior. (2) The Coding Agent inspects and edits the relevant implementation, then launches a candidate application. (3) The Web Test Agent edits R17, saves the new title, and reloads the page. The stacked views show the new title after saving and the old title after reload, with cursor overlays marking interactions. A test report returns this discrepancy, recorded actions, and screenshots to the Coding Agent for repair and retesting under the unchanged expectation. Generated GUI views, code, and test feedback are illustrative.

The issue then defines what to check in this workspace. Its expected behavior supplies the coding agent's repair objective and a test list of browser actions and expected results for the web agent. For example, an issue may report that an edited page title appears to save but disappears after reload. The corresponding test changes the title, saves it, reloads the page, and checks

that the new title remains. Both the code change and its assessment therefore address the same user-reported problem.

The coding agent inspects the relevant implementation using the issue and any feedback from the previous browser test. It then changes the code and launches the revised application. These actions, illustrated by the editor views in Figure 17(2), supply a runnable candidate whose behavior still needs to be tested.

The web agent performs the test actions against that candidate and captures the resulting states. A visible save confirmation establishes what the interface showed immediately after saving. Reloading tests whether the change persists. In the illustrated failure, the edited title appears after save but the old title returns after reload.

The browser test report links these states to the actions that produced them and states the discrepancy directly. The stacked views in Figure 17(3) follow the same R17 record through editing, saving, and reloading. Because the edited title should persist, the observed old title identifies a discrepancy for the coding agent to investigate. This feedback guides its next revision, which the web agent tests again through the running interface. The issue's expected behavior remains fixed across iterations. This loop supplies GUI interaction trajectories and outcome feedback tied to real development requests, with evidence qualification and final acceptance detailed in Appendix F.1.

Case study of a responsive homepage repair. Figure 18 follows a local Mercedes-Benz GLE SUV homepage built from supplied materials. The request called for a responsive page with usable controls and navigation. Two defects emerged when the WebTestAgent followed a button click over time and inspected the page at a narrow width.

In the first browser test (R1), the WebTestAgent clicked *Learn more* and waited. The transient notification disappeared, leaving no stable confirmation. Resizing the page to 320 px then exposed a second defect, with text navigation covering the centered logo. The R1 report on the left links both findings to the actions and screenshots that exposed them, giving the Coding Agent two specific behaviors to repair.

The central patch excerpts show how this feedback changed the code. For the first finding, the Coding Agent added an HTML confirmation element below the button and updated its text through the JavaScript click handler, so the response could remain visible. For the second, a CSS rule hid the text links at widths up to 560 px while retaining the hamburger menu. This freed space around the logo while preserving navigation access. The matching numbers connect each reported defect to its corresponding patch.

With the revised page running, the Coding Agent requested a second browser test (R2). The WebTestAgent clicked the button again, waited three seconds, and changed the carousel slide. The inline confirmation remained visible. At 320 px, the links no longer covered the logo, the menu opened and closed, and the page had no horizontal overflow. Together with a homepage visual check, these observations supported PASS on three selected items from R1's five-item checklist. The two reported defects were resolved in this targeted retest.

Together, these routes expand environment supply from user requests, reference websites, and runnable software projects. Their prepared workspaces support experience collection, and reconstructed sites also feed new task generation through Scaling Intents. Tasks equipped with reliable executable verifiers offer a further route to expansion: the next section uses their goals and outcome checks as seeds for additional task construction.

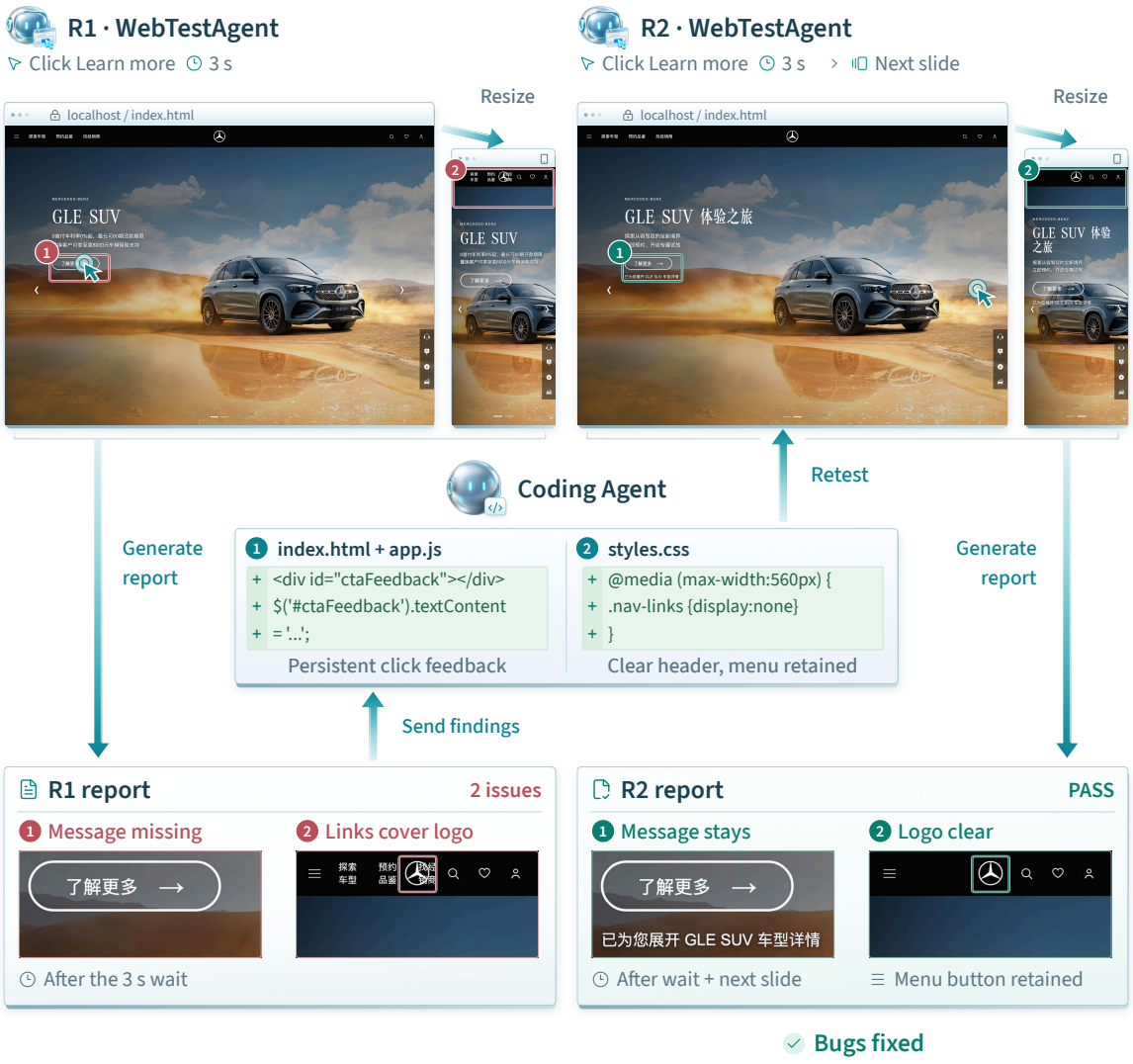


Figure 18. Bug-guided repair and targeted retesting. Desktop and narrow-screen tests produce one report, which guides both code changes before retesting. Screenshots are later replays of archived code, with subtitle line breaks cleaned for presentation. Overlays, report excerpts, and abbreviated patches are explanatory. R2 passed three selected checks, including homepage appearance, without repeating all five R1 items.

4.2.3 Scaling Verifiers

Task datasets and manually authored cases provide user goals already paired with executable verifiers. A reliable verifier makes the intended outcome explicit and supplies a repeatable check on collected trajectories. Creating additional checks is costly, however, because each needs to match its task and distinguish correct results from plausible mistakes. We design Scaling Verifiers to expand this resource by adapting existing task–verifier pairs (Wang et al., 2026a, Lv et al., 2026).

Figure 19 illustrates the construction and validation phases. We first use a seed's goal and success conditions to construct related tasks, changing the request, initial state, and verifier together. The new pair then undergoes a GUI execution trial and tests of the verifier's judgments. Their findings guide repairs before the pair enters collection.

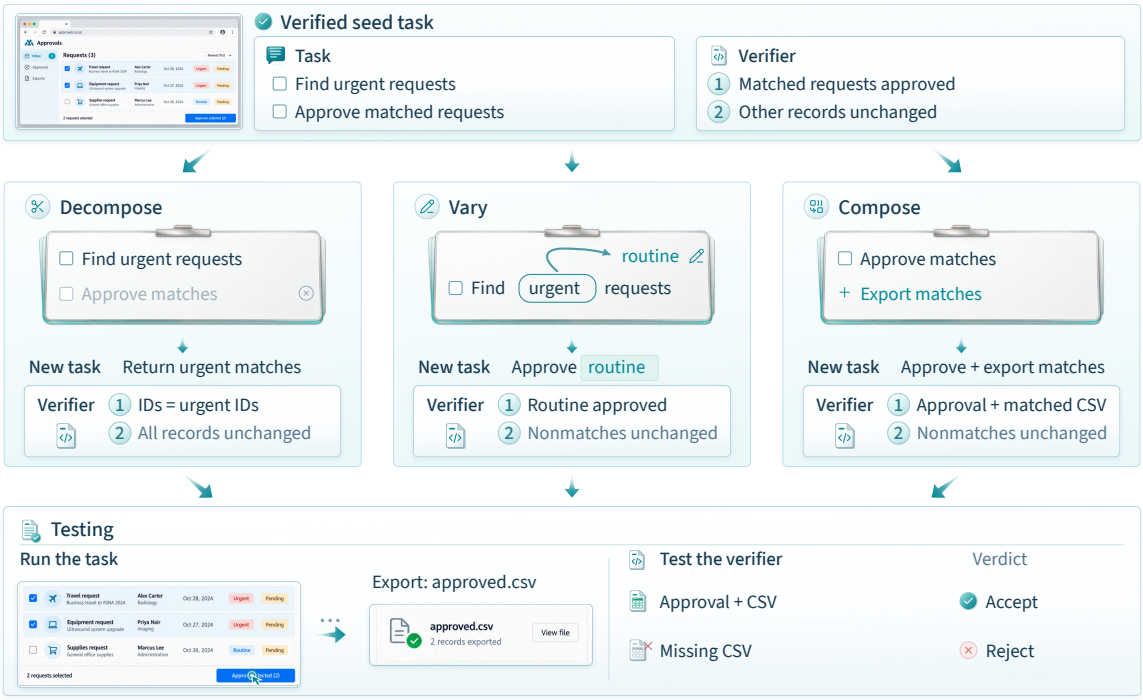


Figure 19. Expanding tasks and their verifiers together. The agent extracts a subtask, changes a selection criterion, or joins approval with export, adapting completion and preservation checks to the new task. The lower composition example separates a GUI execution trial from verifier tests on prepared outcomes. The verdict column gives the requirement-derived response for each probe. Testing checks whether the verifier returns it. The task lists, generated GUI views, and hypothetical judgments illustrate the construction procedure. The ellipsis omits intervening GUI actions, including export. Initial-state preparation and repair iterations are not shown.

This joint treatment matters because changing a request's selection criterion also changes which records the verifier should accept as updated and which it should require to remain intact.

Expanding the work and its success conditions. The construction agent reads the seed request, initialization, and verifier to identify useful subgoals, their prerequisites, and the observable changes that establish success. A subgoal can involve several GUI actions, such as editing a document and exporting the result. We use these subgoals in three different ways. Decomposition extracts a meaningful part of the seed task. Variation changes the condition or operation while retaining its basic structure. Composition connects subgoals into a longer workflow, whose later work uses earlier results. The approval example in Figure 19 shows the effect of each choice on the requested work and the corresponding outcome checks.

- **Decomposition** turns a meaningful part of the workflow into a task of its own. Extracting record selection yields a request to identify the qualifying records, an initial state containing the required data, and a verifier for the returned selection and preservation of all records.
- **Variation** changes the materials, parameters, or operations. A new selection criterion leads to a new target set derived from the corresponding input records. The request and verifier are updated to express and assess that change.

- **Composition** joins compatible subgoals through their inputs and outputs. Adding a summary export to an approval task makes the approved identifiers the contents of the later file. The adapted verifier checks the updates, the exported summary, and preservation of the records the request protects. The added dependency can produce a more demanding multi-stage task.

Across these changes, the agent resolves the actual identifiers and file references used by the request, environment, and verifier. In the composition example, the targets established during selection remain the targets during assessment, even when the update changes a field used for selection. Desktop compositions carry the same relationship through documents and intermediate files. Requirements that matter during a stage are checked there, while final-state requirements are assessed at completion.

Testing the task and its verifier. Adapting the request and verifier together does not by itself establish that the new task is executable or that its check is correct. The setup may prevent the requested work, while an apparently reasonable verifier may accept a wrong result. We therefore test two separate questions. A GUI trial examines whether an agent can carry out the request from the prepared state. Verifier probes examine whether the checker judges known outcomes as the task requires. The lower part of Figure 19 keeps these two sources of evidence separate.

For the execution trial, we run a CUA from the declared initial state with the rewritten request. Its interaction and resulting state reveal problems such as missing inputs, unreachable controls, or instructions inconsistent with the workspace. These findings identify which setup or request needs repair.

To test the verifier itself, we construct outcome probes whose expected verdicts follow from the request and starting data, then compare those verdicts with the verifier's outputs. In the approval-and-export example, a complete result should pass, whereas an otherwise correct update without the required export should fail. Additional probes isolate incomplete updates, changes to protected records, and allowed alternative outputs. A disagreement identifies a checker rule to revise, even when the GUI task itself can be completed.

We collect trial failures and verifier disagreements into a construction report. Setup and instruction failures return to task construction, while incorrect verdicts return to the verifier. After the revision, we rerun the affected tests against the same intended requirements before admitting the pair to collection. Requirements that need contextual judgment use the evaluator in Section 3.3. This feedback loop keeps task expansion connected to executable conditions and meaningful outcome assessment.

The three construction routes supply tasks to the shared collection service (Section 3.4). Executing these tasks produces trajectories and outcome assessments. Obtaining successful experience from the hardest tasks introduces the next challenge.

4.3 Recursive Self-Improvement for Hard-Task Collection

The routes in Section 4.2 supply executable tasks whose rollouts provide experience for learning. During collection, however, we encounter long and difficult tasks that remain unsolved within the ordinary budget. These tasks could provide useful demonstrations of sustained planning and recovery, but task availability alone does not produce those demonstrations. Two recurring obstacles limit this collection. Repeated attempts can lose useful discoveries and repeat

ineffective procedures. A late failure can also force a new attempt to rebuild substantial earlier work.

We design two recursive self-improvement (RSI) pipelines for collecting experience from these tasks. Each carries forward the information needed to address one of these obstacles.

- **Knowledge accumulation.** Skill-guided recursive exploration turns trial-and-error findings into tested procedures that guide subsequent attempts (Figure 20).
- **Progress accumulation.** Milestone-guided trajectory search preserves completed work and resumes exploration from the corresponding software state and agent context (Figure 21).

Both use feedback from earlier trials to shape later exploration (Zhu et al., 2026). They target tasks that pass construction checks but remain unsuccessful in ordinary collection. The original acceptance conditions stay fixed. We check the completed task before admitting an assisted trajectory as a full-task demonstration. Milestone-guided collection can also retain verified subtask traces from attempts that leave the full task unfinished.

4.3.1 Skill-Guided Recursive Exploration

Some GUI tasks require a procedure that the agent has yet to discover. The difficulty may be an unfamiliar application command, a hidden prerequisite, or a sequence whose useful effect becomes clear only after trying it and inspecting the result. Repeated attempts can make partial discoveries, but those discoveries remain scattered across their histories unless the collector carries them forward. We therefore need a way to turn trial-and-error experience into reusable operating guidance.

We design a skill-guided loop with four stages, shown in Figure 20. Task-directed exploration first produces observations about useful operations and unresolved obstacles. A skill creator then reviews that experience and consolidates actionable findings into written guidance. A separate agent tests the candidate instructions before they are supplied to a new task attempt. That attempt can use the validated procedure to make further progress, and its new discoveries feed the next round of skill refinement. The loop accumulates tested knowledge across attempts while retaining full-task verification for the final result.

Explore the task and identify obstacles. We use a computer-use agent to attempt the original query in its prepared workspace. It inspects the relevant applications and files, tries controls, and adjusts its action sequence as it observes their effects. An unsuccessful attempt can still reveal an important prerequisite, an ineffective action to avoid, or a way to reach a previously inaccessible state. Successful operations likewise show procedures worth retaining. We examine these findings in the trajectory and result, using evaluator observations of the live state to help identify the remaining obstacle (Section 3.3) (Shinn et al., 2023).

In the representative report workflow in Figure 20, exploration may reveal several export options. One trial produces a PDF with all the required pages but represents their text as an image. The searchability check then focuses further trials on which export settings preserve text. The initial exploration has thus turned a broad completion goal into a specific procedural question that can be tested in the application.

Consolidate a candidate skill. The skill creator reviews the accumulated trajectories and their outcomes. When the useful procedure is still unclear, further exploration or focused practice

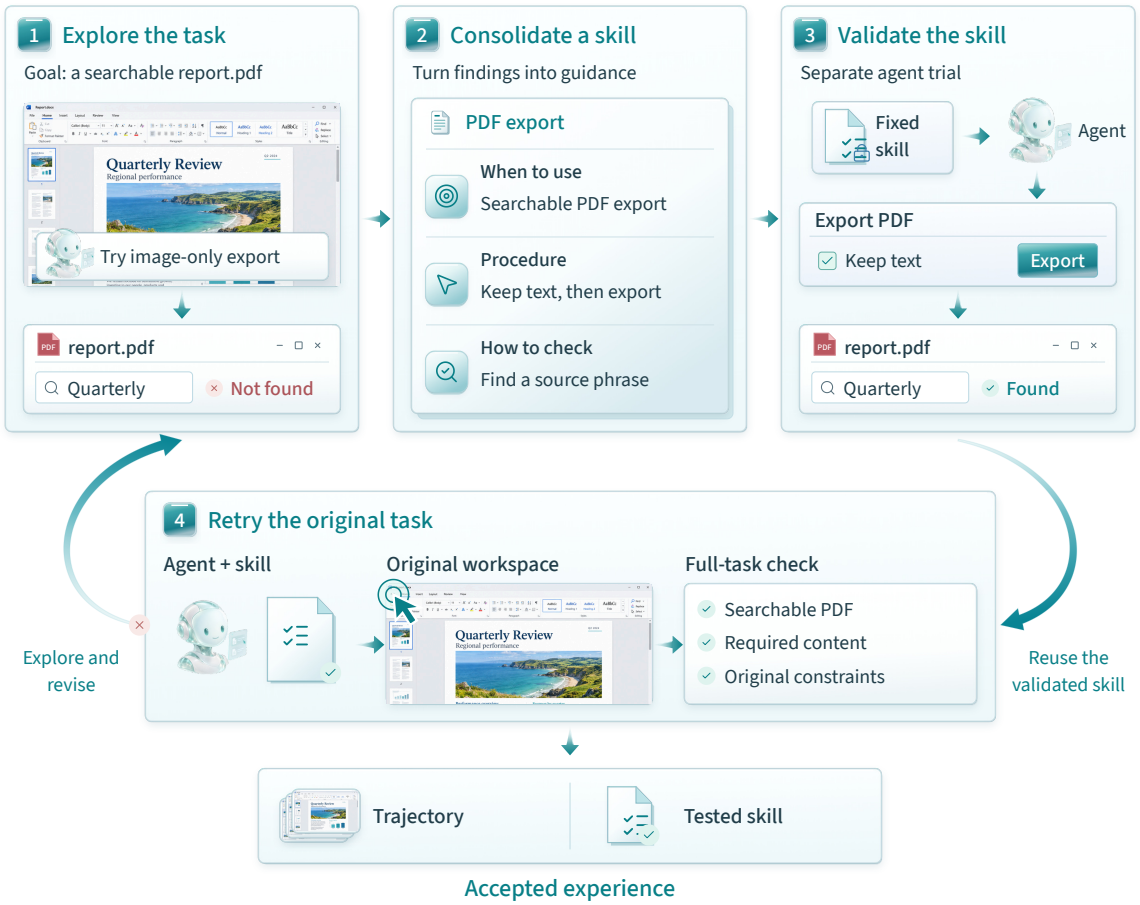


Figure 20. Accumulating operating knowledge through exploration. The source report contains "Quarterly," but an image-only export makes the phrase unsearchable in the PDF. Further exploration yields a text-preserving procedure. An agent follows its fixed instructions in a separate trial and searches the resulting PDF. The validated skill then guides a new attempt in the original workspace, where full-task checks determine trajectory acceptance. The return arrow depicts a failed task attempt, while failed validation also prompts revision. Accepted trajectories retain the supplied skill in context. GUI scenes, controls, actions, and outcomes are illustrative, with intermediate exploratory trials omitted.

tests alternative operations. The creator turns the resulting findings into textual guidance that states when the procedure applies, which actions to take, and how to check its intended effect (Wang et al., 2024, 2026c). For the PDF example, this means recording the text-preserving export settings and a search check, together with the image-only choice to avoid. The candidate gives later attempts a concrete procedure and the relevant pitfall.

Validate the skill independently. Guidance inferred from exploratory trials may omit a prerequisite or misidentify why an operation worked. We therefore freeze the candidate and give it to a separate agent in an independent validation trial. The agent applies the instructions and checks their claimed effect in the environment.

In the PDF example, it exports using the specified settings and searches for a phrase from the source document. Finding the phrase establishes the intended effect of this skill, while a failed

check returns the instructions to exploration and revision. The revised candidate is tested again before admission for reuse.

Reuse the skill and continue exploration. We provide the validated guidance to a new attempt at the original task. The agent can now use the tested procedure when it reaches the relevant operation. If it encounters another obstacle, or discovers a useful way to complete a later stage, that experience feeds the same consolidation and validation loop. Previously validated skills remain available, so each round can build on what earlier attempts established.

The process continues until an attempt satisfies the original task's full acceptance conditions or the exploration budget is exhausted. Accepted attempts supply trajectories with the actual guidance retained in their decision context. The skill library also makes a tested procedure available to later tasks that meet its applicability conditions. This accumulation is designed to improve hard-task collection by carrying useful knowledge between attempts. The TERRA dashboard case in Section 4.3.3 follows how a charting problem becomes revised guidance that a fresh attempt applies (Figure 22).

For a long workflow, however, even a better-informed attempt can spend substantial effort repeating earlier work. The next pipeline addresses that remaining cost by retaining execution progress.

4.3.2 Milestone-Guided Trajectory Search

A long task can be difficult to finish in a single attempt even when its individual stages are within the agent's reach. When a late failure ends an attempt, restarting from the initial state repeats work already completed. This motivates a second RSI strategy in which collection focuses on smaller, verifiable goals and retains the work that reaches them.

We design milestone-guided trajectory search to implement this strategy. We first decompose the original task into dependent milestones and their checks. GUI attempts then work toward one milestone at a time, saving the workspace and agent context when a check passes. Further attempts restore that checkpoint to explore the next stage, following the principle of returning to useful states for exploration (Ecoffet et al., 2021). After the final milestone, the original verifier checks the complete result before the selected path becomes a full-task demonstration. Figure 21 illustrates how this process builds successful continuations on retained progress. It reduces the work each trial needs to repeat while keeping acceptance tied to the original task. A construction case in Section 4.3.3 follows this accumulation of progress from an empty Blender scene to a complete collected trajectory (Figure 23).

Define milestones and their checks. We use a planning agent to read the user request and the original verifier code, identify dependencies, and propose meaningful intermediate goals. Each milestone pairs a subtask with a check on its observable result. The goal should represent useful progress and leave a state from which later work can continue. Reviewing the pairs against the original requirements ensures that local checks support the intended full task. This uses the decomposition principle in Section 4.2.3, here to guide collection within one long task.

For the office workflow in Figure 21, the original goal includes updating a business record and producing a searchable PDF. We organize it into three milestones: complete the record update (M1), prepare the report (M2), and export the searchable PDF (M3). Their checks inspect the corresponding records, document content, and output file. The executing agent receives the permitted subtask guidance, while private verifier constants remain in the checking context.

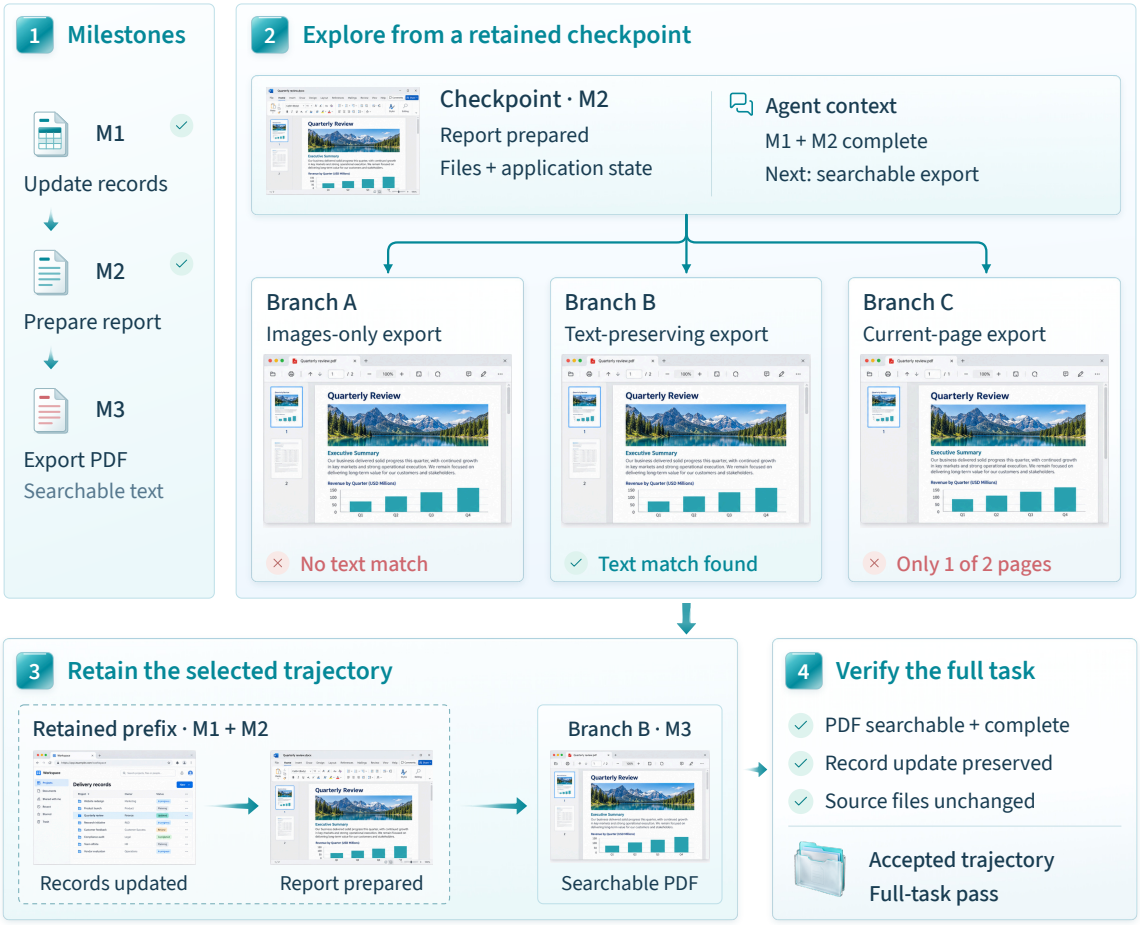


Figure 21. Retaining progress for branching trajectory search. Checkpoint M2 pairs the prepared report and software state with the agent context. Independent continuations restore this checkpoint to try different exports. The example contrasts image-only, text-preserving, and incomplete single-page exports. The lower panel retains the common prefix and the chosen continuation as one candidate path. The original task verifier checks the PDF, earlier record update, and source preservation before accepting the trajectory. Failed continuations leave the parent available for further attempts. GUI states and diagnostic outcomes are illustrative.

Preserve progress and explore continuations. The harness supplies the first subtask to a computer-use agent in the initial workspace and runs repeated attempts within the collection budget. When an attempt passes the milestone check, we save an E2B workspace snapshot together with the agent context about completed work, useful findings, and remaining goals. Restoring both gives the next attempt a consistent starting point for the following milestone (Section 3.2). We retain the full interaction prefix alongside the bounded decision context and can keep several passing states to support different continuations.

Retained checkpoints form the nodes of a search tree, and GUI action sequences toward the next milestone form its branches. Each continuation restores its parent checkpoint in an isolated instance, observes the interface, and attempts the next stage. Passing results become checkpoints for the next milestone. If a continuation stalls or fails its check, its parent remains available so that a new attempt can try a different sequence without replaying the earlier stages.

We can also retain selected recoverable failed states for further exploration. These states keep their unresolved milestone labels. A continuation attempts the necessary correction before advancing to later goals, allowing the collected path to include genuine recovery from an earlier mistake. Thus the tree can support both alternative attempts from verified progress and recovery attempts from a useful but incomplete state.

In Figure 21, branches A and C fail the export requirement, while B produces a searchable PDF. All three begin at M2, retaining the same earlier record update and report preparation. A failed export can also trigger the skill exploration described above, whose validated guidance can be used by a new continuation. The saved checkpoint preserves the work already done, and the skill helps the agent choose how to perform the unresolved operation.

Verify outcomes and retain useful traces. After the final milestone, the original task verifier assesses candidate completions against all requirements, including those established in earlier stages. For the office task, it checks the PDF, the required record update, and source preservation. Search continues until a branch passes this final check or the budget is exhausted. A passing branch yields a *search-assisted trajectory*: the actual parent–child path from the initial state to completion, combining its retained prefix with the selected continuations. It keeps the path's observations, actions, and supplied guidance, including mistakes followed by recovery. Failed sibling branches remain exploration records and are not spliced into the accepted trajectory.

If an attempt completes only part of the milestone sequence, we retain only the trace portions associated with passed milestones as subtask-level demonstrations, preserving their original decision context. This recovers useful supervision from unfinished attempts and makes fuller use of the collection budget. A passed milestone supports a subtask record. Admission to a full-task completion cohort still requires the original task's full acceptance checks.

4.3.3 Case Studies of Knowledge and Progress Accumulation

We use two case studies to explain how the RSI strategies collect experience on difficult tasks. The first follows skill refinement while building a live retail dashboard in Calc. The second follows milestone accumulation while constructing a greenhouse cafe in Blender. In each case, the recorded attempts show what goes wrong and how retaining knowledge or progress helps subsequent attempts complete the task.

Knowledge accumulation in a dashboard task. Figure 22 starts with a request to build a quarterly operations dashboard for TERRA from sales records and store targets. The deliverables are an editable workbook with live calculations and charts, and a matching one-page PDF. The request specifies grouped vertical columns and a navy, teal, and orange palette. The difficulty is translating these visual requirements into Calc's orientation and color settings, whose effects need to be checked in the saved workbook and exported PDF.

In Attempt 1, the agent follows skill v1, which asks for grouped columns and final document inspection. These instructions leave the relevant API settings unspecified. The agent sets `Vertical=True` through the legacy chart interface, producing the horizontal bars shown in the top row. It discovers the correct setting and fixes the direction during this attempt, but the charts retain the default blue and red series colors. The final palette check therefore rejects the delivery.

Reviewing this trajectory identifies the two changes shown in the middle of the figure. Skill v2 records `Vertical=False` as the setting for vertical columns and adds explicit series-color assign-

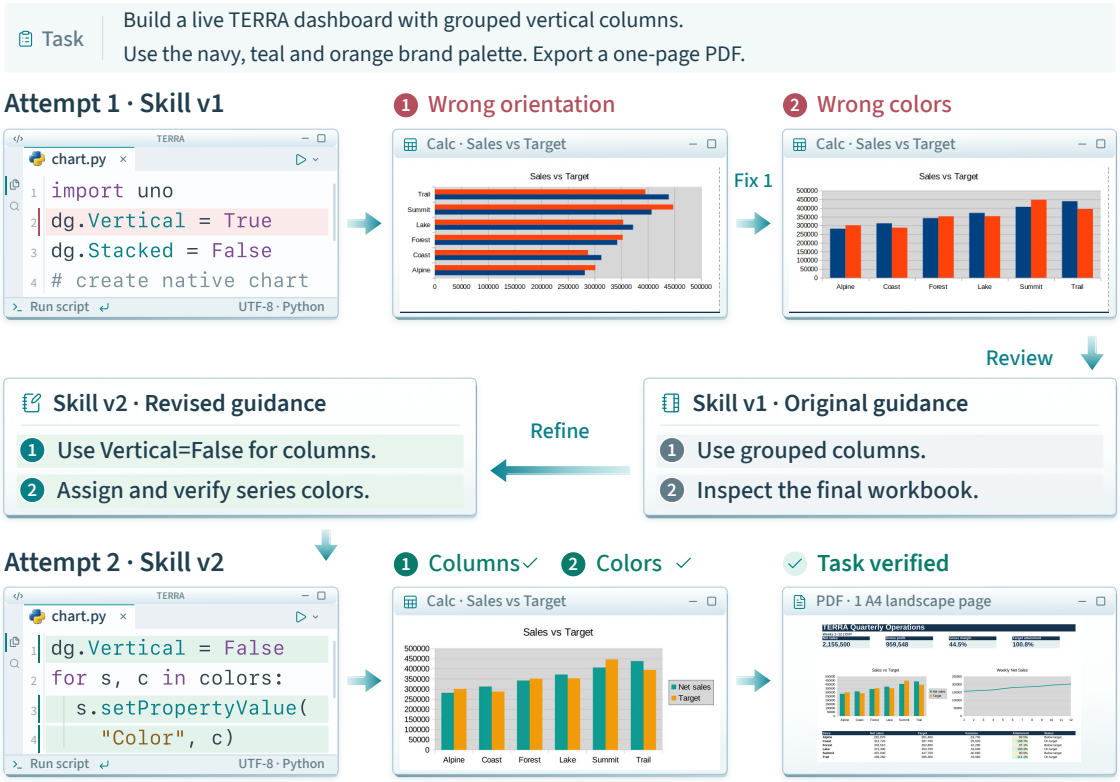


Figure 22. Skill refinement in the TERRA dashboard case. Numbered annotations track chart orientation (1) and series colors (2) through exploration, revised guidance, and a fresh attempt. Attempt 1 repairs the orientation during execution but fails the final palette check. Attempt 2 applies skill v2 and passes all 26 final checks. Chart crops and the final PDF come from recorded artifacts. Window frames and equivalent Python excerpts are composed for presentation.

ment through `chart2`, followed by property readback and visual inspection. Engineering probes check these procedures before the revised skill is fixed in a new attempt's context. Attempt 2 then starts from the original input in a fresh workspace. It applies the orientation setting and assigns teal to net sales and orange to the target series before exporting the dashboard. The bottom row shows the resulting columns, palette, and PDF. The completed task passes all 26 final checks, including input-change tests of the live formulas and charts. Earlier trial-and-error has thus become precise operating guidance that the next attempt can apply to produce a verified demonstration.

Progress accumulation in a 3D modeling task. Figure 23 follows a request to build ORBIT BOTANICA, a futuristic greenhouse cafe, from an empty Blender scene and deliver an editable model with its final render. The task combines structural, equipment, and appearance requirements, so an error in the final render can arise after much of the construction is complete.

We apply the milestone-guided procedure in Section 4.3.2 to organize construction into checkable intermediate goals. The figure groups the recorded checks into three display milestones and shows selected branches. At M1, the agent builds the platform, frame, and roof. The left branch fails because its upper ring misses the posts, whereas the other two displayed branches produce connected frames. Saving these passing scenes with their agent contexts allows exploration to continue from established structures.

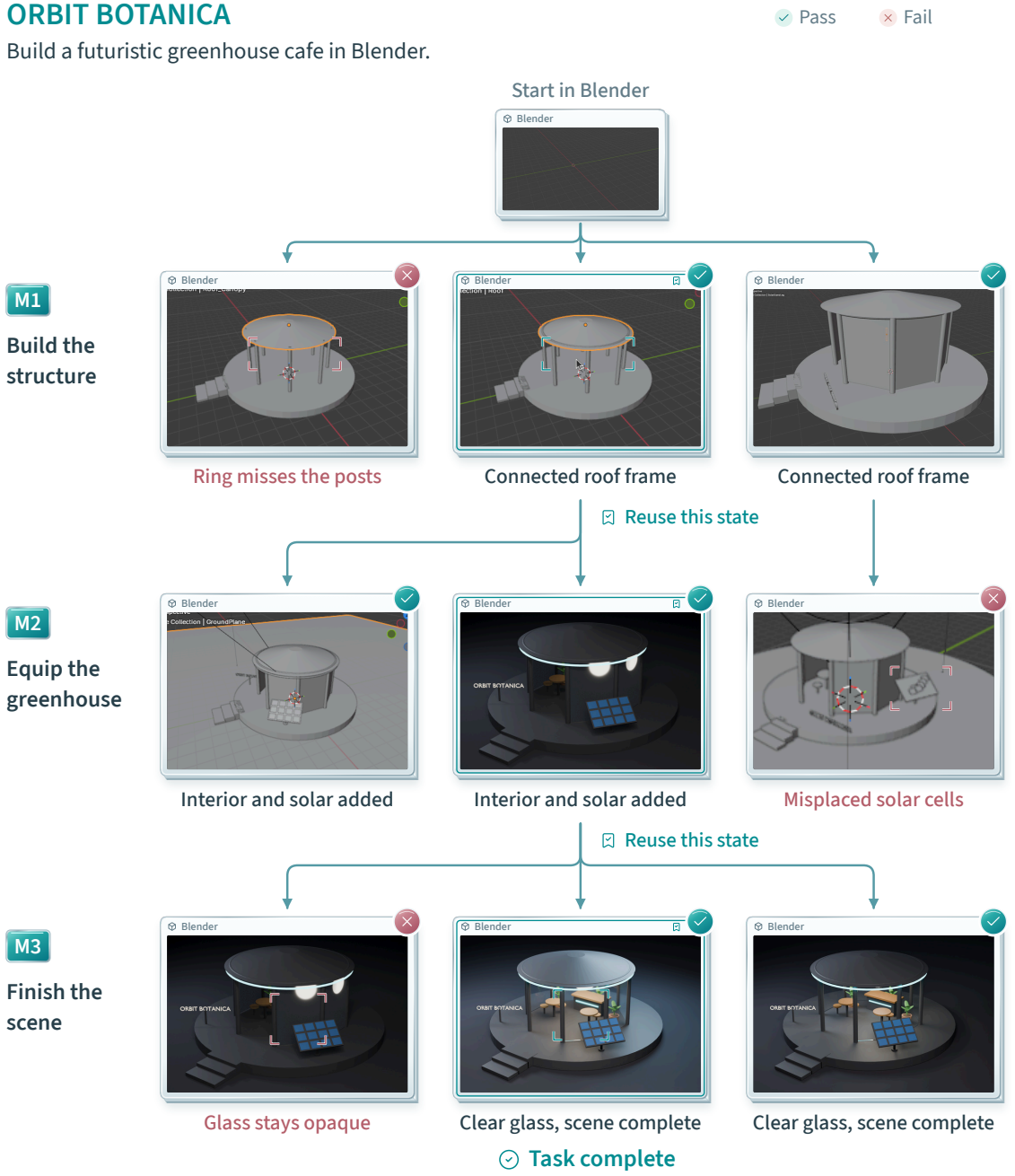


Figure 23. Milestone-guided construction of ORBIT BOTANICA. Three display milestones group the recorded checks. Teal frames mark one retained path, and arrows preserve ancestry through omitted intermediate steps. Branches can complete later work early. The equipped checkpoint and its failed finishing continuation share the unchanged opaque-glass render. Both successful finishing branches pass the full-task checks. Scene crops are unretouched, with frames and diagnostic annotations added for presentation.

At M2, the continuations add the interior and solar equipment. The right branch misplaces solar cells and fails the equipment check. The two continuations from the central M1 checkpoint pass, preserving the completed structure and equipment for finishing work.

At M3, the displayed continuations all start from the central equipped checkpoint. One leaves the glass opaque, obscuring the interior and failing the finishing requirements. The other two produce clear-glass renders and pass the full-task checks. Because the equipped checkpoint remains available, these finishing trials can address the remaining rendering requirements without rebuilding the frame or furnishing the cafe. The teal frames trace one retained path through all three milestones. Once its final result passes the original task verifier, the interaction prefix and successful continuations along that path form a complete demonstration. Failed siblings remain search records. This case shows how preserving checked intermediate work lets collection concentrate further trials on the unresolved part of a long task.

Together, the cases illustrate two ways to carry useful experience forward. TERRA reuses a procedure learned from earlier execution, while ORBIT BOTANICA reuses the work already completed in the software environment. Both routes direct subsequent attempts toward unresolved requirements and retain verified task trajectories for learning. Appendix C.2 details checkpoint contents, search budgets, skill records, and the treatment of complete paths and checkpoint continuations.

Summary. The Experience Flywheel supplies the staged learner with both demonstrations and opportunities for fresh interaction. Curated grounding examples support control localization, while selected trajectories demonstrate complete workflows. RSI expands the candidate pool with attempts at difficult tasks. The same executable tasks can support new rollouts with task-specific outcome feedback as the policy develops. Section 5 explains how the learner uses these complementary resources. Appendix D records corpus scale, provenance, and source-grouped splits, with construction evidence in Appendix C.1.

5 Staged Learning from Scalable GUI Experience

The Experience Library assembled by the Experience Flywheel provides quality-selected demonstrations of software work, alongside grounding examples and executable tasks for further collection (Section 4). The next step is to turn these resources into a policy that can reason about interfaces and carry out user goals across desktop and web applications. We design a three-stage learning pipeline that develops this capability progressively, as outlined in Figure 2.

1. **Broad GUI mid-training** first establishes familiarity with controls, application functions, and their effects through diverse software experience.
2. **Verified long-horizon SFT** builds on that foundation and concentrates learning on high-quality demonstrations that carry dependent work to completion.
3. **Joint desktop–web agentic reinforcement learning** then lets the policy learn from its own decisions and their outcomes in interactive environments.

Each stage uses the preceding policy as its starting point while changing the learning emphasis, from operational breadth to sustained completion and then outcome-driven improvement.

Table 3 summarizes the resources and priorities of the three stages. The shared interfaces of the NorthCUA Harness connect demonstration collection with online execution, so the policy can continue using the same observation and action conventions as training progresses. We describe each stage below. Appendix E defines the supervised prediction units and how recorded RL attempts become learning samples.

Table 3. Learning priorities, data selection, and scale across the three stages. Supervised scale entries are rounded cohort summaries (Table 16). GUI steps count interactions within trajectories. Grounding examples are counted separately, as are auxiliary evaluator demonstrations. RL counts describe the prepared task pool.

Stage	Learning focus	Data and acceptance	Reported scale
Broad GUI mid-training	Localization and broad software use	Grounding, judge-selected task trajectories, and auxiliary evaluator demonstrations	$> 100{,}000$ trajectories $> 4{,}000{,}000$ GUI steps
Verified long-horizon SFT	Sustained completion of dependent workflows	Complete trajectories selected for script acceptance, dependencies, and execution quality	$\approx 3{,}000$ trajectories $\approx 310{,}000$ GUI steps
Joint desktop–web agentic RL	Improvement through the policy's own execution	Human-reviewed tasks with outcome variation, followed by fresh verifier-scored task groups	$\approx 1{,}500$ desktop tasks ≈ 500 Web tasks

For model training, all spatial coordinates, including grounding targets and coordinate-based action arguments, use a shared $[0, 1000]$ representation. For an image of width W and height H, a pixel location (x, y) becomes $(1000x/W, 1000y/H)$, giving the model a consistent coordinate convention across screen resolutions.

5.1 Broad GUI Mid-Training

Computer-use agents encounter many applications, interface layouts, and ways of performing similar operations. To work across these settings, a policy needs experience recognizing controls, interpreting their functions, and observing how actions change software state. We use broad GUI mid-training to build this foundation from diverse demonstrations (Wang et al., 2025c, Jung et al., 2026).

Through the collection routes in the Experience Flywheel, the reported broad task corpus contains over 100,000 trajectories and over 4 million GUI interaction steps, spanning dozens of applications and hundreds of operation families. This range exposes the policy to different tasks, materials, and interfaces within and across applications. We complement the trajectories with approximately 40,000 single-step grounding examples to train spatial recognition and control localization directly. The following collection and selection choices determine what experience enters this mixture.

Broad software coverage. We select commonly used desktop applications and organize their functions into operation families. These profiles guide the task requests and CaseSpec environments used for experience collection, making it possible to vary the operations demonstrated within each application. Sampling across applications and operations, together with variation in source files and user goals, broadens this coverage. Representative coverage in the application profiles and workflow templates includes:

- *Office work.* Calc tasks cover formulas, auditing and reconciliation, filtering, pivot tables, and print layout. Writer adds document formatting and annotation, while Impress covers slide composition, content insertion, styling, and synthesis of source material into presentations.

Table 4. Representative coverage of broad mid-training sources. Dataset scale and sampling conventions appear in Appendix D and Appendix E.1.

Source	Representative coverage
Desktop base tasks	Office operations, visual/media work, and workspace/information management
Desktop long workflows	Cross-source reasoning, conflicting-information resolution, and dependent work toward a shared deliverable
Web tasks	Information retrieval, form filling, human-verification flows, and frontend testing
Single-step grounding	Locating requested controls in desktop and web screenshots
Evaluator trajectories	Tool-assisted inspection and evidence-based assessment of task outcomes

- *Visual and media work.* GIMP tasks include cropping, resizing, annotation, and export. Inkscape adds vector text, color, alignment, and grouping operations. Image viewers, PDF viewers, and VLC provide content inspection and playback tasks.
- *Workspace and information management.* File-manager tasks cover organization and batch processing. Thunderbird, browsers, and VS Code add message handling, browser configuration, information lookup, and GUI-based file editing.
- *CAD and 3D workflows.* Expanded workflow templates define model review and part design in FreeCAD and SolveSpace, together with scene organization in Blender.
- *Medical-image and anatomy visualization.* 3D Slicer scenarios cover inspecting image volumes, checking orientation, annotating landmarks, and saving review scenes.

To extend this coverage beyond individual functions, we also use the long-workflow construction in Section 4.2.1. Its dependent tasks provide experience combining operations, carrying information between applications, and maintaining progress toward a shared deliverable. Mid-training therefore includes sustained workflows as part of its broad foundation, before the next stage concentrates on selected verified examples.

We further enrich web experience with (1) information retrieval, (2) form filling, (3) human-verification flows, and (4) frontend testing. Retrieval and form filling represent common everyday uses of websites. Human-verification flows add visually conditioned interactions encountered during web use, while frontend testing exercises the effects of controls and the persistence of state changes. Together, these settings diversify both what users ask and how an agent must interact, complementing desktop operation with web-specific workflows. The web scenario library also provides browser-game and sliding-puzzle tasks. Grounding examples add direct practice locating the requested control, including specialized controls handled by the documentation-guided route in Section 4.1. Table 4 summarizes the coverage of these sources, while Appendix D gives their rounded counts by domain.

Outcome-based trajectory curation. Broad coverage is useful for learning only when the collected behavior provides a worthwhile demonstration. Large-scale attempts can also leave tasks incomplete or spend many actions without progress. We therefore use the agent-as-a-judge evaluator in Section 3.3 to inspect each candidate's requested outcome, trajectory, and resulting artifacts or state (Zhuge et al., 2025). Its assessment selects completed work with useful execution behavior for the training corpus. This selection limits the inclusion of incomplete or low-quality traces while preserving examples of successful software use across the coverage pool.

Among the trajectories that pass review, selection preserves broad application and operation coverage. Meaningful exploration and recovery remain part of this experience when they lead

to completed work, giving the model examples of how actions and subsequent observations inform the next decision. Outcome-based filtering thus makes quality control part of building a large and varied training corpus.

Learning localization, interaction, and assessment. The selected mixture teaches both where to act and how to continue a task. For grounding examples, the model predicts the target control's coordinates from the request and screenshot (Cheng et al., 2024, Lu et al., 2024). Trajectories train the model to predict each recorded assistant response from the task and the observations and history available at that decision. Targets include recorded reasoning, text responses, and action tool calls, while screenshots and tool results provide context. This response-level objective connects interpreting the interface, deciding what to do, and issuing the operation.

We also include evaluator trajectories as a separate judgment-task source. These examples condition the model on the evaluator's inspection context and train its recorded assessment responses, providing demonstrations of how to check an outcome and support a judgment with evidence. They do not insert post-execution evaluation evidence into the acting agent's earlier decisions. The evaluator therefore serves two distinct purposes in mid-training: its reports select task demonstrations, and its own trajectories teach assessment. Appendix E.1 distinguishes the sample types and their response targets.

Through this broad mixture, mid-training develops familiarity with diverse GUI functions, action effects, and ways of interpreting results. This provides the operational foundation for learning more demanding workflows. The next stage sharpens that foundation using a concentrated set of high-quality, verified long-horizon demonstrations.

5.2 Verified Long-Horizon SFT

Mid-training gives the policy broad experience operating software, including extended workflows. Sustaining correct work over a long and difficult task requires a more concentrated learning signal: later decisions need to use earlier findings, preserve constraints, and complete the final deliverable. We therefore continue from the mid-trained checkpoint with supervised fine-tuning on selected long-horizon trajectories whose complete outcomes pass executable task verifiers (Yuan et al., 2026, Lu et al., 2026). The cohort contains approximately 3,000 trajectories and 310,000 GUI interaction steps.

Selecting verified demonstrations. We start with tasks and executable checks supplied by Scaling Verifiers (Section 4.2.3) and collect successful completions. For difficult tasks that ordinary attempts do not solve, the skill-guided and milestone-guided RSI routes in Section 4.3 provide additional ways to obtain candidate demonstrations. Skills carry useful operating knowledge between attempts, while retained checkpoints let exploration build on completed intermediate work. In either route, a candidate enters this stage only after its final result passes the original task verifier. Appendix D lists the source contributions that remain to be reported separately from this aggregate cohort.

Within this accepted pool, we prioritize difficult workflows with long-range dependencies while maintaining application diversity. We use length to identify sustained work, then inspect how later actions use earlier results to assess its dependencies. We then check whether these demonstrations provide usable training sequences. Data cleaning rejects unusable records and traces with malformed or unmatched tool calls or unproductive loops. This screening retains coherent

recovery when the trace shows an error, its correction, and verified completion. Each retained trace keeps its original action–observation sequence, preserving the context needed to learn that recovery. These checks combine programmatic outcome acceptance with demonstrations of purposeful work across stages.

Learning sustained completion. With this selected corpus, we fine-tune the mid-trained policy to predict the recorded assistant responses from the request and decision-time context. The objective remains response prediction, but the data now emphasize maintaining task state, using intermediate results, coordinating work across applications, and recovering without losing the original goal. For assisted demonstrations, any supplied skill or milestone guidance remains in the context under which the recorded response was produced. Representation and sampling requirements are described in Appendix E.1.

This stage strengthens sustained task execution on top of the broad operational foundation. The resulting policy is the starting point for autonomous rollouts, where its own choices can create states absent from the demonstrations. Reinforcement learning provides the next learning signal by connecting those choices to their task outcomes.

5.3 Joint Desktop–Web Agentic Reinforcement Learning

Mid-training and long-horizon SFT provide a policy that can recognize interfaces and follow demonstrated workflows. To develop its ability further, we let it learn from autonomous interaction. Its own choices determine which states it encounters, and task outcomes provide feedback on whether those choices accomplish the user's goal. We use agentic reinforcement learning to connect this fresh experience to policy improvement (Qi et al., 2025, Lai et al., 2026).

We train a single policy jointly on desktop and web tasks. The shared runtime in the NorthCUA Harness already expresses both as repeated observation, reasoning, and tool-based action, making it natural to learn from them through one training loop. Platform-specific observations and tools retain the differences needed to operate each interface. Task-specific verifiers then provide outcome feedback for the work performed in either domain.

Figure 24 follows an attempt from task preparation to a policy update. The Task and Environment Manager selects executable cases and prepares their isolated workspaces and verifiers, giving NorthCUA Rollout Workers the conditions needed to act with the current policy. Once those attempts have been recorded and scored, the Experience Adapter pairs the model calls with their task outcomes to construct training samples. These task-grouped samples let the Policy Learner compare alternative attempts with GRPO and update the shared policy (Shao et al., 2024, Zhu et al., 2025). Returning the new weights to the workers closes the loop, so subsequent interaction uses the updated model. The following subsections explain how we select useful tasks, obtain comparable feedback, and preserve executed decisions as that feedback enters learning.

5.3.1 Selecting Tasks with Outcome Variation

GRPO obtains its task-level learning signal by comparing the rewards of multiple attempts at the same task. An attempt's relative outcome determines its advantage within that group. If all valid attempts receive the same terminal reward, their terminal group-relative advantages are zero. Since GUI rollouts are costly, we first identify tasks on which the SFT policy produces different outcomes, so initial collection can provide informative comparisons.

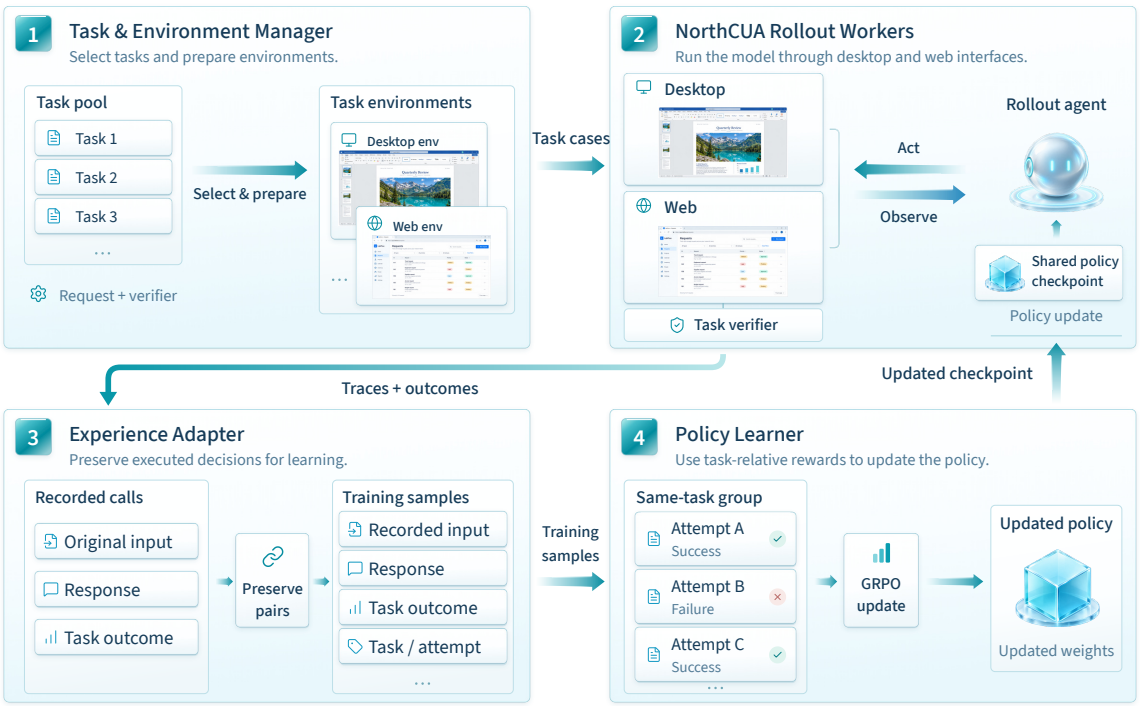


Figure 24. Joint desktop–web agentic reinforcement learning. The manager selects tasks, binds verifiers, and prepares isolated environments. Workers use a model checkpoint to act through desktop or web interfaces. Manager-run checks supply outcomes before workspace release. The adapter binds outcomes to training samples while preserving each recorded input–response pair and task/attempt identity. The learner uses GRPO to compare valid attempts within the same task, initial-state, and verifier group, then returns an updated checkpoint to workers. Worker and learner icons denote execution and training uses of the same model. GUI scenes and binary outcomes are illustrative. The example leaves task-group size unspecified.

For each model, the Task and Environment Manager starts from human-reviewed desktop and web tasks associated with its SFT data. For each candidate case, we collect at least five rollouts under the SFT checkpoint with fixed task and execution settings. The task verifier scores each attempt's outcome, and we select cases with variation among these valid verifier scores. Environment and verifier errors are recorded separately and do not count as unsuccessful task outcomes. This procedure finds tasks that offer observed room for improvement without treating infrastructure problems as policy failures.

The selected cases form the initial RL task pool. Fresh grouped attempts supply the subsequent learning data. Appendix E.2 describes the screening protocol and the fields needed to record valid repeats and the selection rule for each run.

5.3.2 Joint Rollout and Task-Relative Feedback

With the task pool selected, we generate fresh attempts and turn their outcomes into the comparisons used by GRPO. For each case, the manager first establishes equivalent starting conditions. Workers then act in those environments, and the task verifier assesses each completed attempt. The recorded decisions and outcome are converted into samples for policy learning.

The Task and Environment Manager prepares a separate E2B sandbox for each attempt in a task group. All members receive the same request, initial files and application state, verifier, and

execution budget. Their sandboxes are isolated, so one attempt's edits do not change another's starting conditions. This makes differences in outcomes interpretable as alternative executions of the same task.

NorthCUA Rollout Workers use the policy to carry out each multi-turn attempt. Desktop and web execution follow the same runtime loop, using their respective screen or browser observations and allowed tools. After execution, the manager runs the task verifier on the resulting workspace. Its valid verdict supplies the complete attempt's terminal reward. Both successful and unsuccessful attempts contribute to the comparison, while setup or evaluation errors follow the separate invalid-attempt path.

The Experience Adapter binds each outcome to the model calls from that attempt. It reconstructs training samples from the recorded multimodal inputs and responses, retains behavior-policy likelihoods and versions, and marks which response tokens are learning targets. Each sample keeps its task and attempt identity so that the Policy Learner can compute relative advantages within the correct task group and apply the GRPO update. Desktop and web groups contribute to the same shared policy through this procedure.

This completes the path from an executable case to outcome-driven learning. For long GUI attempts, however, converting the record into training samples requires additional care. The next subsection explains the two rules we use to preserve both the original decision conditions and the attempt's learning contribution during that conversion.

5.3.3 Preserving Executed Decisions for Learning

The rollout procedure provides complete attempts and their outcomes, but the learner must also preserve the conditions under which each decision was made. Our harness maintains a sliding window of recent screenshots and can compact earlier history. As a task proceeds, an image may leave the window or a summary may replace earlier messages. Using only the final history to train every response would therefore change the evidence available to some decisions. Preserving the sampled input is also a central concern in harnessed agentic RL (He et al., 2026).

To accommodate this context management, we design the Experience Adapter around two preservation rules. First, each response is learned with its recorded decision-time input, starting a new training segment when the history no longer extends consistently. Second, those segments retain the reward attribution and learning weight of their original complete attempt. This matters because frequent context changes can split one attempt into more segments than another. Figure 25 illustrates input preservation in panel (a) and attempt-level comparison and aggregation in panel (b). Together, the two rules let the harness manage long histories without turning a representation change into a different learning problem.

Preserve decision-time inputs. A response generated before a context change cannot be faithfully trained against a later, rewritten history. We therefore capture each model call's input, delivered response, generation log-probabilities, and behavior-policy version. The adapter joins consecutive calls only when their text and ordered image histories form a confirmed prefix extension under the same version. Image identity includes its content digest, so replacing a clean frame with the historical action overlay from Section 3.2.1 also breaks this prefix. The adapter starts a new segment from the captured input whenever images are replaced or evicted, a summary rewrites history, or the behavior version changes.

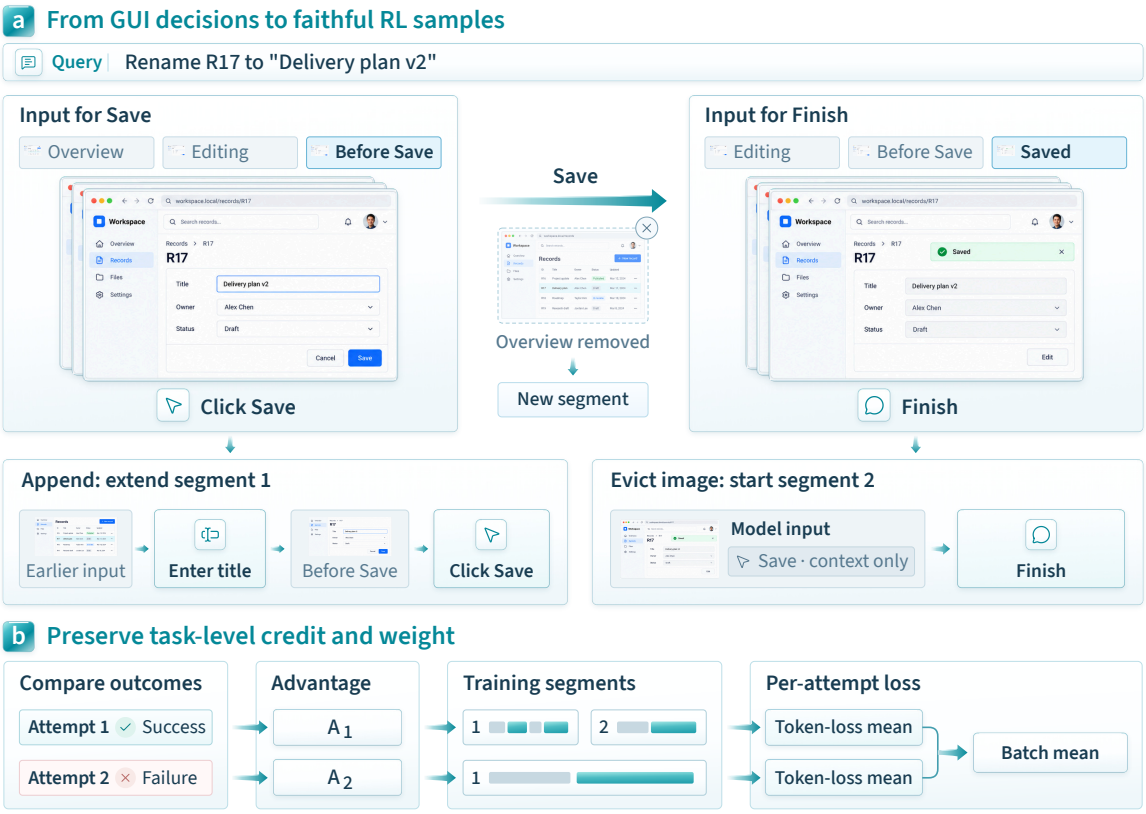


Figure 25. Context-preserving RL sample construction. (a) The earlier input contains Overview and Editing. Appending the typing result preserves history, so Enter title and Save share one segment. After the successful Save action, Overview leaves the visual window (cross), starting a new segment. The highlighted label marks the latest observation. Gray objects supply context, while teal responses and token spans contribute to policy loss. **(b)** Task-group outcomes determine one advantage per attempt, shared across its segments. In the trajectory-mean configuration, masked policy losses are pooled and normalized over each attempt's target tokens before averaging across attempts (Eq. 1). Panel (a) shows one continuous attempt under one behavior-policy version, with earlier images unchanged until eviction. GUI states, responses, and outcomes are illustrative. Thumbnails are input excerpts, strip lengths are schematic, and earlier calls are omitted.

Earlier responses can still appear as history in a later segment, but they serve only as context there, following the history-masking principle used in ScaleCUA (Lv et al., 2026). Response masks assign each learning target to its original occurrence, so retaining Save in the input for Finish does not train Save a second time. This construction lets the learner use the harness's actual decision conditions while keeping only new responses as targets. Appendix E specifies processor alignment and masks.

Preserve the complete attempt's learning weight. Preserving inputs can produce different numbers and lengths of segments across attempts. If these segments were treated as separate reward-bearing attempts, more frequent context changes would alter the group comparison. Averaging segment losses independently would also give an attempt more weight simply because it was split more often. Neither change reflects additional task experience.

We therefore retain the complete attempt as the comparison unit. GRPO computes its relative advantage once, and all of its segments receive that same value. In Figure 25b, the two segments of attempt 1 share A_1, while attempt 2 receives A_2. Loss aggregation then preserves the same

unit. In the supported trajectory-mean configuration, we use the trajectory-level normalization also adopted in Qwen-CUA (Lu et al., 2026). We sum the masked policy losses over all target tokens of an attempt, divide by its total target-token count, and average these per-attempt means:

$$\mathcal{L}_{\text{policy}} = \frac{1}{N}\sum_{i=1}^{N} \frac{\sum_{s\in S_i}\sum_{t} m_{ist}\ell_{ist}}{\max\left(1, \sum_{s\in S_i}\sum_{t} m_{ist}\right)}, \tag{1}$$

where N is the number of complete attempts, S_i contains the segments of attempt i, m_{ist} selects target tokens, and ℓ_{ist} is the token-level policy loss. All segments of an attempt enter the same optimizer step, so the denominator covers the complete attempt across microbatches. Appendix E defines the reduction and the run-specific optimization fields. This normalization prevents segmentation alone from increasing an attempt's contribution when its target tokens and losses are unchanged. Combined with decision-time input preservation, it allows one continuous execution to yield several training segments without creating extra reward comparisons or extra outer weight.

5.3.4 Continuous Rollout and Policy Updates

Training from long GUI rollouts is costly even after the samples are correctly constructed. Overlapping screenshot windows repeat visual preprocessing and carry large tensors to the learner. Meanwhile, task durations vary, so waiting for every active task before updating the policy can delay the next learning cycle. We address these two costs separately: visual-delivery mechanisms reduce redundant input preparation and transfer, while asynchronous coordination lets policy updates proceed without restarting ongoing tasks.

Deliver visual experience efficiently. Adjacent decision windows often contain unchanged screenshots. Processing every window from scratch repeats work on those images. For a supported per-image processor, the Experience Adapter therefore reuses the preprocessing tensors of unchanged images within an attempt and processes only new images. It reconstructs the requested image order under the same processor settings. Changed images or settings require fresh processing. The reuse saves CPU preprocessing while retaining every image in the model input. The neural visual encoder still processes the complete visual input.

Large image tensors also add delivery cost after preprocessing. The object-store path supports lossless tensor compression, retaining a compressed payload only when it is smaller. The learner decodes the microbatch it is about to consume, recovering the same values, shape, and type. Bounded CPU workers prepare samples, and group admission counts completed work awaiting delivery so that preparation does not create an unaccounted queue. These mechanisms target redundant processing and payload transfer while preserving the sample's contents. Appendix E.4 describes the implemented reuse and transport paths, their configuration, and the checks used to assess sample equivalence and delivery cost.

Coordinate policy updates asynchronously. In asynchronous mode, completed groups can supply the learner while other workspaces remain active. To publish an update, the system pauses new model calls, waits for calls already in progress, commits the new weights, and resumes calls. Ongoing tasks keep their workspace and completed work, so later decisions can use the updated policy.

The captured behavior version and generation probabilities identify the policy that produced each response. A version change starts a new segment, so each policy ratio uses the corresponding behavior probabilities. For attempts spanning several versions, the clipped objective is an off-policy approximation. Preserving the recorded inputs keeps the sampled decisions identifiable, while policy lag affects the resulting update. Appendix E.2 defines policy lag and mixed-version attempts and distinguishes asynchronous coordination from synchronous collection. This coordination lets learning continue while ongoing tasks retain their environments and completed work.

Together, these mechanisms address the cost of supplying visual experience and coordinating updates, while the preceding preservation rules keep learning tied to the executed decisions and complete task outcomes. Section 6.5 reports the Mid-to-SFT checkpoint changes. Section 6.6 then examines joint RL through reported checkpoint comparisons, experience-delivery measurements, and a concrete example of task-relative feedback.

6 Experiments and Analysis

Our evaluation follows the progression from identifying a control to completing a software task. We first compare grounding, desktop, and Web performance, then examine how results change through long-horizon SFT and joint desktop–Web RL. Trace analyses and selected cases connect these scores to concrete decisions, while an engineering profile measures the cost of delivering visual experience to the learner. Finally, we study NorthCUA-Ultrafast to assess how direct action selection changes the balance between task success and execution latency.

6.1 Evaluation Setup

Computer use requires both finding the right control and maintaining the user's objective through successive state changes. We select the six benchmarks in Figure 1 to cover these complementary requirements. ScreenSpot-Pro tests dense professional interfaces, while OSWorld-G adds functional and spatial instructions. Desktop evaluation extends these decisions into application workflows, and Web evaluation covers both user tasks and defect detection. Table 5 summarizes what each benchmark measures.

Models and checkpoints. We evaluate the NorthCUA family at three sizes, Nano (9B), Mini (35B), and Pro (397B), built on Qwen3.5 (Qwen Team, 2026a). We call the checkpoint after broad GUI mid-training *Mid* and the checkpoint after verified long-horizon supervision *SFT*. Thus, Mid already contains NorthCUA training. Mini and Pro use nominal 35B/3B and 397B/17B total/active language-model parameter counts, while Nano is dense. We keep the publicly released Qwen3.5 checkpoints separate as backbone-family references. The main grounding results use the long-horizon SFT checkpoints, while the main desktop, WebArena, and WebTestBench results use the final RL checkpoints. Ultrafast starts from the pre-RL Nano model. Dashes indicate unreported metrics. WebTestBench includes models with at least one measured metric. We report result metrics to one decimal, computing differences before rounding.

Grounding protocol. We evaluate SSP by direct inference on the full screenshot, without iterative zoom or tool-assisted refinement, using all 1,581 examples. OSWorld-G uses 564 Refined

Table 5. The six-benchmark suite and the capabilities and metrics covered by each evaluation.

Benchmark	Coverage	Metric (%)
Grounding		
ScreenSpot-Pro (Li et al., 2025b)	High-resolution professional software and small controls	Localization accuracy
OSWorld-G (Xie et al., 2025a)	Element semantics, layout, precise targeting, and absent targets	Grounding accuracy
Desktop		
OSWorld-Verified (Xie et al., 2025b)	Application, file, and cross-application workflows	Task success
OSWorld 2.0 (Yuan et al., 2026)	Longer workflows with dependent requirements	Full completion, partial score
Web		
WebArena-Verified (El Hattami et al., 2025)	Stateful websites and outcome-checked user tasks	Task success
WebTestBench (Kong et al., 2026)	Interactive defect detection with a gold checklist	Precision, recall, F1

instructions and normalized 0–1000 click coordinates. In addition to the main accuracy comparison, repeated predictions characterize consistency and target coverage. Appendix I gives the sampling settings, metric definitions, and repeated-run results.

Interactive evaluation. We use a maximum of 100 steps per task for OSWorld-Verified, WebArena-Verified, and WebTestBench, and 500 steps for OSWorld 2.0. OSWorld-Verified uses the officially supported 361-task setting, excluding eight Google Drive tasks that require separate account configuration.[1] We evaluate OSWorld 2.0 on the full set of 108 tasks and retain binary completion, requiring a normalized score of 1, and mean partial score where available. We keep these measures separate because completing some requirements can still leave a dependent workflow unfinished. WebArena-Verified uses all 812 tasks and official outcome checks, with screenshots and structured page references as the agent's observations. WebTestBench uses the golden-checklist setting, supplying the reference test items directly to the agent. This isolates interactive defect detection from checklist generation. We treat Fail as the positive class. Precision, recall, and F1 are computed within each task and then averaged separately across tasks, giving each task equal weight regardless of its checklist length.

Comparison models. We select representative GUI specialists and general-purpose models from original papers and official model cards, including explicitly attributed evaluations of other providers' models. Source markers distinguish our evaluations from published scores within each model group. Our OSWorld-G evaluation includes nine general-purpose baselines. Model-name citations identify original papers, reports, or official model pages, while Appendix I collects score sources and protocol details, including benchmark versions and action interfaces (Tong et al., 2026, Zhou et al., 2026a).

[1] https://timothyxxx.github.io/OSWorld/user_guides/run_public_evaluation.html

6.2 GUI Grounding Results

NorthCUA achieves state-of-the-art grounding results among the systems compared in Table 6, with Pro reaching 93.3% on SSP and 91.0% on OSWorld-G. The two benchmarks test complementary requirements, from locating small controls in dense professional interfaces to following functional and spatial instructions. We report the final long-horizon SFT checkpoints, using direct SSP inference and the Refined OSWorld-G instructions described in Section 6.1.

Claude Opus 5.5 is the strongest reference on both benchmarks in this comparison, scoring 88.1% on SSP and 89.9% on OSWorld-G in our evaluation. Pro exceeds these scores by 5.2 and 1.1 percentage points, respectively. On SSP, Pro also leads Qwen3.8-Max's reported 84.5% by 8.8 points. On OSWorld-G, it exceeds Nex-N2.5-Pro's reported 87.4% by 3.6 points.

The smaller tiers retain much of this capability. Mini reaches 92.4% on SSP and 87.9% on OSWorld-G with 35B total and 3B active language-model parameters. Its SSP score is within one percentage point of Pro. Nano reaches 72.9% and 83.0% with 9B parameters, with SSP accuracy close to GUI-Owl-1.5-32B-Instruct's reported 72.9%. To understand what these aggregate scores capture, we next separate geometric precision, control semantics, and target availability.

Broad coverage and fine localization. Mini and Pro both exceed 90% SSP accuracy in all six application groups. Pro reaches 95.0% in CAD and 94.5% in scientific software. This broad coverage also includes controls with little room for a coordinate error. To measure that difficulty directly, we pair the three models' predictions on all 1,581 examples and group targets by size, measured as the smaller of the target box's width and height in the original screenshot (Table 7). A target in the smallest group is less than 16 pixels wide or tall. On these narrow targets, accuracy rises from Nano's 53.4% to Mini's 85.2% and Pro's 92.1%. Pro's 6.8-point gain over Mini is larger than its 1.2-point gain on targets at least 32 pixels wide and tall. Its stronger aggregate result therefore also extends to the smallest target regions in this breakdown.

Interpreting specialized controls. A small target must also be matched to the user's intended function. Pro reaches 90.2% on SSP's icon targets, followed by Mini at 88.6% and Nano at 55.0%. The difference persists within the most populous size band of 16–$<$ 32 pixels, where Pro and Mini both locate 89.0% of the 418 icon targets and Nano locates 53.1%. Their corresponding text-target scores are closer at 94.6%, 94.1%, and 84.6%, respectively. Even after this coarse size grouping, the gap between Nano and the larger models remains greater for icon targets than for text targets.

This distinction helps explain why our data construction addresses both target geometry and control meaning. The broad route retains the image coordinates, while documentation supplies functional descriptions for specialized controls.

Figure 26 illustrates this construction with the Legato Slide control in Ample Bass P, a virtual instrument hosted in FL Studio. Its articulation buttons encode different playing techniques through small symbols, so the icon alone gives little indication of the requested function.

The construction record links this control to the manual's Legato Slide description (Ample Sound, n.d.) and generates the instruction *Activate the Legato Slide articulation*, retaining the original target location.

This documentation-grounded package was used in Nano and Mini training. In the SSP example, NorthCUA correctly locates the documented control from the full screenshot and query alone, without access to the manual during inference.

Table 6. Grounding accuracy (%) on ScreenSpot-Pro (SSP) and OSWorld-G across general-purpose models, GUI specialists, and the three NorthCUA tiers.

Model	Parameters	SSP	OSWorld-G
General-purpose models			
Qwen3.5-35B-A3B† (Qwen Team, 2026a)	35B / 3B	64.5	63.6
Qwen3.5-27B† (Qwen Team, 2026a)	27B	68.4	67.9
Qwen 3.7 Plus† (Qwen Team, 2026b)	397B / 17B	68.9	78.2
Qwen3.8-Max* (Qwen Team, 2026c)	2.4T / 95B	84.5	71.6
Seed 2.1 Pro† (ByteDance Seed, 2026)	Undisclosed	65.3	78.0
Claude Opus 4.8 (Anthropic, 2026c)	Undisclosed	82.3	–
Claude Opus 5.5* (Anthropic, 2026d)	Undisclosed	88.1	89.9
Claude Fable 5.1* (Anthropic, 2026a)	Undisclosed	81.0	87.2
GPT-5.6 Sol (OpenAI, 2026b)	Undisclosed	76.9	77.7
Kimi-K3* (Moonshot AI, 2026)	2.8T / 104B	67.7	73.8
GPT-6 Astra* (OpenAI, 2026d)	Undisclosed	80.0	77.0
GPT-6.1 Sol* (OpenAI, 2026c)	Undisclosed	81.2	73.6
DeepSeek-V4.1-Flash* (DeepSeek-AI, 2026a)	552B / 16B	60.1	60.5
GLM-5.3-Flash* (Z.ai, 2026)	320B / 18B	56.7	68.8
Gemini 4 Argon* (Google DeepMind, 2026d)	Undisclosed	77.6	79.1
Nex-N2.5-mini (Nex-AGI, 2026a)	35B	–	82.9
Nex-N2.5-Pro (Nex-AGI, 2026b)	397B	–	87.4
GUI-specialized models			
GUI-Owl-1.5-8B-Instruct (Xu et al., 2026b)	8B	71.1	69.3
GUI-Owl-1.5-32B-Instruct (Xu et al., 2026b)	32B	72.9	69.7
UI-Venus-1.5-8B (Venus Team, 2026a)	8B	68.4	74.1
UI-Venus-1.5-30B-A3B (Venus Team, 2026a)	30B / 3B	69.6	76.4
MAI-UI-32B (Zhou et al., 2025)	32B	67.9	73.9
Qwen-UI-Agent-4B (Zhou et al., 2026b)	4B	67.8	70.5
Qwen-UI-Agent-35B-A3B (Zhou et al., 2026b)	35B / 3B	76.1	74.6
Qwen-UI-Agent-27B (Zhou et al., 2026b)	27B	76.6	78.5
Our NorthCUA			
NorthCUA-Nano	9B	72.9	83.0
NorthCUA-Mini	35B / 3B	92.4	87.9
NorthCUA-Pro	397B / 17B	**93.3**	**91.0**

*Our evaluation, applying only to OSWorld-G for Qwen3.8-Max. †Evaluated by the Qwen-UI-Agent authors. Score sources are listed in Appendix I. Parameters give reported total/active counts. Bold marks column maxima. Dashes indicate unavailable results.

Target availability and repeated predictions. Reliable grounding also requires recognizing when a requested target is absent. OSWorld-G tests this distinction alongside visible-target localization. Across three draws, Pro averages 91.7% on the 510 visible-target items, compared with Mini's 88.6%. Its correct-refusal rate reaches 75.9% on the 54 absent-target items, compared with Mini's 74.1% (Table 7). The larger gain is therefore in locating visible controls, while recognizing when to decline a location request remains harder across all three tiers. This distinction matters when a predicted coordinate becomes an agent's next action.

Repeated sampling then reveals whether correct predictions persist across draws. Pro is correct in all three draws on 489 of 564 items, compared with 469 for Mini and 445 for Nano. Its pass@3 reaches 94.0%, covering 530 items, while its three complete runs range from 89.4% to 91.0%. Pro also succeeds at least once on 36 of Mini's 53 persistently missed items, including 26 that Pro gets

Table 7. Trace-based grounding accuracy (%). SSP uses one prediction per item. Target size is the smaller of the annotated box's width and height, measured in original-image pixels. OSWorld-G averages three draws per item. Subgroup definitions and repeated-draw breakdowns appear in Appendix I.

Benchmark	Target subset	Items	NorthCUA-Nano	NorthCUA-Mini	NorthCUA-Pro
SSP	< 16 px	88	53.4	85.2	92.1
	16–< 32 px	1,079	72.4	92.1	92.4
	≥ 32 px	414	78.3	94.7	95.9
OSWorld-G	Visible target	510	84.0	88.6	91.7
	Absent target	54	67.9	74.1	75.9

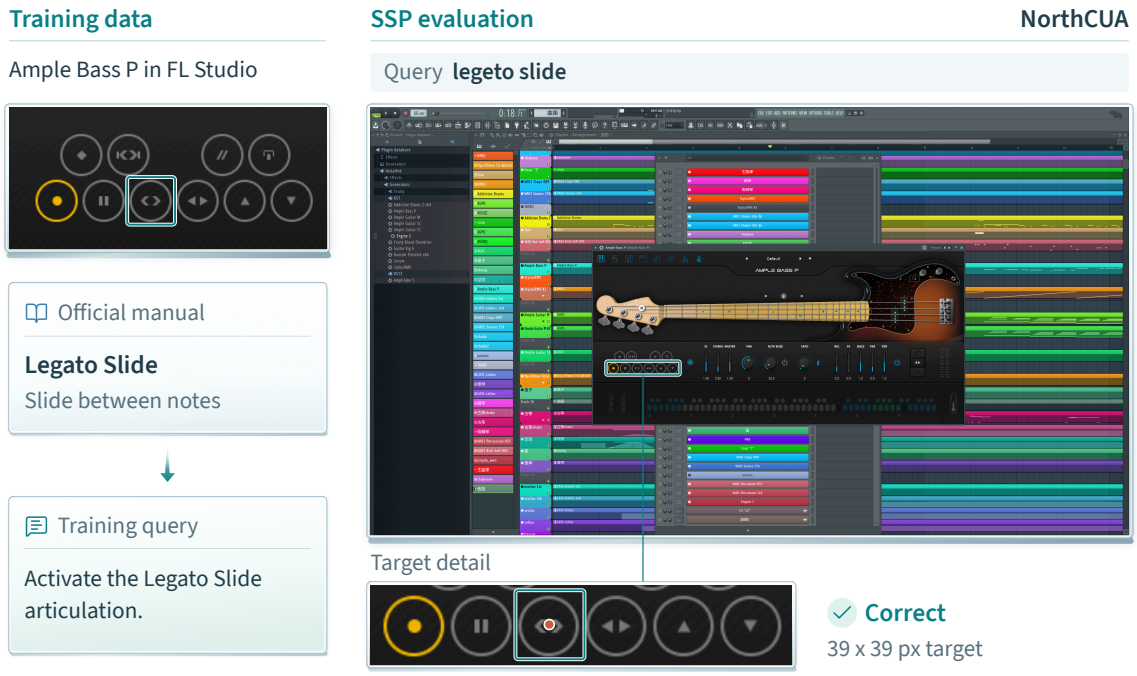


Figure 26. Grounding a documented control. Left, an actual documentation-grounded annotation pairs an Ample Bass P control with its manual-defined articulation and a generated training query. Right, NorthCUA localizes this control in an SSP screenshot with a different surrounding GUI state, following the recorded query *legeto slide*. The target's local appearance and position match the construction view. In the enlarged detail, the teal box marks the reference target and the orange point is the recorded prediction. Enlargements and annotations are reader aids. Documentation is used in data construction, not supplied during direct evaluation.

right in every draw. Thus, its stronger aggregate result accompanies both more stable successes and coverage of additional difficult items.

6.3 Desktop Agent Results

Grounding accuracy is useful only if the policy can sustain correct decisions until the requested software state is reached. Tables 8 and 9 therefore compare complete application workflows and longer dependent tasks. NorthCUA-Pro reaches 88.6% on OSWorld-Verified, the highest score among the compared systems. Mini reaches 85.3% with 35B total parameters, while the

Table 8. OSWorld-Verified results (%). NorthCUA uses RL checkpoints on 361 tasks. Reference protocols, including partial-credit variants, are detailed in Appendix I. An asterisk marks our evaluation. Bold marks the best available NorthCUA result.

Model	Parameters	Score
General-purpose models		
Qwen3.5-9B (Qwen Team, 2026g)	9B	41.8
Qwen3.5-35B-A3B (Qwen Team, 2026e)	35B / 3B	54.5
Qwen3.5-397B-A17B (Qwen Team, 2026f)	397B / 17B	62.2
Qwen3.8-Max (Qwen Team, 2026c)	2.4T / 95B	86.1
Seed 2.1 Pro (ByteDance Seed, 2026)	Undisclosed	78.8
Kimi-K3 (max) (Kimi Team, 2026)	2.8T / 104B	84.8
MiniMax M3 (MiniMax, 2026b)	∼428B / 23B	75.2
GLM-5.3-Flash (Z.ai, 2026)	320B / 18B	84.7
DeepSeek-V4.1-Flash* (DeepSeek-AI, 2026a)	552B / 16B	71.7
DeepSeek-V4-Flash-Vision (DeepSeek-AI, 2026b)	Undisclosed	76.7
GPT-5.6 Sol (OpenAI, 2026b)	Undisclosed	83.2
Claude Opus 5 [1m] (Anthropic, 2026f)	Undisclosed	83.4
Claude Fable 5 [1m] (Anthropic, 2026b)	Undisclosed	86.0
Gemini 3.6 Flash (Google DeepMind, 2026b)	Undisclosed	83.0
Nex-N2.5-mini (Nex-AGI, 2026a)	35B	71.2
Nex-N2.5-Pro (Nex-AGI, 2026b)	397B	82.2
GUI-specialized models		
OpenCUA-72B (Wang et al., 2025c)	72B	45.0
GUI-Owl-1.5-8B-Instruct (Xu et al., 2026b)	8B	52.3
GUI-Owl-1.5-32B-Instruct (Xu et al., 2026b)	32B	56.5
UI-Venus-2-9B (Venus Team, 2026b)	9B	70.8
UI-Venus-2-27B (Venus Team, 2026b)	27B	80.5
ScaleCUA-Qwen3.5-9B (Lv et al., 2026)	9B	68.7
MAI-Gym-9B (Tong et al., 2026)	9B	76.2
Qwen-UI-Agent-27B (Zhou et al., 2026b)	27B	79.5
Qwen-CUA (Lu et al., 2026)	397B / 17B	86.2
Qwen-CUA-Max (Lu et al., 2026)	$>$ 1T	87.6
Our NorthCUA		
NorthCUA-Nano	9B	78.9
NorthCUA-Mini	35B / 3B	85.3
NorthCUA-Pro	397B / 17B	**88.6**

9B Nano reaches 78.9%. These final RL results extend the grounding comparison from selecting individual controls to completing software tasks.

Pro exceeds Qwen-CUA-Max's reported 87.6% by 1.0 percentage points and Qwen3.8-Max's 86.1% by 2.5 points, using 397B total parameters against their reported trillion-scale sizes. At a smaller scale, Mini exceeds Nex-N2.5-Pro's 82.2% by 3.1 points, while Nano exceeds MAI-Gym-9B's 76.2% by 2.7 points. The comparison covers both general-purpose and GUI-specialized systems under the source-specific protocols in Appendix I. Section 6.5 examines how performance changes within the NorthCUA training pipeline.

OSWorld 2.0 extends the comparison to longer workflows with dependent requirements (Table 9). Binary completion tests whether all requirements are met, while partial credit records intermediate progress. Both measures increase from Nano through Mini to Pro in the final RL

Table 9. OSWorld 2.0 results (%). Binary requires full completion, while Partial measures earned task credit. NorthCUA uses all 108 tasks. Reference blocks identify reported task scope, with release dates, budgets, and scoring details in Appendix I. An asterisk marks a baseline evaluated by us. Dashes denote unreported results.

Model	Parameters	Binary	Partial
Reference models, 108 tasks			
GPT-5.5 (OpenAI, 2026a)	Undisclosed	13.0	49.5
GPT-5.6 Sol (OpenAI, 2026b)	Undisclosed	27.3	62.7
Claude Opus 4.8 (Anthropic, 2026c)	Undisclosed	20.6	54.8
Claude Opus 5 (Anthropic, 2026f)	Undisclosed	31.4	68.3
Claude Fable 5 (Anthropic, 2026b)	Undisclosed	36.1	72.9
Claude Fable 5.1 (Anthropic, 2026a)	Undisclosed	41.7	77.9
GLM-5.3-Flash* (Z.ai, 2026)	320B / 18B	14.8	48.7
MiniMax M3 (MiniMax, 2026b)	∼428B / 23B	4.6	22.3
UI-Venus-2-9B (Venus Team, 2026b)	9B	0.0	7.5
UI-Venus-2-27B (Venus Team, 2026b)	27B	2.8	13.2
MAI-Gym-9B (Tong et al., 2026)	9B	3.7	11.3
Published references, task scope unspecified			
Qwen3.8-Max (Qwen Team, 2026c)	2.4T / 95B	19.4	46.7
Qwen-UI-Agent-27B (Zhou et al., 2026b)	27B	13.9	40.0
Qwen-CUA (Lu et al., 2026)	397B / 17B	18.5	48.4
Qwen-CUA-Max (Lu et al., 2026)	$>$ 1T	21.2	53.3
Kimi-K3 (Kimi Team, 2026)	2.8T / 104B	–	58.3
MiniMax M3.1 Flash (MiniMax, 2026a)	Undisclosed	–	61.7
DeepSeek-V4.1-Flash (DeepSeek-AI, 2026a)	552B / 16B	–	54.3
Gemini 3.8 Flash (Google DeepMind, 2026c)	Undisclosed	–	59.0
Nex-N2.5-mini (Nex-AGI, 2026a)	35B	–	30.5
Nex-N2.5-Pro (Nex-AGI, 2026b)	397B	–	56.4
Our NorthCUA, 108 tasks			
NorthCUA-Nano	9B	13.0	45.6
NorthCUA-Mini	35B / 3B	18.5	50.1
NorthCUA-Pro	397B / 17B	**30.6**	**66.8**

checkpoints. Pro reaches 30.6% binary completion and 66.8% partial credit on the full 108-task set. Within the 108-task reference block, these scores exceed GPT-5.6 Sol by 3.2 and 4.1 percentage points, respectively, under the listed evaluation protocols. The result combines higher earned credit with a higher rate of fully completed workflows. Section 6.5 examines how both measures develop through the NorthCUA training pipeline.

6.4 Web Agent Results

We next test whether accurate interaction supports complete requests on stateful websites. NorthCUA-Pro's RL checkpoint reaches 73.9% on WebArena-Verified, the highest score in Table 10. Mini reaches 70.4%, while Nano reaches 67.2%, also after RL. We compare these results with published references and six additional general-purpose models evaluated by us. We then use the available trajectory exports to examine which website workflows are completed and where difficulties remain.

Table 10. Task success (%) on WebArena-Verified across reference systems and NorthCUA models.

Model	Parameters	Success
General-purpose models		
Qwen3.7-Plus (Qwen Team, 2026b)	397B / 17B	55.3
Qwen3.8-Max (Qwen Team, 2026c)	2.4T / 95B	66.8
Claude Opus 4.8 (Anthropic, 2026c)	Undisclosed	67.9
Claude Fable 5 (Anthropic, 2026b)	Undisclosed	71.3
Gemini 3.1 Pro (Google DeepMind, 2026a)	Undisclosed	64.3
GPT-5.6 Sol (OpenAI, 2026b)	Undisclosed	69.7
Kimi-K3 (Kimi Team, 2026)	2.8T / 104B	71.6
DeepSeek-V4-Flash-Vision (DeepSeek-AI, 2026b)	Undisclosed	62.3
GPT-6 Astra* (OpenAI, 2026d)	Undisclosed	71.6
GPT-6.1 Sol* (OpenAI, 2026c)	Undisclosed	69.7
Claude Opus 5.5* (Anthropic, 2026d)	Undisclosed	70.3
Gemini 4 Argon* (Google DeepMind, 2026d)	Undisclosed	65.8
GLM-5.3-Flash* (Z.ai, 2026)	320B / 18B	65.8
DeepSeek-V4.1-Flash* (DeepSeek-AI, 2026a)	552B / 16B	67.5
Nex-N2.5-mini (Nex-AGI, 2026a)	35B	63.4
Nex-N2.5-Pro (Nex-AGI, 2026b)	397B	67.6
Our NorthCUA		
NorthCUA-Nano	9B	67.2
NorthCUA-Mini	35B / 3B	70.4
NorthCUA-Pro	397B / 17B	**73.9**

*Our evaluation. All NorthCUA entries use RL checkpoints. Bold marks the highest NorthCUA score. Appendix I records score sources, evaluation settings, and task-level breakdowns.

Pro's score is 2.3 points above Kimi-K3's published 71.6%, the strongest reference in the table. Mini reaches 70.4% with 35B parameters, exceeding Qwen3.8-Max's reported 66.8% and Nex-N2.5-Pro's 67.6%. Nano's 67.2% provides a 9B alternative. These results provide strong Web task execution across three model scales, extending the family from the compact Nano to the highest-performing Pro.

Coverage across websites. Figure 27 examines trajectories across six website groups, using Nano and Mini after SFT and Pro after RL. Pro exceeds 70% success in each of the five single-site groups, reaching 84.0% on forums and 83.0% in store administration. This coverage spans consumer-facing websites and operational interfaces with editable records.

Cross-site tasks remain more demanding. Pro completes 17 of 48 tasks (35.4%), compared with 583 of 764 single-site tasks (76.3%). These workflows require carrying information or requirements between websites, making them a useful test of the sustained execution targeted by our long-horizon experience pipeline. The gap identifies cross-site coordination as a continuing challenge despite broad single-site coverage.

A complementary breakdown by requested outcome shows that Pro completes 274 of 374 state-changing tasks (73.3%), alongside 245 retrieval tasks and 81 navigation tasks. Thus, the result includes workflows that must leave a correct persistent site state, beyond locating or reading a page. Mini completes 253 state-changing and 223 retrieval tasks, compared with Nano's 240 and 204. Its additional completions are concentrated in these two categories, while navigation counts are similar across the two models. Appendix I gives the group counts and the decision-length distributions for all three checkpoints.

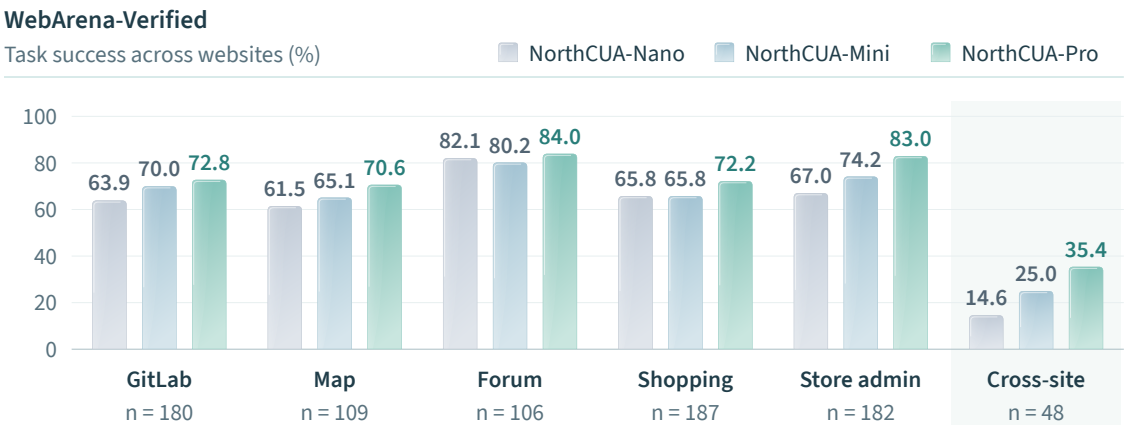


Figure 27. Web success by website scope. The six groups partition all 812 WebArena-Verified tasks using retained attempts. Nano and Mini use SFT checkpoints, while Pro uses RL. Pro exceeds 70% in every single-site group and completes 17 of 48 cross-site tasks. Appendix I gives the selection and group details.

Execution length and local setbacks. The Pro export contains 14,604 model decisions across its 812 retained attempts. Successful attempts have a median of ten decisions, yet 24 successful attempts contain at least 50 decisions, with the longest reaching 93. This range shows completion across both short interactions and extended workflows within the same benchmark.

For Pro, unsuccessful attempts have a median of 15.5 decisions and contribute 38.4% of the exported decisions despite accounting for 26.1% of tasks. Mini shows a similar concentration of work in failed tasks, whose median is 18 decisions compared with 10 for successes. Its 260 failures account for 50.6% of the 15,224 recorded decisions. This difference between successful and unsuccessful execution motivates our emphasis on feedback-guided correction during long workflows. Counts include terminal responses, with the record boundaries defined in Appendix I.

Section 6.7 illustrates such a correction in a Pro trajectory. The agent uses interface feedback to recover from an unavailable option while preserving the user's date constraint.

Web testing. Completing a user request and detecting defects require different judgments. We therefore evaluate defect detection on WebTestBench with the gold checklist supplied to each agent (Kong et al., 2026). Given these test items, the agent must exercise the application and distinguish genuine defects from correctly implemented behavior. Table 11 reports task-averaged precision, recall, and F1. NorthCUA-Pro achieves the highest F1 at 60.3%, exceeding the strongest comparison model, GPT-6 Astra at 58.2%, by 2.1 percentage points. At 35B, NorthCUA-Mini reaches 54.4% F1, improving on the same-size Nex-N2.5-mini by 5.8 points. Its precision and recall both increase by 6.3 points in this comparison, indicating gains in both the correctness of reported defects and the fraction of defects found.

Table 11. WebTestBench defect detection with a gold checklist. All scores (%) are task-level macro averages from our 100-step evaluation.

Model	Parameters	Precision	Recall	F1
General-purpose models				
Nex-N2.5-mini (Nex-AGI, 2026a)	35B	54.7	51.4	48.6
Nex-N2.5-Pro (Nex-AGI, 2026b)	397B	59.5	55.8	52.8
Kimi-K3 (Moonshot AI, 2026)	2.8T / 104B	67.3	48.2	52.4
GPT-5.6 Sol (OpenAI, 2026b)	Undisclosed	54.8	70.3	56.1
GPT-6 Astra (OpenAI, 2026d)	Undisclosed	55.1	**74.6**	58.2
GPT-6.1 Sol (OpenAI, 2026c)	Undisclosed	53.0	73.8	56.8
GLM-5.3-Flash (Z.ai, 2026)	320B / 18B	60.6	54.0	53.1
Qwen3.8-Max (Qwen Team, 2026c)	2.4T / 95B	59.6	54.2	52.3
DeepSeek-V4.1-Flash (DeepSeek-AI, 2026a)	552B / 16B	59.7	48.9	50.1
Our NorthCUA				
NorthCUA-Nano	9B	53.6	53.7	48.0
NorthCUA-Mini	35B / 3B	61.0	57.7	54.4
NorthCUA-Pro	397B / 17B	**67.8**	63.7	**60.3**

Bold marks the best reported value in each column. Each metric is averaged over tasks independently.

The separate metrics clarify these detection profiles. NorthCUA-Pro also achieves the highest precision at 67.8%, alongside 63.7% recall. GPT-6 Astra combines the highest recall of 74.6% with 55.1% precision. GPT-5.6 Sol shows a similar emphasis on recall, at 70.3% against 54.8% precision, with an F1 of 56.1%. These differences distinguish finding a larger fraction of defects from making more accurate defect reports. Each reported F1 is the average of per-task F1 scores, so it is not reconstructed from the table's mean precision and recall. The development case in Figure 18, separate from the scored WebTestBench evaluation, illustrates how such findings can guide a test–repair–retest cycle. Appendix F describes the browser-evidence qualification protocol.

6.5 Progress Across Training Stages

The final-checkpoint comparisons summarize benchmark performance. We next examine how the reported scores develop through training, beginning with the transition from broad mid-training to verified long-horizon SFT (Table 12). Desktop success increases across all three tiers, by 7.4 points for Nano, 15.0 for Mini, and 10.0 for Pro. Pro reaches 85.9% after this transition, while Mini shows the largest absolute desktop gain.

The Web results show a different distribution of gains. Nano improves by 9.2 points and Mini by 2.6, narrowing their gap from 10.5 to 3.8 points. Pro improves from 69.1% to 69.6%, a smaller gain of 0.5 points. The largest gains therefore depend on both model size and task domain. Mini gains most on desktop tasks, while Nano gains most on Web tasks. These descriptive checkpoint comparisons cover the complete training transition, including data selection and additional optimization. Appendix I records the available evaluation settings and aggregation boundaries for the stage comparisons.

The longer OSWorld 2.0 tasks expose the gap between partial progress and complete workflows at the Mid stage. Across the three tiers, partial scores range from 24.2% to 30.6%, while binary completion ranges from 0.9% to 9.3% (Table 12). Long-horizon SFT increases partial scores by 16.7, 20.7, and 33.0 points for Nano, Mini, and Pro, respectively. Binary completion also

Table 12. Results (%) before and after long-horizon SFT across three interactive benchmarks. OSWorld 2.0 separates full-task Binary completion from Partial credit. Table 13 continues the comparison from SFT to RL.

Model	Stage	OSWorld-Verified	OSWorld 2.0		WebArena-Verified
		Success	Binary	Partial	Success
NorthCUA-Nano	Mid	68.7	0.9	24.2	54.9
	SFT	**76.1**	**12.0**	**40.9**	**64.2**
NorthCUA-Mini	Mid	68.3	2.8	25.6	65.4
	SFT	**83.3**	**18.5**	**46.3**	**68.0**
NorthCUA-Pro	Mid	75.8	9.3	30.6	69.1
	SFT	**85.9**	**27.3**	**63.6**	**69.6**

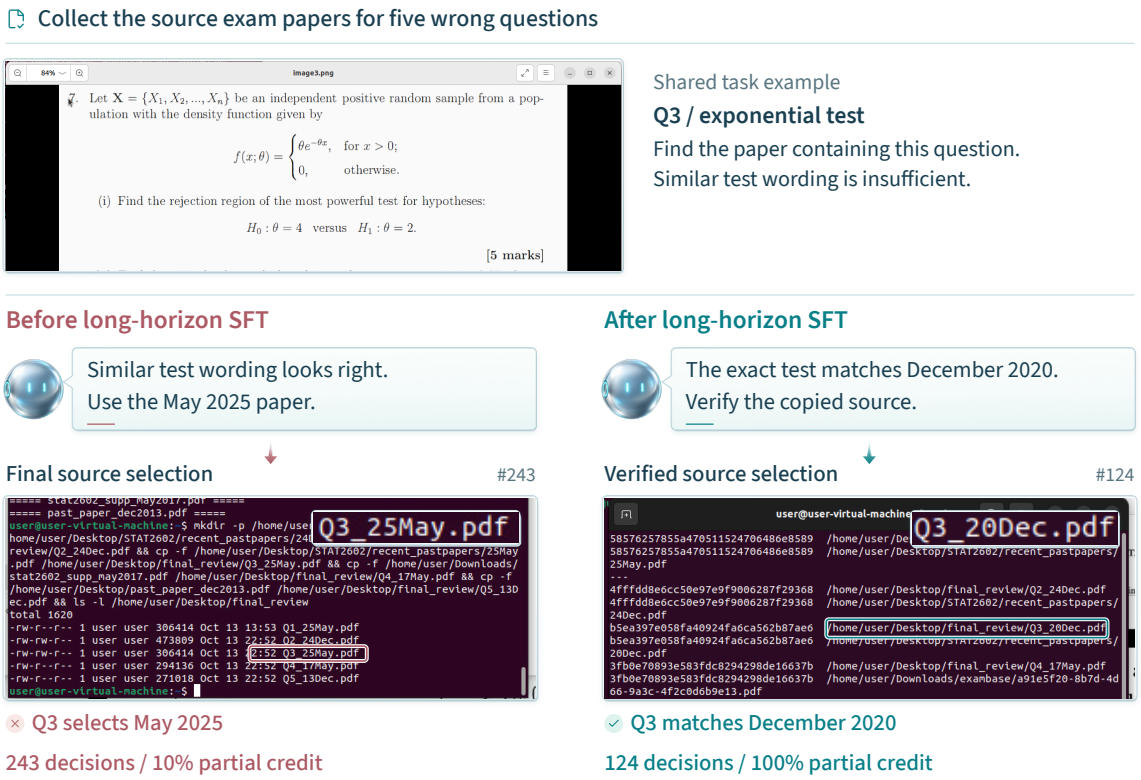


Figure 28. Source-paper selection before and after long-horizon SFT. A shared question anchors the two equal-width result views. Agent callouts paraphrase recorded selection and checking decisions. Highlighted filenames show the different Q3 sources, while scores cover each complete attempt. Screenshots are cropped observations before the indexed decisions, with enlarged filenames. The runs differ in dataset revision and OCR availability, so the comparison illustrates behavior without isolating the effect of SFT.

increases in every tier, with Pro rising from 9.3% to 27.3%. The gains therefore include more fully completed tasks as well as higher earned credit. These SFT checkpoints provide the starting points for the joint-RL comparison in Table 13, allowing the two training transitions to be examined separately.

From similar questions to verified source papers. Figure 28 compares the same OSWorld 2.0 request in two NorthCUA-Mini runs before and after long-horizon SFT. The runs differ in dataset revision and OCR availability, so we examine the observed verification behavior without attributing the full difference to training. A student has collected five question images and asks the agent to find their source exam papers. Similar statistical topics recur across sessions, so recognizing the topic alone can lead to a convincing but incorrect match. The third question, shown at the top of the figure, asks for an exponential-model test with specific null and alternative parameters.

The earlier run reads the question images and searches the candidate papers. It treats related exponential-test wording as confirmation and assigns Q3 to May 2025. Its final folder contains all five named PDFs, but the complete attempt receives 10% partial credit. The later run matches Q3's specific test to December 2020 and checks that the copied file agrees with its source. This more discriminating search also appears elsewhere in the task. For Q5, it rejects a 15-mark candidate that differs from the requested 22-mark question, continues into Trash, and recovers the matching original paper. The resulting attempt receives full credit after 124 decisions, compared with 243 in the earlier run.

The contrast concerns what counts as completion. The earlier run produces the requested collection while accepting an incorrect source match. The later run carries the question-specific criteria through search, rejection and final file checking. These observed behaviors illustrate the sustained verification targeted by long-horizon training. Appendix J gives the run-provenance details.

6.6 Joint Desktop–Web Reinforcement Learning

Long-horizon SFT teaches the policy to follow verified demonstrations. RL adds feedback on the states reached by the policy's own actions, allowing successful and unsuccessful attempts at the same task to inform the next update. We examine this transition across desktop and Web execution under the shared training interface in Section 5.3. The prepared RL task pool contains approximately 1,500 desktop tasks and 500 Web tasks. Each model starts from its long-horizon SFT checkpoint and learns from fresh attempts on these tasks.

Task completion across platforms. Table 13 compares each long-horizon SFT checkpoint (Before RL) with its subsequent joint-training checkpoint (After RL). Within each tier, the task set, tools, observation settings, and execution budget stay fixed, allowing the comparison to track performance through the transition from supervised learning to joint RL. OSWorld 2.0 retains Binary and Partial separately to distinguish complete workflows from progress on individual requirements.

After joint RL, reported success increases on both platforms at all three model sizes. OSWorld-Verified success rises by 2.9 points for Nano, 2.0 for Mini, and 2.8 for Pro. The corresponding WebArena-Verified gains are 3.0, 2.5, and 4.3 points. Pro reaches 88.6% and 73.9% on the two benchmarks, respectively. The desktop gains are similar across tiers, whereas Pro has the largest reported Web increase. There is thus no uniform relationship between model size and the magnitude of the checkpoint gain.

Table 13. Before- and after-RL results on three interactive benchmarks (%). OSWorld-Verified uses 361 tasks, OSWorld 2.0 uses 108 tasks, and WebArena-Verified uses 812 tasks. Scores retain their reported aggregation rules (Appendix I).

Model	Stage	Desktop			Web
		OSWorld-Verified	OSWorld 2.0		WebArena-Verified
		Success	Binary	Partial	Success
Our NorthCUA					
NorthCUA-Nano	Before RL	76.1	12.0	40.9	64.2
	After RL	78.9	13.0	45.6	67.2
NorthCUA-Mini	Before RL	83.3	18.5	46.3	68.0
	After RL	85.3	18.5	50.1	70.4
NorthCUA-Pro	Before RL	85.9	27.3	63.6	69.6
	After RL	88.6	30.6	66.8	73.9

OSWorld 2.0 distinguishes gains in earned credit from gains in full-task completion. Partial scores increase by 4.7, 3.8, and 3.2 points for Nano, Mini, and Pro, respectively. Nano and Pro also improve binary completion by 0.9 and 3.3 points, with Pro reaching 30.6% binary completion and 66.8% partial credit. Mini's binary score stays at 18.5% as its partial score rises from 46.3% to 50.1%. Its higher earned credit therefore does not entail a higher full-task completion rate. Together, the two metrics show where the final checkpoints improve requirement satisfaction and where that improvement also appears in complete workflows.

The remaining analyses address two different parts of the learning process. We first measure how the runtime delivers long visual records without changing their contents, then examine how task rewards distinguish two superficially similar interaction paths.

Delivering long visual experience efficiently. The learning interface preserves the screenshots seen at each decision. As a result, long attempts can contribute repeated image windows across training segments. We use a separate archived 9B desktop run for the engineering profile and trajectory example below. To evaluate the lossless delivery path in Section 5.3.4, we examine 28 batches. Pairing the raw and encoded counters for each batch holds the underlying image tensors fixed while measuring their representation size.

Across this workload, encoding reduces the summed image-tensor payload by 97.2%, leaving 2.8% of the original representation size. The median raw and encoded batch sizes are 110.9 and 2.7 GiB, respectively. Encoding preserves all screenshots and their values. The counters cover eligible image tensors, including repeated segment occurrences, and exclude non-image fields and transport overhead.

Figure 29(b) reports delivery and policy-synchronization timings. Image encoding takes a median of 21.0 seconds, while object-store put calls take 1.9 seconds. The recorded policy-sync lifecycle takes 87.7 seconds and includes the 5.0-second in-flight call wait, weight export, rollout-engine reload, and receipt work. The in-flight wait is nested within the synchronization lifecycle, so these component timings are non-additive. They characterize delivery and synchronization costs, while the payload comparison measures the reduction in representation size for the same visual experience.

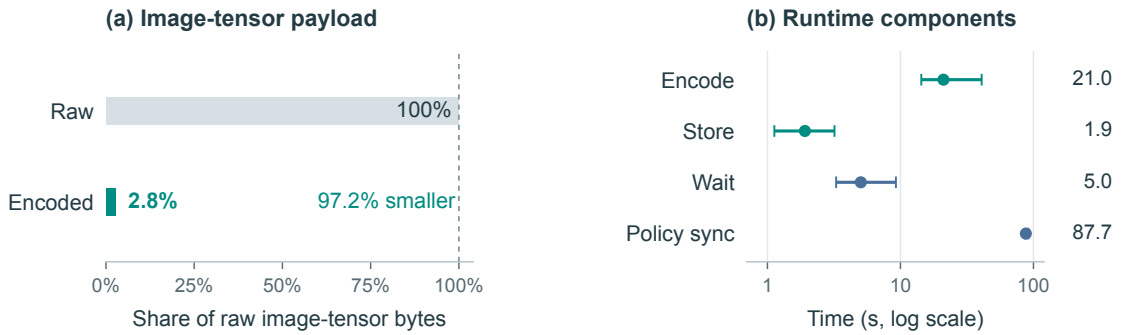


Figure 29. Desktop visual-experience delivery profiling. The profile uses a separate desktop workload. (a) Encoded image-tensor bytes as a percentage of raw bytes, calculated from the summed paired counters across 28 batches. The dashed reference is the original representation size. (b) Median and interquartile range of component timings for the same update range. Policy synchronization includes the in-flight wait, so the displayed timings are not additive. Payload sizes exclude non-image fields and transport overhead.

From interface choices to learning feedback. Figure 30 examines what the terminal reward distinguishes in an actual task. The user asks the agent to switch Thunderbird to Tags view, hide the Get Messages button, and close the application to save the settings. This requires selecting the intended control and preserving its configuration after the application exits.

The two attempts follow the same first five actions and reach Tags view. Their next choices differ in scope. Attempt A disables the entire Folder Pane Header, removing all its controls from view. Attempt B opens the header's own menu and unchecks Show Get Messages, leaving the remaining header controls visible. Both then close Thunderbird and report completion. The task verifier checks the saved Tags mode and the specific button's hidden setting, returning reward 0 for A and reward 1 for B. Thus, the reward distinguishes the saved button configuration from hiding its entire parent header.

The complete group contains five successful and three unsuccessful attempts. Under the group-relative rule in Eq. 3, the successful attempts receive positive advantages and the unsuccessful attempts receive negative advantages, although their task reward is zero. These advantages carry the verifier's distinction between saved configurations into the policy update. The recorded attempts also cross a policy-publication boundary. The learning interface in Section 5.3.3 preserves each segment's input and behavior version while keeping its attempt reward and group identity. Checkpoint-level improvement is evaluated separately in Table 13.

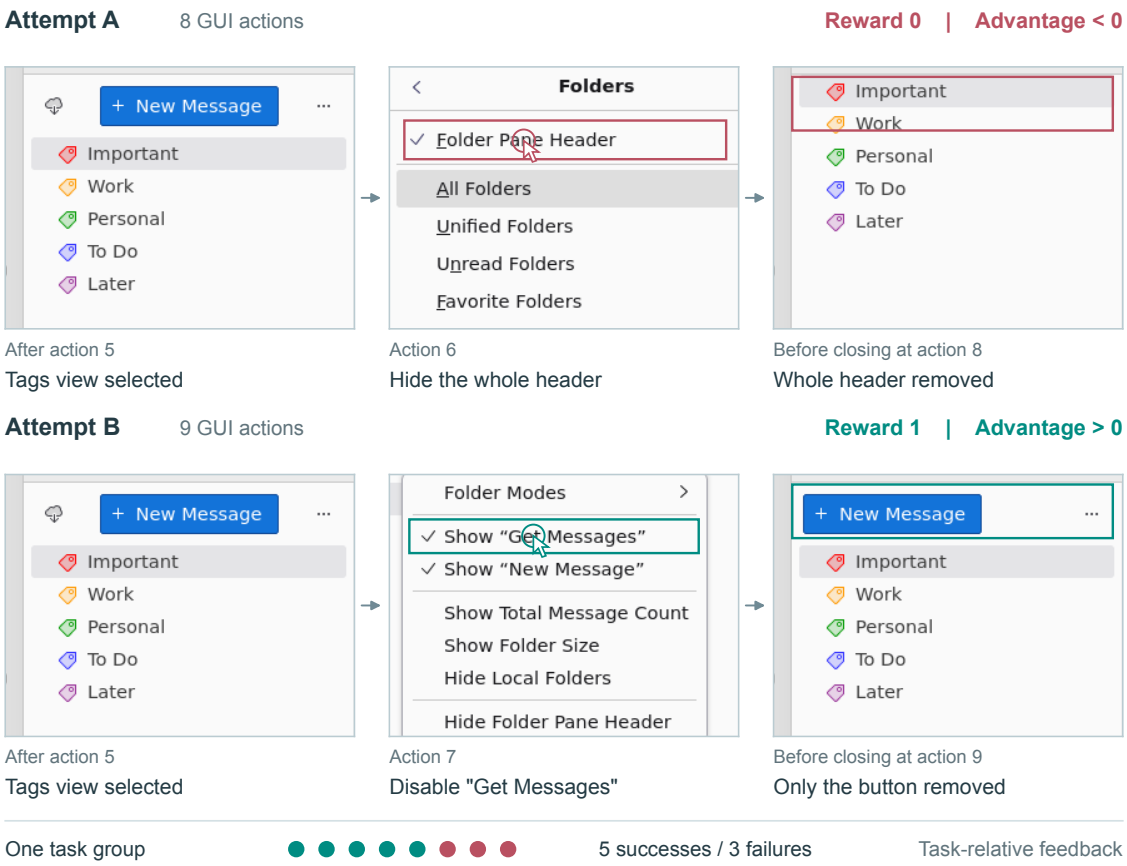


Figure 30. Executed trajectories with contrasting task rewards. Both Thunderbird attempts reach Tags view. A hides the entire header, while B disables Get Messages specifically. Native screenshot crops show executed click positions in the middle panels and the state before application closure on the right. A's first two panels crop one observation. B opens the header menu between those panels. The archived configuration verifier returns rewards 0 and 1. Both attempts belong to one eight-member GRPO group containing five successes and three failures, which determines their advantage signs. They span an asynchronous policy update. This is a within-group reward comparison, not a before/after checkpoint evaluation.

6.7 An Evaluated Browser Recovery Case

Aggregate scores summarize completion, but a recorded interaction can show how the policy handles a mismatch between its plan and the interface. In the Pro episode in Figure 31, NorthCUA is asked to identify the best-selling brand in the first quarter of 2022. Reaching the answer requires navigating to the appropriate report, applying the correct time interval, and reading the returned product rows.

The first attempted period selection requests an unavailable Quarter option and times out. To identify the mismatch, the agent opens the dropdown. At decision 6, it observes that only Day, Month, and Year are available. It then uses Month for aggregation and enters January 1 and March 31 as the date bounds, preserving the requested quarter through the two date fields. By decision 10, the visible form contains the intended interval and is ready to run.

After running the report, the agent scrolls through its rows. The state at decision 12 shows two Sprite products among the highest-quantity entries. The agent checks the remaining rows

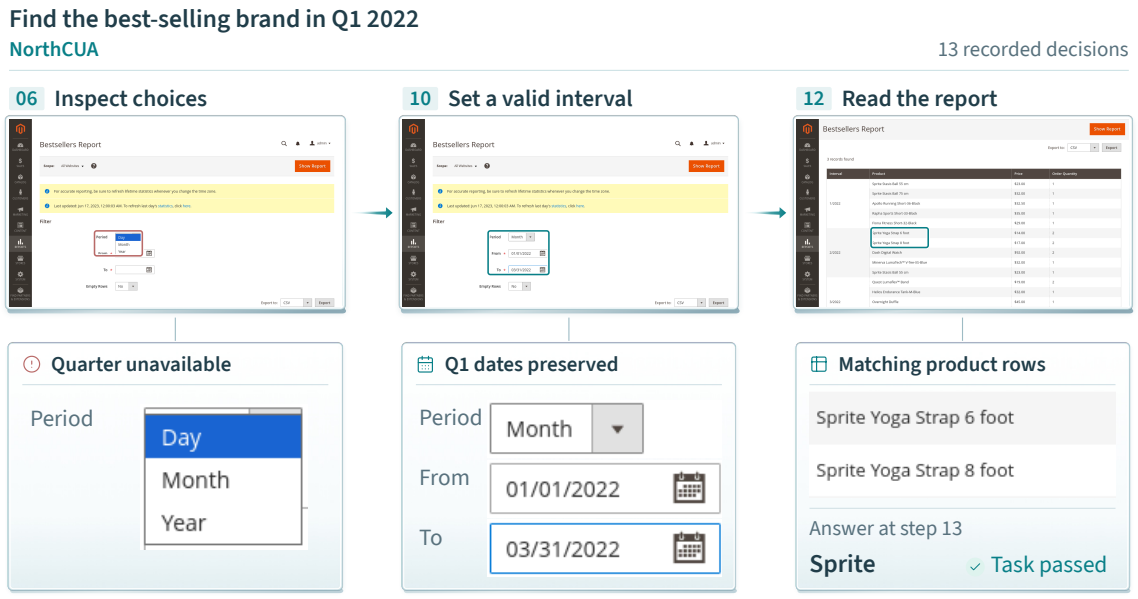


Figure 31. Recovering from an unavailable browser action. In a retained Pro evaluation attempt, NorthCUA first attempts to select an unavailable Quarter period. It inspects the dropdown, switches to Month with the exact Q1 date range, and reads the resulting table before answering Sprite. The saved evaluator marks the task successful. Numbered panels show observations before decisions 6, 10, and 12 of a 13-decision attempt, with intermediate actions omitted. Detail views enlarge original control crops, with field labels arranged for readability. The answer at decision 13 and the external evaluator's success marker are annotations outside the webpage.

before returning Sprite, and the saved task evaluator records success. The episode contains 13 model decisions, including the terminal response. Using the date fields with the available Month aggregation mode lets the agent correct its execution plan while preserving the requested quarter.

6.8 NorthCUA-Ultrafast

Long computer-use tasks accumulate the latency of many successive model decisions. Some decisions require open-ended content, but others reduce to choosing an operation and a control already present on the page. Predicting these finite choices directly could remove the decoding cost of routine actions. Jev provides a relevant interface for this purpose by answering structured choice questions without generating their answers token by token (Almeida, 2026). We adapt this idea to NorthCUA-Nano through NorthCUA-Ultrafast, which combines a choice head for finite decisions with the original model's generation path for open-ended outputs.

Figure 32 illustrates the resulting division of work. The classifier receives the task and recorded interaction history, including page observations, screenshots, and action overlays. Candidate controls are bound to the latest valid page observation, so historical references cannot be executed as current targets. One backbone forward pass supplies the features used to score the operation and its applicable targets. A selected click, hover, or checkbox change can then be converted directly into a tool call.

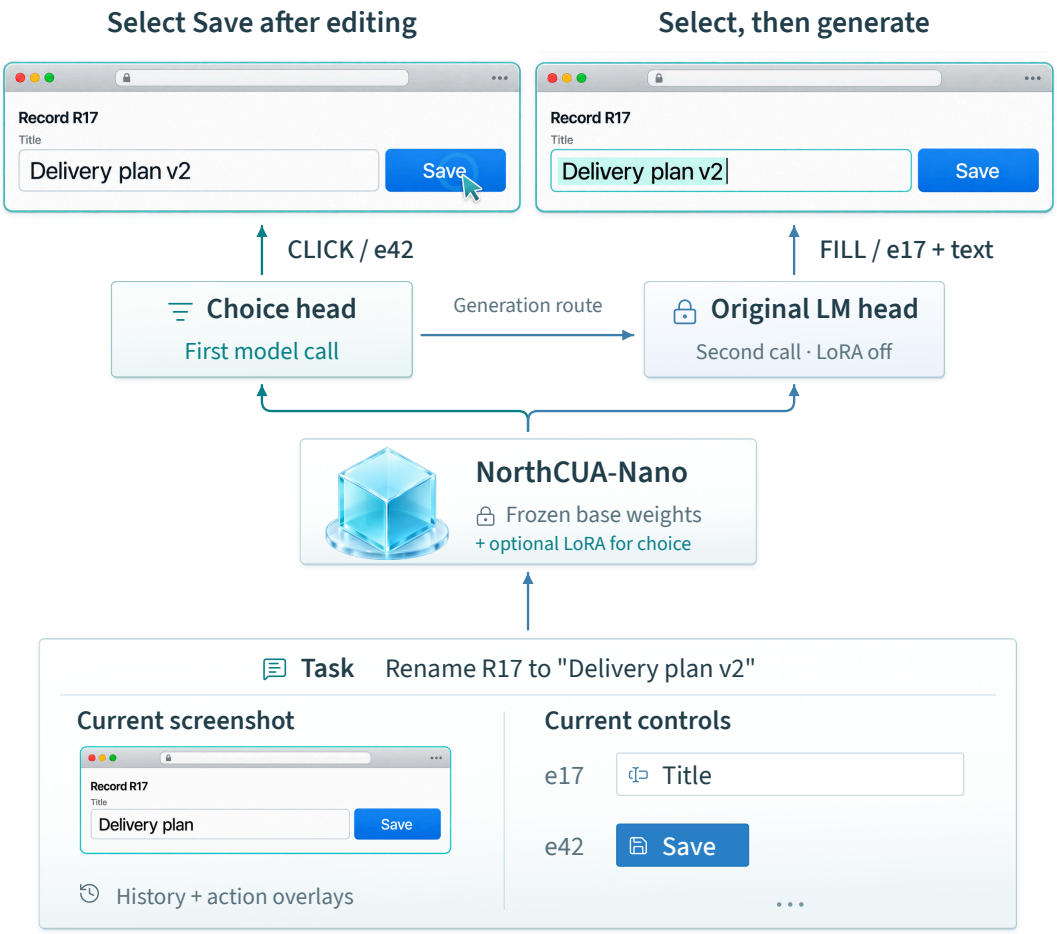


Figure 32. Choice-based Web execution. The input shows the title before editing. Generated GUI views illustrate filling the new title (right) and a subsequent Save click (left). At each decision, current controls support operation and target selection. A direct choice executes without decoding, while a generation route triggers a second full-model call with the original prompt and a fresh cache. Optional LoRA adapters are active only for classification. Route and target checks are omitted.

Filling a field additionally requires text, so it triggers a new call using the original model prompt and generation head. Other tool actions and final responses use the same generation path. The executor validates the returned tool and arguments, and uncertain finite choices can fall back to native generation. This preserves the original output interface while allowing routine finite actions to bypass decoding. Appendix E.7.4 specifies the routing and validation profiles.

We study how much adaptation this choice interface requires, starting from the long-horizon-SFT NorthCUA-Nano checkpoint before joint RL. Following Valen's head-training approach (Liuziyu77, 2026), *Head-only* freezes all original weights and trains a 4.21M-parameter choice head, about 0.045% of the complete multimodal checkpoint. *Head + LoRA* also trains low-rank updates in the language layers (Hu et al., 2022), bringing the trainable total to 47.49M parameters. These adapters are enabled for classification and disabled for native generation, which keeps the original generator common to both variants. Both use about 226,000 training decision states, with fixed validation and test sets of 5,203 and 6,087 states. Supervision covers the recorded operation and its required target. Appendix E.7 gives the readout, training objective, and recipe.

Table 14. Online quality and execution cost of NorthCUA-Ultrafast. Success uses all 812 tasks. Mean task time and steps use the fixed 320-task success-selected subset. Decoding bypass uses dispatched actions. Appendix E.7.4 defines the populations and deployment settings.

Method	Task success (%)	Mean task time (s)	Mean steps	Decoding bypass (%)
Full generation	**64.2**	616.8	**9.5**	0.0
Ultrafast, Head-only	59.2	288.8	10.2	**40.2**
Ultrafast, Head + LoRA	60.6	**266.8**	10.6	31.1

We first evaluate whether the choice interface retains the recorded decisions. Head-only and Head + LoRA reach validation route-and-target accuracies of 86.1% and 88.2%, respectively. On the held-out test set, Head-only reaches 85.4% route-and-target accuracy against Full generation's 85.0%, while their full-action exact-match scores are 64.0% and 64.6%. Generated parameters account for much of the remaining full-action error. The LoRA result is a validation comparison. Appendix E.7 gives the separate metrics and checkpoint-selection details.

To assess complete task execution, we compare the three policies on all 812 WebArena-Verified tasks, deploying each inference replica on a single GPU. Table 14 reports success and measured execution cost under the definitions in Appendix E.7.4. Success uses the full benchmark, while timing and step counts compare the same 320 successfully completed tasks across policies. Decoding bypass measures the fraction of dispatched actions produced directly by the choice head.

Head-only completes 481 tasks using only the choice head for adaptation. Head + LoRA completes 492 (60.6%), gaining 11 tasks or 1.4 percentage points. With about 0.5% trainable parameters, LoRA retains 94.4% of Full generation's 64.2% success rate, leaving a gap of 3.6 percentage points.

The direct-action path is used frequently during execution. Head-only and Head + LoRA bypass generation for 40.2% and 31.1% of recorded actions, respectively. We next examine the task-level cost of these execution policies on the fixed success-selected subset.

Mean task time falls from 616.8 s to 288.8 s for Head-only and 266.8 s for Head + LoRA, despite slightly longer trajectories of 10.2 and 10.6 steps versus 9.45. On the fixed timing subset, this is a factor-of-2.3 reduction in mean task time for Head + LoRA. Its shorter task time despite a lower bypass rate also shows why direct-action frequency alone is insufficient to characterize execution efficiency. Each replica uses one GPU. Appendix E.7.4 gives the decoding and serving settings for these measured execution costs. The following case illustrates how fast selection and flexible generation cooperate within one successful task.

From search recovery to a saved product update. Figure 33 follows a Head + LoRA attempt at a WebArena-Verified task that asks the agent to disable Karmen yoga pants because of quality issues. The request requires finding the correct catalog entry and saving its disabled state. Searching for the full phrase returns no records. The native generator responds to this observation by shortening the query to *Karmen*, after which the search exposes the configurable *Karmen Yoga Pant* and its individual variants.

The choice path opens the configurable product, switches *Enable Product* off, and selects *Save*, all without decoding action text. The final page shows the saved disabled state. The independent evaluator verifies the corresponding product-1819 save request and awards full credit.

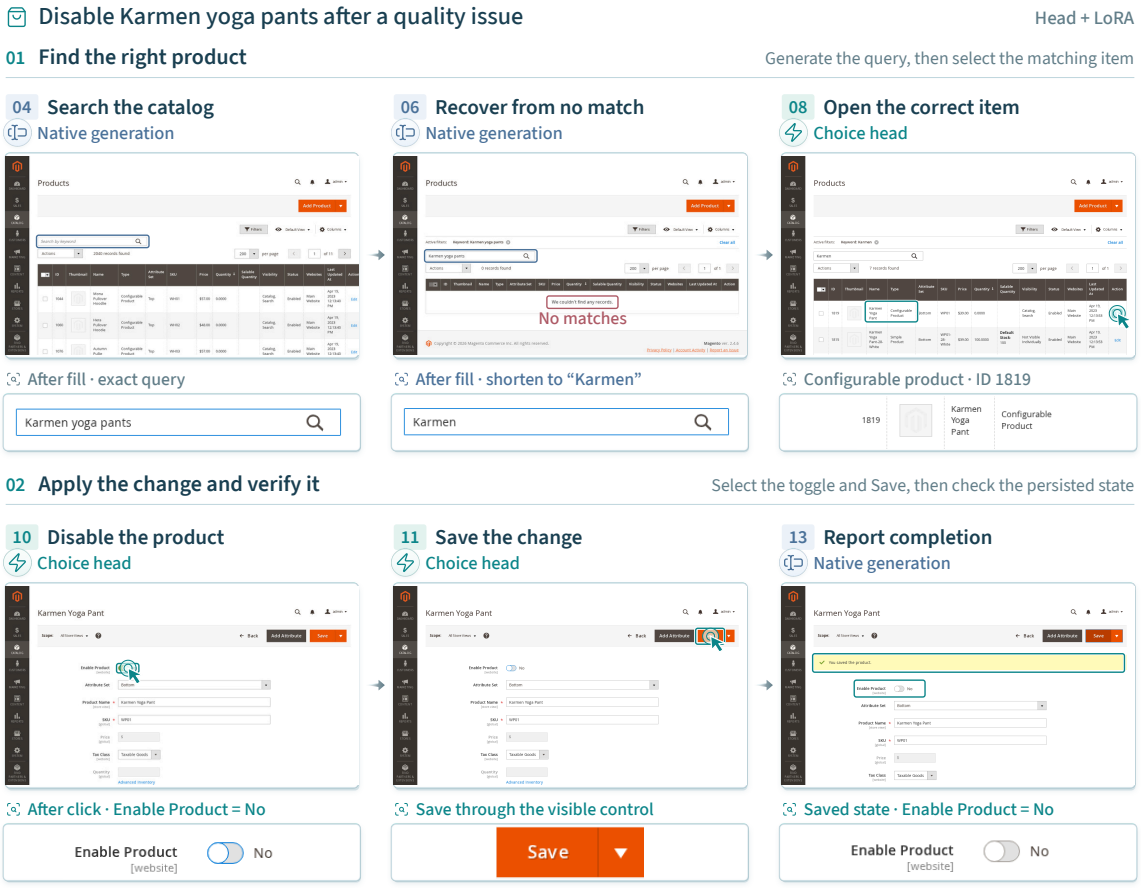


Figure 33. Ultrafast product-update case. Selected Head + LoRA decisions from one WebArena-Verified attempt. Lightning and text-entry icons distinguish direct choice from native generation. Numbers retain the original decision indices, with intermediate steps omitted. Insets enlarge recorded details, with post-action states labeled.

Across the complete attempt, six of twelve executed browser actions bypass generation. The other actions handle navigation, query text, and waiting, and a thirteenth decision generates the final response. This execution illustrates the intended division of work. Generation remains available when the search needs revision, while discrete selections carry the recovered task through to a verified update. The aggregate bypass rates in Table 14 measure how often this direct execution path is used across the benchmark.

7 Related Work

NorthCUA connects computer-use learning with the tasks and execution systems that supply its experience. We review these foundations in three parts. We first discuss policies and learning methods, then the construction of tasks and experience, and finally the harnesses and infrastructure that support their collection and use in training.

Computer-use policies and learning. Accurate GUI actions depend on linking instructions to screen elements. SeeClick studies grounding pre-training (Cheng et al., 2024). GUI-Actor develops attention-based pointing, while GUI-Cursor refines cursor movements through visual

feedback (Wu et al., 2025, Zhao et al., 2026). End-to-end policies build on this capability to coordinate actions over time. UI-TARS integrates perception, reasoning, and action prediction, and OpenCUA develops reflective supervision from human demonstrations (Qin et al., 2025, Wang et al., 2025c). Learning from outcomes offers a further source of improvement. ComputerRL scales live-desktop RL (Lai et al., 2026), while VEM learns offline value estimates to guide environment-free policy exploration (Chen et al., 2026b). UI-TARS-2 and Qwen-CUA connect staged training with interactive learning (Wang et al., 2025a, Lu et al., 2026). Context handling also affects how interaction experience is learned. ScaleCUA masks reused history and shares rollout rewards and advantages across segments (Lv et al., 2026). Qwen-CUA preserves episode credit across slices and normalizes each trajectory's objective over active policy tokens (Lu et al., 2026). NorthCUA combines broad GUI mid-training, verified long-horizon SFT, and joint desktop–web RL. Our emphasis is on integrating decision-time input preservation and complete-attempt weighting with the Harness and experience-production pipeline.

Scalable tasks and experience. Scaling workflow learning creates a corresponding demand for executable tasks and reliable outcomes. OS-Genesis derives tasks from exploratory interaction (Sun et al., 2025), while GUI-360 combines sourced requests, environment templates, execution, and filtering for Windows data collection (Mu et al., 2025). ProCUA-SFT grounds tasks in real materials and checks preconditions before rollout (Jung et al., 2026). For outcome-driven learning, CUA-Gym and Qwen-UI-Agent synthesize compatible tasks, initial states, and verifiers (Wang et al., 2026a, Zhou et al., 2026b). WEFT extends this whole-system view to tool-use post-training, using execution feedback to revise interacting components and fresh rollouts to evaluate each revision (Mao et al., 2026). NorthCUA organizes construction around available software and content, sourced requests, or verified seed tasks. Each route produces an executable `CaseSpec`. A separate grounding route uses application documentation to identify control functions and construct screenshot–query–location supervision.

Hard tasks also motivate retaining procedural knowledge across executions. Work2Skill turns synchronized human GUI demonstrations into reusable skills through on-demand inspection of action, visual, and available accessibility evidence (Huang et al., 2026). WebXSkill extracts executable procedures with language guidance from trajectories (Wang et al., 2026c). Search offers a complementary way to use exploratory computation. Computer-Using World Model simulates UI transitions for action selection (Guan et al., 2026), and Everything of Thoughts combines learned guidance with tree search for reasoning (Ding et al., 2024). Our RSI collection retains validated textual skills and real workspace checkpoints, so later attempts can reuse procedures or resume partial progress before full-task verification.

Agent harnesses and learning infrastructure. Such collection and learning require an execution layer that preserves both software state and training evidence. UFO introduces hierarchical Windows interaction (Zhang et al., 2025c). UFO2 adds application-specialized agents and GUI–API integration (Zhang et al., 2026), and UFO3 extends orchestration to heterogeneous devices through a task DAG (Zhang et al., 2025d). For reuse across training systems, Agent Lightning v1.0 separates harness execution from RL optimization (He et al., 2026). Orchard provides shared sandbox lifecycle services (Peng et al., 2026), while Polar reconstructs token-faithful training trajectories from recorded model calls (Xu et al., 2026a). NorthCUA focuses on integrating these system responsibilities for stateful visual interaction. Its shared desktop/web runtime, live outcome inspection, and parallel lifecycle service support experience collection. The Experience Adapter then preserves the recorded decisions when converting online attempts

into learning samples, connecting execution and assessment to policy training through the same task and outcome interfaces.

8 Limitations and Responsible Use

Generalization and coverage. NorthCUA operates across diverse applications, while unfamiliar software and changing interface conventions remain challenging. Success-based filtering can underrepresent difficult or unusual workflows. Documentation-guided grounding expands the functional knowledge available during data construction, and skill- and milestone-guided collection revisit workflows that ordinary attempts leave unfinished. Extending this coverage to new software and maintaining reliable execution as applications evolve remain important directions for the experience pipeline.

Feedback and optimization. Executable verifiers and live-workspace inspection tie learning feedback to the work actually produced. Controlled verifier probes and evidence-linked judgments help check this feedback. Incomplete predicates can still miss errors, and model judgments of open-ended content can be inconsistent or biased (Zheng et al., 2023). Long workflows also concentrate many decisions into a terminal outcome. More informative intermediate feedback could improve credit assignment and help agents reach correct results with fewer unnecessary actions.

Responsible deployment. Isolated workspaces, explicit tool profiles, and retained execution records provide controls for deployment. Operating real software also requires authorized access, least-privilege credentials, and confirmation for consequential actions. External pages and documents can carry indirect prompt injections (Debenedetti et al., 2024), motivating adversarial testing appropriate to each application. Released materials should undergo licensing, privacy, and credential review alongside the operational checks.

9 Conclusion

We presented NorthCUA, an integrated system for developing computer-use agents through experience at scale. By connecting the NorthCUA Harness, Experience Flywheel, and Staged Learning through shared tasks and outcome checks, the system supports learning from both verified demonstrations and fresh policy interaction. Improved policies can return to the same infrastructure to collect further experience. Recursive self-improvement extends collection to difficult workflows by preserving useful procedures and verified milestones, allowing later attempts to build on earlier progress.

Across six benchmarks, the Nano, Mini, and Pro family achieves state-of-the-art results among open-source models in GUI grounding, desktop execution, and Web interaction. At 397B parameters, Pro outperforms the trillion-parameter general-purpose baselines on both grounding benchmarks, OSWorld-Verified, and WebArena-Verified. Stage comparisons further show improvements in desktop and Web execution across all three model sizes following long-horizon SFT and joint RL. Alongside these gains in task completion, NorthCUA-Ultrafast explores a complementary route to faster decision-making.

We release the three models, the full Harness code, and this technical report so that the work can be extended at both the policy and system levels. Together, they provide an open foundation for adapting agents to new software and developing better ways to collect and learn from experience. The broader goal is to make reliable software execution a general AI capability, allowing users to delegate complete workflows from intent to verifiable outcome.

NorthCUA Team

Shanghai Qiji Zhifeng Co., Ltd.

Chaoyun Zhang[★♦], Bingxuan Li[★♦], Pinxue Guo[★♦], Yangfan Gao[♦], Zikun Cai[♦], Tianci Xie[♦], Hui Li[♦], Yuqi Huang[♦], Yifei Cao[♦], Bo Mao, Tian Tang, Qichang Zheng, Tongtong Liu, Xiaoxing Wang, Jinxiu Liu, Lizhi Lin, Zhou Shao, Maosen Zhou, Miao Zheng, Shuo Zhang, Yangji He, Kepeng Lei, Ning Zhao, Wei Shen, Zhenjiang Jin, Xinyue Zhang, Peng Sun, Zhenhua Han, Rui Zheng, Hang Yan

★ Equal contribution ♦ Core contributors

References

Diogo Almeida. Introducing system one models and Jev. TypeSafe AI blog, 2026. URL https://typesafe.ai/blog/introducing-system-one-models-and-jev. September 15, 2026.

Ample Sound. *Ample Bass P Main Panel Manual*. Ample Sound Technology, n.d. URL https://www.amplesound.net/en/Main_Panel_Manual-ABP.pdf. Accessed 2026-10-18.

Anthropic. Claude Fable 5.1 and Claude Mythos 5.1 system card, 2026a. URL https://www.anthropic.com/claude-fable-5-1-mythos-5-1-system-card. September 2026.

Anthropic. Claude Fable 5 and Claude Mythos 5 system card. Technical report, Anthropic, June 2026b. URL https://www.anthropic.com/claude-fable-5-mythos-5-system-card.

Anthropic. System card: Claude Opus 4.8. Technical report, Anthropic, May 2026c. URL https://www-cdn.anthropic.com/0b4915911bb0d19eca5b5ee635c80fef830a37ea.pdf.

Anthropic. Introducing Claude Opus 5.5, 2026d. URL https://www.anthropic.com/claude-opus-5-5. Released September 22, 2026.

Anthropic. Claude Opus 5.5 system card, 2026e. URL https://www.anthropic.com/claude-opus-5-5-system-card. September 2026.

Anthropic. Claude Opus 5 system card. Technical report, Anthropic, July 2026f. URL https://www.anthropic.com/claude-opus-5-system-card.

ByteDance Seed. Seed2.1: A next-generation agent for real-world productivity, June 2026. URL https://seed.bytedance.com/en/seed2_1.

Hyungjoo Chae, Jungsoo Park, and Alan Ritter. Safe and scalable web agent learning via recreated websites. *arXiv preprint arXiv:2603.10505*, 2026.

Liangyu Chen, Hanzhang Zhou, Chenglin Cai, Jianan Zhang, et al. UI-Ins: Enhancing GUI grounding with multi-perspective instruction as reasoning. In *International Conference on Learning Representations*, pp. 54418–54444, 2026a.

Mengzhuo Chen, Jiani Zheng, Lu Wang, Fangkai Yang, Chaoyun Zhang, Lingrui Mei, Wenjie Yin, Qingwei Lin, Dongmei Zhang, and Saravan Rajmohan. VEM: Environment-free exploration for training GUI agent with value environment model. *Transactions on Machine Learning Research*, 2026b.

Kanzhi Cheng, Qiushi Sun, Yougang Chu, Fangzhi Xu, et al. SeeClick: Harnessing GUI grounding for advanced visual GUI agents. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 9313–9332. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.505.

Edoardo Debenedetti, Jie Zhang, Mislav Balunović, Luca Beurer-Kellner, Marc Fischer, and Florian Tramèr. AgentDojo: A Dynamic Environment to Evaluate Prompt Injection Attacks and Defenses for LLM Agents. In *Advances in Neural Information Processing Systems*, volume 37, pp. 82895–82920. Curran Associates, Inc., 2024. doi: 10.52202/079017-2636.

DeepSeek-AI. DeepSeek-V4.1-Flash: Pushing the limits of KV cache compression, 2026a. URL https://huggingface.co/deepseek-ai/DeepSeek-V4.1-Flash. Accessed 2026-10-19.

DeepSeek-AI. DeepSeek-V4-Flash-Vision-Exp release, August 2026b. URL https://api-docs.deepseek.com/updates/.

Ruomeng Ding, Chaoyun Zhang, Lu Wang, Yong Xu, Minghua Ma, Wei Zhang, Si Qin, Saravan Rajmohan, Qingwei Lin, and Dongmei Zhang. Everything of thoughts: Defying the law of Penrose triangle for thought generation. In *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 1638–1662. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.findings-acl.95.

Rotem Dror, Gili Baumer, Segev Shlomov, and Roi Reichart. The Hitchhiker's Guide to Testing Statistical Significance in Natural Language Processing. In Iryna Gurevych and Yusuke Miyao (eds.), *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 1383–1392, Melbourne, Australia, July 2018. Association for Computational Linguistics. doi: 10.18653/v1/P18-1128.

E2B. E2B Desktop Sandbox: Open source virtual computer for computer use, 2026a. URL https://github.com/e2b-dev/desktop. Official software and documentation snapshot, revision 17ddc44f3108. Accessed 2026-09-16.

E2B. E2B Runtime, 2026b. URL https://github.com/e2b-dev/runtime. Official software and documentation snapshot, revision e7b22501988c. Accessed 2026-09-16.

Adrien Ecoffet, Joost Huizinga, Joel Lehman, Kenneth O. Stanley, and Jeff Clune. First return, then explore. *Nature*, 590:580–586, 2021. doi: 10.1038/s41586-020-03157-9.

Amine El Hattami, Megh Thakkar, Nicolas Chapados, and Christopher Pal. WebArena Verified: Reliable evaluation for web agents. In *Workshop on Scaling Environments for Agents*, 2025.

Tao Ge, Baolin Peng, Hao Cheng, and Jianfeng Gao. Synthetic computers at scale for long-horizon productivity simulation. *arXiv preprint arXiv:2604.28181*, 2026.

Google DeepMind. Gemini 3.1 Pro model card, 2026a. URL https://deepmind.google/models/model-cards/gemini-3-1-pro/.

Google DeepMind. Gemini 3.6 Flash model card, July 2026b. URL https://deepmind.google/models/model-cards/gemini-3-6-flash/.

Google DeepMind. Gemini 3.8 Flash model card, September 2026c. URL https://deepmind.google/models/model-cards/gemini-3-8-flash/.

Google DeepMind. Gemini 4 Argon model card, October 2026d. URL https://deepmind.google/models/model-cards/gemini-4-argon/.

Yiming Guan, Rui Yu, John Zhang, Lu Wang, Chaoyun Zhang, Liqun Li, Bo Qiao, Si Qin, He Huang, Fangkai Yang, et al. Computer-using world model. *arXiv preprint arXiv:2602.17365*, 2026.

Zhiyuan He, Siwei Zhang, Zhiwen Zhou, Yuqing Yang, Yu Kang, Yuge Zhang, Luna K. Qiu, Tin Yan Tsui, Jiahang Xu, and Chong Luo. Agent Lightning v1.0: Towards Harnessed Agentic RL. *arXiv preprint arXiv:2608.17528*, 2026.

hojuna. Sharing GLM-5.3-Flash evaluation results, September 2026. URL https://github.com/xlang-ai/OSWorld/issues/591. Independent evaluation with public configuration and trajectories.

Ari Holtzman, Jan Buys, Li Du, Maxwell Forbes, and Yejin Choi. The Curious Case of Neural Text Degeneration. In *International Conference on Learning Representations*, 2020.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

Yuqi Huang, Hui Li, Yifei Cao, Zikun Cai, Tianci Xie, Yangfan Gao, Pinxue Guo, Chaoyun Zhang, Bingxuan Li, Rui Zheng, et al. Work2Skill: Capturing human GUI experience for reusable agent skill induction. *arXiv preprint arXiv:2610.12984*, 2026.

Jaehun Jung, Ximing Lu, Brandon Cui, Muhammad Khalifa, et al. ProCUA-SFT technical report. *arXiv preprint arXiv:2606.17321*, 2026.

Kimi Team. Kimi K3: Open frontier intelligence. *arXiv preprint arXiv:2607.24653*, 2026.

Fanheng Kong, Jingyuan Zhang, Yang Yue, Chenxi Sun, Yang Tian, Shi Feng, Xiaocui Yang, Daling Wang, Yu Tian, Jun Du, Wenchong Zeng, Han Li, and Kun Gai. WebTestBench: Evaluating computer-use agents towards end-to-end automated web testing. *arXiv preprint arXiv:2603.25226*, 2026.

Hanyu Lai, Xiao Liu, Yanxiao Zhao, Han Xu, et al. ComputerRL: Scaling end-to-end online reinforcement learning for computer use agents. In *International Conference on Learning Representations*, pp. 23553–23591, 2026.

Hongxin Li, Jingfan Chen, Jingran Su, Yuntao Chen, et al. AutoGUI: Scaling GUI grounding with automatic functionality annotations from LLMs. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 10323–10358. Association for Computational Linguistics, 2025a. doi: 10.18653/v1/2025.acl-long.510.

Kaixin Li, Ziyang Meng, Hongzhan Lin, Ziyang Luo, Yuchen Tian, Jing Ma, Zhiyong Huang, and Tat-Seng Chua. ScreenSpot-Pro: GUI grounding for professional high-resolution computer use. In *Workshop on Reasoning and Planning for Large Language Models*, 2025b.

Liuziyu77. Valen. Software repository, 2026. URL https://github.com/Liuziyu77/Valen/tree/61529d134a1c8a8c2e7d165e1050f81fe7905277. Revision 61529d1, accessed 2026-10-08.

Dunjie Lu, Shuai Bai, Tianyi Bai, Sicheng Fan, et al. Qwen-CUA: Native computer use for (almost) everything. *arXiv preprint arXiv:2608.02352*, 2026.

Yadong Lu, Jianwei Yang, Yelong Shen, and Ahmed Awadallah. OmniParser for pure vision based GUI agent. *arXiv preprint arXiv:2408.00203*, 2024.

Bowen Lv, Xiao Liu, Yanyu Ren, Hanyu Lai, et al. SCALECUA: Scaling computer use agents with verifiable task synthesis and efficient online RL. *arXiv preprint arXiv:2607.11185*, 2026.

Bo Mao, Hang He, Linting Wang, Lizhi Lin, Maosen Zhou, Guanming Liu, Jinxiu Liu, Tianyu Huai, Chaoyun Zhang, Bingxuan Li, et al. WEFT: Scaling tool-use post-training for general-purpose agents. *arXiv preprint arXiv:2609.36887*, 2026.

MiniMax. MiniMax M3.1 Flash: Redefining agent performance and cost, October 2026a. URL https://www.minimax.io/blog/minimax-m31f. Released 2026-10-19.

MiniMax. MiniMax-M3 model card, 2026b. URL https://huggingface.co/MiniMaxAI/MiniMax-M3.

Model Context Protocol. Model Context Protocol Specification: Tools, 2025. URL https://modelcontextprotocol.io/specification/2025-11-25/server/tools. Protocol revision 2025-11-25. Accessed 2026-09-17.

Moonshot AI. Kimi-K3 model card, 2026. URL https://huggingface.co/moonshotai/Kimi-K3. Accessed 2026-10-19.

Jian Mu, Chaoyun Zhang, Chiming Ni, Lu Wang, Bo Qiao, Kartik Mathur, Qianhui Wu, Yuhang Xie, Xiaojun Ma, Mengyu Zhou, et al. GUI-360: A comprehensive dataset and benchmark for computer-using agents. *arXiv preprint arXiv:2511.04307*, 2025.

Nex-AGI. Nex-N2.5-mini model card, 2026a. URL https://huggingface.co/nex-agi/Nex-N2.5-mini.

Nex-AGI. Nex-N2.5-Pro model card, 2026b. URL https://huggingface.co/nex-agi/Nex-N2.5-Pro. Accessed October 18, 2026.

OpenAI. Introducing GPT-5.5, 2026a. URL https://openai.com/index/introducing-gpt-5-5/.

OpenAI. Previewing GPT-5.6 Sol: A next-generation model, June 2026b. URL https://openai.com/index/previewing-gpt-5-6-sol/.

OpenAI. Introducing GPT-6.1 Sol, 2026c. URL https://openai.com/index/introducing-gpt-6-1-sol/.

OpenAI. GPT-6 Astra: A new generation of intelligence, 2026d. URL https://openai.com/index/gpt-6-astra/.

Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. MemGPT: Towards LLMs as operating systems. *arXiv preprint arXiv:2310.08560*, 2024.

Baolin Peng, Wenlin Yao, Qianhui Wu, Hao Cheng, Xiao Yu, Rui Yang, Tao Ge, Alessandro Sordoni, Xingdi Yuan, Yelong Shen, Pengcheng He, Tong Zhang, Zhou Yu, and Jianfeng Gao. Orchard: An Open-Source Agentic Modeling Framework. *arXiv preprint arXiv:2605.15040*, 2026.

Zehan Qi, Xiao Liu, Iat Long Iong, Hanyu Lai, et al. WebRL: Training LLM web agents via self-evolving online curriculum reinforcement learning. In *International Conference on Learning Representations*, pp. 79791–79821, 2025.

Yujia Qin, Yining Ye, Junjie Fang, Haoming Wang, et al. UI-TARS: Pioneering automated GUI interaction with native agents. *arXiv preprint arXiv:2501.12326*, 2025.

Qwen Team. Qwen3.5: Towards native multimodal agents, February 2026a. URL https://qwen.ai/blog?id=qwen3.5.

Qwen Team. Qwen API platform, 2026b. URL https://qwen.ai/apiplatform.

Qwen Team. Qwen3.8-Max: A new bar for coding and cowork, August 2026c. URL https://qwen.ai/blog?id=qwen3.8.

Qwen Team. Qwen3.8-2.4T-A95B model card, 2026d. URL https://huggingface.co/Qwen/Qwen3.8-2.4T-A95B. Accessed 2026-10-19.

Qwen Team. Qwen3.5-35B-A3B model card, 2026e. URL https://huggingface.co/Qwen/Qwen3.5-35B-A3B.

Qwen Team. Qwen3.5-397B-A17B model card, 2026f. URL https://huggingface.co/Qwen/Qwen3.5-397B-A17B.

Qwen Team. Qwen3.5-9B model card, 2026g. URL https://huggingface.co/Qwen/Qwen3.5-9B.

Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 8748–8763. PMLR, 2021.

Corby Rosset, Pratyusha Sharma, Andrew Zhao, Miguel Gonzalez-Fernandez, and Ahmed Awadallah. The art of building verifiers for computer use agents. *arXiv preprint arXiv:2604.06240*, 2026.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*, 2024.

Wentao Shi, Yu Wang, Yuyang Zhao, Yuxin Chen, Fuli Feng, Xueyuan Hao, Xi Su, Qi Gu, Hui Su, Xunliang Cai, and Xiangnan He. AJ-Bench: Benchmarking Agent-as-a-Judge for environment-aware evaluation. In *Findings of the Association for Computational Linguistics: ACL 2026*, pp. 25371–25413. Association for Computational Linguistics, 2026. doi: 10.18653/v1/2026.findings-acl.1269.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. Reflexion: language agents with verbal reinforcement learning. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine (eds.), *Advances in Neural Information Processing Systems*, volume 36, pp. 8634–8652. Curran Associates, Inc., 2023. doi: 10.52202/075280-0377.

Qiushi Sun, Kanzhi Cheng, Zichen Ding, Chuanyang Jin, et al. OS-Genesis: Automating GUI agent trajectory construction via reverse task synthesis. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics* (*Volume 1: Long Papers*), pp. 5555–5579. Association for Computational Linguistics, 2025. doi: 10.18653/v1/2025.acl-long.277.

Panrong Tong, Quyu Kong, Pengxiang Li, Tianyu Xia, Hanzhang Zhou, Lei Wang, Chenglin Cai, Long Li, Long Chen, Jianan Zhang, Zhengwen Zeng, Xu Zhang, Gongjie Zhang, Yue Wang, and Steven Hoi. MAI-Gym: An open all-in-one platform for gui agent evaluation and rl training across mobile and computer environments. *arXiv preprint arXiv:2610.20939*, 2026.

Venus Team. UI-Venus-1.5 technical report. *arXiv preprint arXiv:2602.09082*, 2026a.

Venus Team. UI-Venus-2 technical report. *arXiv preprint arXiv:2609.00028*, 2026b.

Yuxuan Wan, Tingshuo Liang, Jiakai Xu, Jingyu Xiao, Yintong Huo, and Michael R. Lyu. From runnable to shippable: Multi-agent test-driven development for generating full-stack web applications from requirements. *arXiv preprint arXiv:2605.17242*, 2026.

Bowen Wang, Dunjie Lu, Junli Wang, Tianyi Bai, et al. CUA-Gym: Scaling verifiable training environments and tasks for computer-use agents. *arXiv preprint arXiv:2605.25624*, 2026a.

Guanzhi Wang, Yuqi Xie, Yunfan Jiang, Ajay Mandlekar, Chaowei Xiao, Yuke Zhu, Linxi Fan, and Anima Anandkumar. Voyager: An open-ended embodied agent with large language models. *Transactions on Machine Learning Research*, 2024.

Haoming Wang, Haoyang Zou, Huatong Song, Jiazhan Feng, et al. UI-TARS-2 technical report: Advancing GUI agent with multi-turn reinforcement learning. *arXiv preprint arXiv:2509.02544*, 2025a.

Lu Wang, Fangkai Yang, Chaoyun Zhang, Junting Lu, Jiaxu Qian, Shilin He, Pu Zhao, Bo Qiao, Ray Huang, Si Qin, et al. Large action models: From inception to implementation. *Transactions on Machine Learning Research*, 2025b.

Xin Wang, Liangtai Sun, Yaoming Zhu, Shuang Zhou, Jiaxing Liu, Fengjiao Chen, Lin Qiu, Xuezhi Cao, Xunliang Cai, Licheng Zhang, and Zhendong Mao. Asuka-Bench: Benchmarking code agents on underspecified user intent and multi-round refinement. *arXiv preprint arXiv:2606.05920*, 2026b.

Xinyuan Wang, Bowen Wang, Dunjie Lu, Junlin Yang, et al. OpenCUA: Open foundations for computer-use agents. In *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pp. 139756–139806. Curran Associates, Inc., 2025c. doi: 10.52202/085713-4669.

Zhaoyang Wang, Qianhui Wu, Xuchao Zhang, Chaoyun Zhang, Wenlin Yao, Fazle Elahi Faisal, Baolin Peng, Si Qin, Suman Nath, Qingwei Lin, et al. WebXSkill: Skill learning for autonomous web agents. *arXiv preprint arXiv:2604.13318*, 2026c.

Qianhui Wu, Kanzhi Cheng, Rui Yang, Chaoyun Zhang, Jianwei Yang, Huiqiang Jiang, Jian Mu, Baolin Peng, Bo Qiao, Reuben Tan, Si Qin, Lars Liden, Qingwei Lin, Huan Zhang, Tong Zhang, Jianbing Zhang, Dongmei Zhang, and Jianfeng Gao. GUI-Actor: Coordinate-Free Visual Grounding for GUI Agents. In D. Belgrave, C. Zhang, H. Lin, R. Pascanu, P. Koniusz, M. Ghassemi, and N. Chen (eds.), *Advances in Neural Information Processing Systems*, volume 38, Main Conference, pp. 15101–15128. Curran Associates, Inc., 2025. doi: 10.52202/085713-0508.

Tianbao Xie, Danyang Zhang, Jixuan Chen, Xiaochuan Li, et al. OSWorld: Benchmarking multimodal agents for open-ended tasks in real computer environments. In *Advances in Neural Information Processing Systems*, volume 37, pp. 52040–52094. Curran Associates, Inc., 2024. doi: 10.52202/079017-1650.

Tianbao Xie, Jiaqi Deng, Xiaochuan Li, Junlin Yang, Haoyuan Wu, Jixuan Chen, Wenjing Hu, Xinyuan Wang, Yuhui Xu, Zekun Wang, Yiheng Xu, Junli Wang, Doyen Sahoo, Tao Yu, and Caiming Xiong. Scaling computer-use grounding via user interface decomposition and synthesis. In *Advances in Neural Information Processing Systems*, volume 38, 2025a. doi: 10.52202/085713-0817.

Tianbao Xie, Mengqi Yuan, Danyang Zhang, Xinzhuang Xiong, Zhennan Shen, Zilong Zhou, Xinyuan Wang, Yanxu Chen, Jiaqi Deng, Junda Chen, Bowen Wang, Haoyuan Wu, Jixuan Chen, Junli Wang, Dunjie Lu, Hao Hu, and Tao Yu. Introducing OSWorld-Verified. *XLANG Lab*, July 2025b. URL https://xlang.ai/blog/osworld-verified.

Tianci Xie, Zikun Cai, Chaoyun Zhang, Pinxue Guo, Bingxuan Li, Rui Zheng, and Hang Yan. WebAnything: From the wild web to scalable agent environments. *arXiv preprint arXiv:2610.37786*, 2026.

XLANG Lab. OSWorld 2.0 official leaderboard, 2026a. URL https://osworld-v2.xlang.ai/. Results updated 2026-10-15. Accessed 2026-10-19.

XLANG Lab. OSWorld-Verified: Official evaluation results, 2026b. URL https://osworld-v1.xlang.ai/static/data/osworld_verified_results.xlsx. Accessed 2026-10-21.

Binfeng Xu, Hao Zhang, Shaokun Zhang, Songyang Han, Mingjie Liu, Jian Hu, Shizhe Diao, Zhenghui Jin, Yunheng Zou, Michael Demoret, Jan Kautz, and Yi Dong. Polar: Agentic RL on Any Harness at Scale. *arXiv preprint arXiv:2605.24220*, 2026a.

Haiyang Xu, Xi Zhang, Haowei Liu, Junyang Wang, et al. Mobile-Agent-v3.5: Multi-platform fundamental GUI agents. *arXiv preprint arXiv:2602.16855*, 2026b.

Haolong Yan, Jia Wang, Xin Huang, Yeqing Shen, et al. Step-GUI technical report. *arXiv preprint arXiv:2512.15431*, 2025.

Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, and Jianfeng Gao. Set-of-Mark prompting unleashes extraordinary visual grounding in GPT-4V. *arXiv preprint arXiv:2310.11441*, 2023.

John Yang, Carlos E. Jimenez, Alex L. Zhang, Kilian Lieret, Joyce Yang, Xindi Wu, Ori Press, Niklas Muennighoff, Gabriel Synnaeve, Karthik R. Narasimhan, Diyi Yang, Sida I. Wang, and Ofir Press. SWE-bench Multimodal: Do AI systems generalize to visual software domains? In *International Conference on Learning Representations*, pp. 2794–2829, 2025.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. ReAct: Synergizing Reasoning and Acting in Language Models. In *International Conference on Learning Representations*, 2023.

Rui Ye, Zhongwang Zhang, Kuan Li, Huifeng Yin, Zhengwei Tao, Yida Zhao, Liangcai Su, Liwen Zhang, Zile Qiao, Xinyu Wang, Pengjun Xie, Fei Huang, Jingren Zhou, Siheng Chen, and Yong Jiang. AgentFold: Long-horizon web agents with proactive context folding. In *International Conference on Learning Representations*, pp. 154754–154776, 2026.

Mengqi Yuan, Zilong Zhou, Xinzhuang Xiong, Weiming Wu, et al. OSWorld 2.0: Benchmarking computer use agents on long-horizon real-world tasks. *arXiv preprint arXiv:2606.29537*, 2026.

Z.ai. GLM-5.3-Flash model card, 2026. URL https://huggingface.co/zai-org/GLM-5.3-Flash. Accessed 2026-10-20.

Chaoyun Zhang, Shilin He, Liqun Li, Si Qin, Yu Kang, Qingwei Lin, Saravan Rajmohan, and Dongmei Zhang. API agents vs. GUI agents: Divergence and convergence. *arXiv preprint arXiv:2503.11069*, 2025a.

Chaoyun Zhang, Shilin He, Jiaxu Qian, Bowen Li, Liqun Li, Si Qin, Yu Kang, Minghua Ma, Guyue Liu, Qingwei Lin, et al. Large language model-brained GUI agents: A survey. *Transactions on Machine Learning Research*, 2025b.

Chaoyun Zhang, Liqun Li, Shilin He, Xu Zhang, Bo Qiao, Si Qin, Minghua Ma, Yu Kang, Qingwei Lin, Saravan Rajmohan, Dongmei Zhang, and Qi Zhang. UFO: A UI-Focused Agent for Windows OS Interaction. In Luis Chiruzzo, Alan Ritter, and Lu Wang (eds.), *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (Volume 1: Long Papers)*, pp. 597–622, Albuquerque, New Mexico, April 2025c. Association for Computational Linguistics. ISBN 979-8-89176-189-6. doi: 10.18653/v1/2025.naacl-long.26.

Chaoyun Zhang, Liqun Li, He Huang, Chiming Ni, Bo Qiao, Si Qin, Yu Kang, Minghua Ma, Qingwei Lin, Saravan Rajmohan, et al. UFO3: Weaving the digital agent galaxy. *arXiv preprint arXiv:2511.11332*, 2025d.

Chaoyun Zhang, He Huang, Chiming Ni, Jian Mu, Si Qin, Shilin He, Lu Wang, Fangkai Yang, Pu Zhao, Bo Qiao, Chao Du, Liqun Li, Yu Kang, Zhao Jiang, Suzhen Zheng, Rujia Wang, Jiaxu Qian, Minghua Ma, Jian-Guang Lou, Qingwei Lin, Saravan Rajmohan, and Dongmei Zhang. UFO2: The desktop AgentOS. *Transactions on Machine Learning Research*, 2026.

Yu Zhao, Wei-Ning Chen, Huseyin A Inan, Samuel Kessler, Lu Wang, Lukas Wutschitz, Fangkai Yang, Chaoyun Zhang, Pasquale Minervini, Saravan Rajmohan, and Robert Sim. Learning GUI grounding with spatial reasoning from visual feedback. In *Proceedings of the 43rd International Conference on Machine Learning*, volume 306 of *Proceedings of Machine Learning Research*, pp. 161193–161218. PMLR, 2026.

Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. In *Advances in Neural Information Processing Systems*, volume 36, pp. 46595–46623. Curran Associates, Inc., 2023. doi: 10.52202/075280-2020.

Hanzhang Zhou, Xu Zhang, Panrong Tong, Jianan Zhang, et al. MAI-UI technical report: Real-world centric foundation GUI agents. *arXiv preprint arXiv:2512.22047*, 2025.

Hanzhang Zhou, Long Li, Quyu Kong, Lei Wang, Panrong Tong, Long Chen, Tianyu Xia, Gongjie Zhang, Xu Zhang, Chen Liu, Jianan Zhang, Chenglin Cai, Zhengwen Zeng, Steven Hoi, and Yue Wang. OpenGCUA: Toward a fully open stack for generalist computer use. *arXiv preprint arXiv:2610.22200*, 2026a.

Hanzhang Zhou, Panrong Tong, Xu Zhang, Quyu Kong, Chenglin Cai, Tianyu Xia, Gongjie Zhang, Jianan Zhang, Long Li, Long Chen, Lei Wang, Gaole Dai, Pengxiang Li, Liangyu Chen, Yue Wang, and Steven Hoi. Qwen-UI-Agent technical report: Toward next-generation real-world centric foundation GUI agents. *arXiv preprint arXiv:2607.28227*, 2026b.

Shuyan Zhou, Frank F. Xu, Hao Zhu, Xuhui Zhou, et al. WebArena: A realistic web environment for building autonomous agents. In *International Conference on Learning Representations*, pp. 15585–15606, 2024.

Sibo Zhu, Shicheng Fan, Xinyue Wang, Wenyi Wu, Kun Zhou, and Biwei Huang. Rsiagent: Autonomous exploration for recursive self-improvement in new environments. *arXiv preprint arXiv:2609.15364*, 2026.

Zilin Zhu, Chengxing Xie, Xin Lv, and slime Contributors. slime: An LLM post-training framework for RL scaling, 2025. URL `https://github.com/THUDM/slime`. GitHub repository. Corresponding author: Xin Lv. Accessed 2026-09-10.

Mingchen Zhuge, Changsheng Zhao, Dylan R. Ashley, Wenyi Wang, Dmitrii Khizbullin, Yunyang Xiong, Zechun Liu, Ernie Chang, Raghuraman Krishnamoorthi, Yuandong Tian, Yangyang Shi, Vikas Chandra, and Jürgen Schmidhuber. Agent-as-a-Judge: Evaluate agents with agents. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, pp. 80569–80611. PMLR, 2025.

A Implementation and Reproducibility

The executable task, its workspace, and the resulting learning samples share an attempt identity. CaseSpec binds the request to its materials, initial state, and execution settings. The environment service manages the workspace, the NorthCUA Harness records interaction, and the configured verifier or evaluator establishes the outcome. The Experience Adapter retains the policy's decision-time inputs when constructing samples, and the learner associates those samples with their attempt-level feedback.

A.1 Information and Permission Boundaries

Construction, execution, evaluation, and learning use different information. The generator can inspect construction metadata, while the acting policy receives the request and permitted task inputs. Host-only acceptance criteria remain with evaluation. Tool capability checks control permitted operations, while input construction and sample extraction separately control which information each role receives.

Actor and evaluator records remain separate so that inspection cannot silently change the actor's result or its earlier inputs. Actor imitation excludes privileged evaluation calls. Evaluator demonstrations used in mid-training are separate judgment tasks with their own inspection context (Appendix E.1). A construction dependency graph likewise does not become an answer plan for ordinary execution. Runtime profiles, live-inspection rules, and resource ownership are detailed in Appendices H.1, H.1, and H.3.

B Grounding Data Details

Section 4.1 describes visual, structural, and documentation-grounded labeling. The common annotation unit links an unannotated screenshot, a visible control, its target geometry, and one or more queries. Documentation supplies functional evidence, while the GUI image supplies the target location.

Table 15. Annotation records for tracing a grounding label from source evidence to the original screenshot.

Record	Contents and alignment rule
GUI target	Original image dimensions, target point or region, and any crop/resize transform. Predictions and labels refer to the same coordinate system.
Documentation match	Application and available version evidence, manual source, candidate illustrations and function descriptions, and the context supporting the selected match.
Query variants	Function-, name-, appearance-, or position-based queries linked to the same visible control. Variants remain grouped with their source target.
Candidate disposition	Accepted, rejected, or unresolved matches, keeping semantic ambiguity separate from geometric invalidity.

Crop and resize transforms preserve correspondence with the original screen. A point target lies inside the actionable region, without an occluding foreground element. A menu-item target requires the item to be visible in the captured menu state. Queries requiring unprovided hidden state are excluded. Diagnostic boxes and action annotations are kept separate from the unannotated grounding input.

Documentation matching checks the control's appearance, nearby controls, visible application state, and the manual's applicability. Ambiguous or version-incompatible matches remain outside the accepted set. Label checks distinguish a wrong function from a spatial miss, even when both refer to similar-looking controls. We count source images, target controls, and query variants separately, grouping paraphrases with their original target. Source grouping and split rules are shared with Appendix D.

C Task Construction Details

The construction routes in Section 4 bind a request, usable materials, an initial workspace, and acceptance conditions into an executable task. A material record combines file format, bounded content summaries, and provenance. After selection, the workspace builder copies the inputs and records their content hashes and initialization procedure. The task distinguishes read-only inputs, distractors, mutable targets, and required outputs.

For dependent work, the plan connects producers to the artifacts or services consumed by later stages. Structural checks reject unknown predecessors, cycles, disconnected stages, incompatible references, and required outputs without production stages. Policy stages cannot consume host-only evaluation artifacts. These constraints define valid dependencies while leaving the actor free to choose GUI actions within the task's prerequisites. The following details cover environment preparation, assisted collection, and verifier test states.

C.1 Environment Construction and Binding

The request-conditioned desktop builder first filters candidate materials by format and application requirements, then ranks them by content. Request refinement uses the selected inputs to make references concrete while preserving the source goal and constraints. A mismatch between the request and available materials sends the task back to construction. Content summaries support selection without replacing inspection of the files being packaged.

EnvAgent finalizes the workspace against a current report of file evidence and query alignment. Material changes require renewed inspection before finalization. The compiler retains packaged paths, checksums, environment configuration, and startup actions, with ownership and permissions that make the initialized files accessible to the desktop user. The specified live-validation procedure additionally checks GUI opening and required operations in an isolated attempt. Collection begins from a fresh initialization so that validation edits do not alter the actor's starting state.

Dependency checks distinguish data flow from required workflow order. For example, the office plan in Figure 14 uses content from A and B in its final PDF, while the record update in C has an independent acceptance condition. The C-to-D edge imposes order without making the record a source of PDF content. A designated terminal artifact therefore need not be the task's only required result.

Reference-based web construction specifies both the visible interface and the state supporting its interactions. Its environment description records the reference source, supported pages and operations, observed versus generated content, backend schema and seed state, reset procedure, and external dependencies. Rendering, navigation, persistence, initialization, and task feasibil-

ity are separate checks. Successful operation of the reconstruction does not by itself establish fidelity to the source service.

The available material index and workflow primitives support package construction, while request-conditioned desktop retrieval and finalization are author-confirmed implementations. The web-reconstruction, verifier-expansion, and recursive-collection descriptions specify their respective procedures. These implementation and design descriptions are distinct from the route-specific consumed-data quantities discussed in Appendix D.

C.2 Hard-Task Collection Protocol

The two collection routes in Section 4.3 retain different products between attempts: validated operating guidance and recoverable execution progress. Selection starts in the training partition after ordinary attempts expose a useful obstacle. Invalid environments, ambiguous requirements, and unreliable verifiers are repaired or excluded before further exploration. The task's final acceptance conditions remain fixed as candidate skills and intermediate milestones evolve.

A skill record identifies its task family, applicability, guidance, version, source attempts, validation cases, and known failures. Validation uses a frozen candidate, separate from the attempts that produced it. Task-specific assistance, reuse within a family, and transfer to a new application remain distinct uses of the record.

For milestone search, a planning agent may inspect the request and verifier to define dependent subtask–check pairs. Actor-visible guidance uses only permitted task information. Hidden predicate constants and expected answers stay in evaluation context. A checkpoint binds the environment snapshot to its parent attempt and action prefix, prompt-visible history, summary, notes, skill and milestone guidance, model/tool configuration, and remaining budget. Restoration re-establishes device bindings and obtains fresh observations. External service state needs its own isolation and restoration mechanism.

A candidate actor state is retained before state-changing inspection. Such inspection uses an evaluation copy or separate branch and receives its own action record. Actor continuations resume from the retained state, so a checker cannot silently complete or repair the path credited to the actor. The checkpoint record retains enough ancestry to reconstruct the selected path and the information available at each decision.

Search configuration specifies candidate counts, state selection, diversity and retention limits, uncertainty handling, and stopping rules. Its budget includes failed attempts, skill validation, snapshot/branch operations, inspection, and live-resource costs. A selected recoverable failure retains its unresolved milestone label until its continuation repairs that requirement. Local checks guide expansion, while the original task verifier determines full-task acceptance.

Training extraction follows the selected actor path. Search-controller decisions, snapshot operations, and failed sibling branches are excluded from actor targets. Supplied skills and milestone guidance remain in the context that conditioned those targets. Cross-branch knowledge transfers are recorded at their point of use. Deleting this guidance would change the decision inputs, so an unguided demonstration requires recollection.

A complete assisted path, a checkpoint-conditioned continuation, and a verified subtask from an incomplete attempt are different sample types. A claim of unaided completion requires a separate fresh-start run without task-specific skill or milestone assistance. Their counting units and source-family separation follow Appendix D.

C.3 Controlled Verifier States

Verifier tests hold the intended obligations fixed while varying the submitted state. An incomplete initial state is a negative control, a correct result is a positive control, a near miss isolates one violation, and a valid alternative tests sensitivity to presentation. Each test retains the initial and final state, protected inputs, acceptance conditions, and the expected verdict. Testing score composition with stipulated predicate values is separate from executing the predicates on actual artifacts.

Figure 34 gives a concrete regional-report example. Expected gross sales are North 420 and South 380, totaling 800. Acceptance also requires complete regional content, template compliance, and unchanged protected source files. The two accepted presentations satisfy all of those conditions, not just the displayed arithmetic.

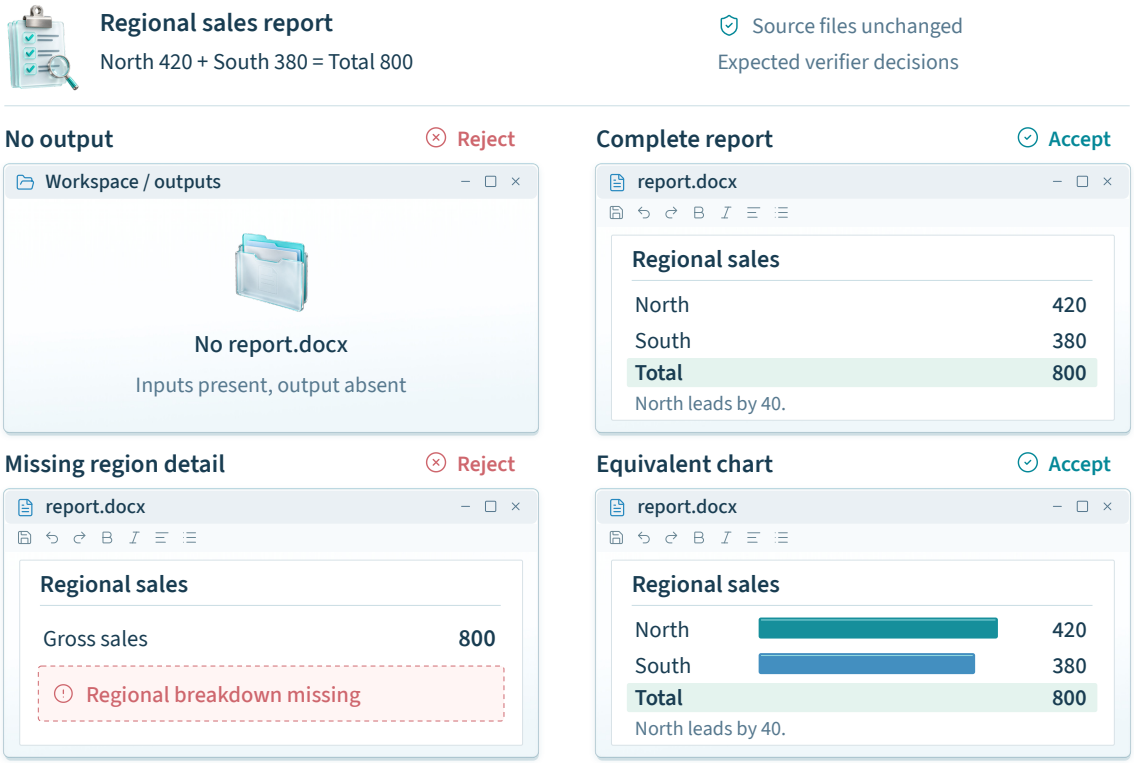


Figure 34. Four states of the same report task. Missing output and missing regional detail should be rejected. A complete table and an equivalent chart should be accepted. Both accepted examples are stipulated to satisfy the full contract, including regional discussion, template compliance, and unchanged source files. The GUI excerpts and inspection asset are illustrative. The dashed red region annotates missing content. Accept/Reject denote expected decisions, not measured verifier outcomes.

D Corpus Accounting and Provenance

Corpus accounting distinguishes source materials, constructed tasks, execution attempts, accepted trajectories, extracted samples, and training consumption. A task can produce several attempts, and a trajectory can produce several prediction units. Grounding images, target controls, and query variants use their own units. Table 16 summarizes the supervised data by stage and domain using rounded counts.

Table 16. Approximate supervised data scale. Counts are rounded independently. SFT domain step totals and the web mean are estimated from the rounded query counts, total steps, and desktop mean. Grounding steps denote coordinate targets and are separate from trajectory interactions.

Source	Queries	Steps	Mean steps/query
Broad GUI mid-training			
Desktop	$\approx 100{,}000$	$\approx 4{,}200{,}000$	≈ 40
Web	$\approx 10{,}000$	$\approx 280{,}000$	≈ 30
Grounding	$\approx 40{,}000$	$\approx 40{,}000$	1
Verified long-horizon SFT			
Desktop	$\approx 2{,}000$	$\approx 240{,}000$	≈ 120
Web	$\approx 1{,}000$	$\approx 70{,}000$	≈ 70

Together, the desktop and web sources provide over 100,000 trajectories and over 4 million GUI interactions for mid-training. The long-horizon SFT cohort contains approximately 3,000 queries and 310,000 interaction steps.

These rounded counts summarize source cohorts by stage and domain before model-specific sampling and repeated consumption. Auxiliary evaluator demonstrations are accounted for separately from task-execution trajectories.

A sample's lineage connects source and task family, environment and fixture identity, collection policy, permitted tools, outcome check, and extraction version. A consumed training sample additionally identifies its training shard and stage. Actor episodes and evaluator judgment episodes retain separate roles and permissions. Complete assisted paths, checkpoint continuations, and verified subtask demonstrations also retain distinct sample types.

Related screenshots, documents, material sets, task seeds, source sites, repository/issue families, and derived skills are grouped before splitting. Source-policy disclosure, licensing, and access permissions apply to each supervised source.

E Model Identities and Learning Details

The Nano, Mini, and Pro tiers use the Qwen3.5-9B, Qwen3.5-35B-A3B, and Qwen3.5-397B-A17B backbone variants, respectively (Qwen Team, 2026a). Table 17 separates the language-model parameter convention from the identity of a trained checkpoint. The mixture-of-experts variants have distinct total and activated counts.

The results use *Mid* for broad GUI mid-training and *SFT* for verified long-horizon supervision. Mid is already a NorthCUA checkpoint, distinct from an upstream pre-trained release. The supervised stages differ in their source mixture and selection criteria. We first describe the super-

Table 17. Model-family identities and language-model parameter counts following Qwen3.5 (Qwen Team, 2026a). These are not counts of all tensors in a multimodal checkpoint.

Field	NorthCUA-Nano	NorthCUA-Mini	NorthCUA-Pro
Backbone variant	Qwen3.5-9B	Qwen3.5-35B-A3B	Qwen3.5-397B-A17B
Language-model parameters	9B, dense	35B total 3B activated	397B total 17B activated

vised prediction units, then explain how task-group feedback and recorded RL attempts become learning samples.

E.1 Supervised Representation and Sampling

The supervised stages share response prediction but use different sources and selection rules. An extraction map Φ converts a grounding example, task trajectory, or evaluator episode into prediction units $x \sim \mathcal{D}$. A unit can be a decision or a context-bounded slice. For a binary target mask m_t with $\sum_t m_t > 0$, the conditional objective is

$$\mathcal{L}_{\text{sup}} = -\mathbb{E}_{x\sim\mathcal{D}}\left[\frac{1}{\sum_t m_t}\sum_t m_t \log p_\theta(x_t \mid x_{<t})\right]. \tag{2}$$

Here $x_{<t}$ includes the recorded multimodal input at that decision. The expectation is over extracted units after source sampling, so an episode's weight depends jointly on extraction, sampling, and the within-unit mean.

Grounding examples target coordinates. Actor trajectories target recorded assistant responses, including reasoning, text, and tool calls according to the target mask. Evaluator episodes are separate judgment tasks targeting inspection and assessment responses from their own permitted context. Grounding, actor trajectories, and evaluator examples enter mid-training. Long-horizon SFT uses task trajectories. The three sources have separate counts and sampling weights.

An actor input contains the request and the observations, tool results, summaries, notes, and guidance available at that decision. Post-execution judgments and hidden acceptance values do not enter its earlier context. A post-action screenshot or action overlay is available only to later decisions. Actor-issued note calls remain in the response stream with the mask specified by the recipe. Construction-agent and privileged evaluator calls are outside actor targets.

Sampling determines how source and episode lengths influence learning. Uniform decision sampling, uniform trajectory sampling, and token-weighted sampling give different weights to long episodes. The recipe therefore distinguishes extraction, source/domain weighting, packing, target deduplication, and loss reduction. A response repeated as context has one target occurrence. Assisted traces retain the skill or milestone guidance that conditioned them. These supervised choices are specified separately from online RL assembly.

Selection precedes sampling. Broad trajectories are judge-selected completed tasks. Long-horizon trajectories require original full-task script acceptance followed by execution-quality and dependency checks. Skill- and milestone-assisted trajectories undergo the same checks. Cleaning rejects unusable records. Retained episodes preserve action–observation alignment, including coherent recoveries. Source attribution and cohort overlap follow Appendix D.

E.2 Task Groups and Terminal Feedback

The initial RL pool follows Section 5.3.1: human-reviewed desktop and web tasks are screened with at least five attempts under each model's SFT checkpoint and fixed execution settings. Selection uses variation among valid verifier scores. Environment or verifier errors do not count as negative task outcomes. Fresh grouped attempts provide the learning data. Periodic rescreening is outside this specified recipe.

Members of a group attempt the same case package from equivalent initial states under the same verifier and budget. A task-script result can be a finite numeric reward or an interpretable Boolean/status verdict, with Boolean fallback mapped to zero or one. Reward ranges and preprocessing remain task-specific. Script errors, timeouts, nonzero exits, and uninterpretable results follow the invalid-evaluation path. Valid negative verdicts remain learning outcomes.

Retries retain logical member identity under the configured retry policy. Complete-group checks prevent missing members from silently changing the comparison. Domain sampling weights specify scheduled work, while completed, admitted, and consumed members describe successive stages of actual use. The logical attempt remains the unit of group membership, independently of infrastructure retries or the number of extracted training segments.

For $G > 1$ valid members, terminal advantages are

$$A_i = \frac{R_i - \bar{R}}{s_R + \epsilon}, \qquad \bar{R} = \frac{1}{G}\sum_{i=1}^{G} R_i, \qquad s_R^2 = \frac{1}{G-1}\sum_{i=1}^{G}(R_i - \bar{R})^2, \tag{3}$$

where R_i is the attempt's terminal reward and $\epsilon = 10^{-6}$. The baseline is computed within the task group, independently of other tasks or domains. Identical rewards produce zero terminal advantages. Each attempt contributes one advantage shared by its segments. The trajectory-mean configuration uses the full attempt's target-token denominator in Eq. 1.

Synchronous and asynchronous execution differ in when work can overlap policy updates. For asynchronous runs, policy lag is the learner version at consumption minus the behavior version of each call. Lag and the fraction of mixed-version attempts describe the executed configuration.

E.3 Clipped Policy Objective

Let $b(s)$ denote segment s's captured behavior version, and let h_{ist} contain its multimodal context and generated prefix before token x_{ist}. The learner uses the PPO-style clipped surrogate (Schulman et al., 2017) with the member advantage from Eq. 3:

$$\begin{aligned} \rho_{ist} &= \exp\big(\log \pi_\theta(x_{ist} \mid h_{ist}) - \log \pi_{b(s)}(x_{ist} \mid h_{ist})\big), \\ \ell_{ist}^{\text{clip}} &= -\min\{\rho_{ist} A_i, \text{clip}(\rho_{ist}, 1-\delta_-, 1+\delta_+) A_i\}. \end{aligned} \tag{4}$$

The likelihoods use the temperature and recorded-support convention in Appendix E.6. The parameters δ_- and δ_+ define the lower and upper clipping widths, respectively. In the trajectory-mean configuration, Eq. 1 aggregates token losses within each complete attempt before averaging attempts.

A behavior version is constant within a segment, but an asynchronous attempt can span several versions. Captured likelihoods identify the denominator for each response without correcting all effects of policy lag or changing state distributions. The clipped update is consequently an

off-policy approximation for those attempts. Terminal feedback applies across the episode without establishing each action's separate causal contribution. Optional local process rewards and distillation are outside this terminal-reward recipe.

E.4 Visual Experience Delivery

The delivery mechanisms in Section 5.3.4 address repeated CPU image preparation and visual-tensor transfer. They preserve the learner's inputs, unlike semantic compaction, which changes those inputs. Lossless tensor encoding reduces transfer volume. CPU preprocessing reuse is a separate mechanism whose enablement depends on processor compatibility.

The preprocessing-reuse path supports the Hugging Face Qwen2-VL processor's independent per-image outputs. Within a member, it retains `pixel_values` and `image_grid_thw` under fixed processor settings, reconstructs the requested image order, and processes new content. A clean-to-overlay replacement or a configuration change requires fresh processing. Unsupported processors or incompatible output structures use ordinary processing. Reuse concerns CPU tensors, not neural visual features.

Owned per-image storage avoids retaining an obsolete batch through a view. The member's window is cleared on exit, and cancellation cannot publish into another member's state. Tensors already staged for prompt projection can be reused during training materialization under the same content and processor identity.

The selected transport path uses Zstandard level 1 on contiguous CPU image tensor bytes after data-parallel partitioning. An encoding replaces the original only when smaller. A failed encoding leaves the original available. The consumer checks the expected byte count, decodes its selected microbatch, and transfers tensors to the accelerator without retaining a decoded full-batch cache. Dtype, shape, resolution, and floating-point values remain unchanged.

Payload accounting includes delivery-pending groups as well as reserved, active, and ready work. This is a group-count bound, not a byte-level memory limit. Successful compact object-store transfer allows the producer to release visual references, while consumption records keep sample identities without those payloads. The data-parallel delivery barrier remains in place.

Equivalence checks compare prompt tokens, image order, preprocessing tensors, dtype, shape, masks, rewards, and behavior likelihoods on the same captured calls. Full-update comparison additionally fixes model, optimizer, random state, and partition schedule. Component measurements cover preprocessing, codec, transfer, and memory costs. The complete optimizer interval is a separate measurement boundary.

E.5 Sequence Integrity and Update Boundaries

Before assembling an actor RL segment, the adapter checks the delivered-response receipt, policy/tokenizer/template/processor identities, call order, and response digests. It admits main-agent calls only. Terminal Controller evidence confirms delivery by linking ordered execution steps and their artifact digests to the captured responses. Task success is established separately by the outcome check.

Calls can be appended when prompt tokens and ordered media bindings confirm a prefix extension under the same behavior version. Media identity includes a content digest. Replacing a clean frame with an action overlay, evicting an image, rewriting history, or changing behavior version therefore starts a new segment from the newly captured input. Responses repeated

as history provide context and are masked out of the new segment's targets. Their original target occurrences remain available for learning. Processor-specific visual-position projection preserves captured image content and order.

Segments from the same logical attempt share its identity and reward provenance. The adapter checks group completeness before computing advantages. In the trajectory-mean configuration, the learner sums target token counts over the full attempt before partitioning. Every segment uses this denominator within the same optimizer step. Partition invariance holds when conditioning, masks, and token losses stay fixed.

Weight publication pauses new model calls, drains calls already in progress, commits the new weights, and resumes calls with the new behavior version. The tasks themselves can remain active in their workspaces. After abnormal recovery, execution restarts from committed checkpoint and cursor state. Warm queues and unfinished rollouts are rebuilt, so recovery is not a lossless restoration of in-flight work.

E.6 Sampling Likelihood Convention

The policy ratio requires compatible learner and behavior likelihoods. For temperature $T > 0$ and the recorded behavior support $S_b(h)$, the likelihood convention is

$$p_\theta^{(T,S_b)}(x \mid h) = \frac{\mathbf{1}\{x \in S_b(h)\} \exp(z_\theta(x \mid h)/T)}{\sum_{u \in S_b(h)} \exp(z_\theta(u \mid h)/T)}.$$

Here h contains the captured context and generated prefix, and z_θ denotes the learner's logits. With top-p equal to one and no other vocabulary restriction, $S_b(h)$ is the full vocabulary. For nucleus sampling (Holtzman et al., 2020), replay uses the recorded token identities and offsets at each target position rather than constructing a new nucleus from current learner logits. Temperature, support capture, and replay settings belong to the run configuration. Entropy regularization has a separately configured support mask.

E.7 NorthCUA-Ultrafast Adaptation and Evaluation

This section details the choice-based execution mode introduced in Section 6.8. Its training unit is a decision state extracted from a WebAgent trajectory. The original causal prefix supplies the input, and the following recorded action supplies the label. The classifier and native generator use this same underlying evidence through different prompts.

E.7.1 Multimodal Inputs and Action Routing

The classification prompt retains the task, preceding actions, tool results, screenshots, and action overlays. It replaces the original system instruction with a finite-choice instruction and appends the current operation and target candidates. Native generation instead uses the original messages, tool schemas, and chat template. Classification therefore does not overwrite the generator's input with a summary or an additional selected-target instruction.

Targets come from the latest valid set-of-mark observation. We exclude explicitly disabled elements and restrict filling to input-like roles and checkbox changes to checkbox roles. At most 150 elements are retained per observation, with the number omitted recorded. A finite operation is offered when its tool and eligible targets are available. Selection operates over this retained

candidate set. Table 18 gives the seven routes. The operation and each available target type form separate variable-size choice questions. Only the chosen operation's target is used for execution.

Table 18. Ultrafast routing and completion. Current references are resolved through the latest observation. Completion describes strict routing. The original runner performs tool execution and checks the final response format.

Route	Target choice	Completion
CLICK	Element	Construct a click call directly.
HOVER	Element	Construct a hover call directly.
CHECK	Checkbox	Construct a check call directly.
UNCHECK	Checkbox	Construct an uncheck call directly.
FILL	Input field	Generate the original fill action and require its field to agree with the selected target.
NATIVE	None	Generate a complete offered tool call and its arguments.
RESPOND	None	Generate the original structured final response.

In the offline full-action evaluation below, generation starts with a fresh input encoding and cache, with any LoRA adapters disabled. Decoding is greedy with thinking disabled and at most 4,096 new tokens. A valid result requires model termination and the relevant tool schema or response structure. Strict routing rejects FILL target disagreements. NATIVE accepts a valid offered tool, including a finite-action tool if the original generator selects one. This matters when comparing route labels with complete actions below.

E.7.2 Readout and Training

The last hidden layer supplies four vectors for each candidate. These are the mean context representation before the appended choice block, the mean question-instruction representation, the mean candidate representation, and the final decision-marker representation. Each vector has dimension 4,096. A shared head normalizes and projects these roles to width 512, adds learned role embeddings, and applies two Mixer blocks. Each block mixes the four roles through a $4 \to 16 \to 4$ MLP and the channels through a $512 \to 1024 \to 512$ MLP. A scalar readout from the updated candidate role supplies its score. Softmax is applied within each question, and argmax selects its answer. The Mixer operates within each candidate's four roles, while the backbone has already encoded the full candidate list.

For state i, let Q_i contain the supervised operation question and, when required, the corresponding target question. With correct candidate y_{iq}, score z_{iqk}, and candidate set C_{iq}, the objective for a minibatch of decision states $\mathcal{B}$ is

$$\mathcal{L}_{\text{choice}} = \frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \frac{1}{|Q_i|} \sum_{q \in Q_i} -\log \frac{\exp z_{iqy_{iq}}}{\sum_{k \in C_{iq}} \exp z_{iqk}}. \tag{5}$$

Averaging within each state keeps states with an operation and target equally weighted with operation-only states. Other target questions receive no label, and the original output tokens receive no generation loss.

The dataset contains 225,710 training, 5,203 validation, and 6,087 test states from 25 trajectory collections. Normalized task goals define groups, and the fixed holdouts remain unchanged when training sources are expanded. Filtering removes stale or missing targets, conflicting labels, invalid actions, and known target-tool errors. The native multimodal prefixes are rejoined to their

source records before training. Both input paths are checked against a 73,728-token budget after image-token expansion, without truncation. Grouping removes identified request duplicates. Semantically related paraphrases and task templates may still occur across groups.

Table 19. Ultrafast adaptation settings. Original model weights remain frozen in both variants. Parameter percentages use the complete 9.410B-parameter multimodal checkpoint as the denominator.

Setting	Head-only	Head + LoRA
Trainable parameters	4.214M (0.0448%)	47.492M (0.5047%)
Choice head	Width 512, two blocks	Same
LoRA	None	Rank 16, scale 2, dropout 0
Adapter scope	None	All 248 language-layer linear modules
Optimizer	AdamW	AdamW
Learning rate	10^{-4}, constant	Same
Global batch	128 states	Same
Precision	BF16 backbone, FP32 head	BF16 backbone, FP32 head/adapters
Gradient clipping	Global norm 1.0	Same
Seed	42	42
Completed updates	63,504 (36 epochs)	24,521 (13 epochs plus part of 14)
Selected checkpoint	Epoch 22	Epoch 1

AdamW uses $\beta = (0.9, 0.999)$, $\epsilon = 10^{-8}$, and weight decay 0.01, with no learning-rate warmup or schedule. Candidates retain deterministic ID order, without shuffling or class reweighting. Head + LoRA adds 43.278M adapter parameters to the same 4.214M head. The adapters remain unmerged so that generation can disable them. The two variants share data, logical batch size, and seed, but use different completed budgets. Checkpoints are selected by full-validation joint accuracy, then loss and earlier epoch.

E.7.3 Offline Decision and Action Results

Route-and-target joint accuracy requires all labeled questions for a state to be correct. It excludes generated text and other open-ended parameters. Full-action exact matching instead compares the tool name and complete arguments, or the final response object, with the reference action. It normalizes reference IDs to strings but does not accept alternative wording or equivalent action sequences. Because NATIVE can return a finite-action tool, these two correctness sets need not be nested.

Table 20. Offline Ultrafast results in percent. Validation and test contain 5,203 and 6,087 states. Original generated actions are projected into the same route labels. Missing results are left as dashes.

Policy	Validation joint	Test joint	Test action exact
Full generation	–	85.0	64.6
Head-only	86.1	85.4	64.0
Head + LoRA	88.2	–	–

Head-only correctly selects the route and target in 5,196 test states, compared with 5,172 for projected full generation. Complete-action matching is lower for the hybrid policy, at 3,897 versus

3,933 states. In the paired comparison, 3,757 states match under both policies, 140 only under Head-only, and 176 only under full generation.

The hybrid gains 46 exact matches among 2,576 CLICK states, while losing 25 among 906 FILL states and 30 among 2,087 NATIVE states. Losses on generation-dependent FILL and NATIVE actions exceed the gains on CLICK.

The diagnosis is clearest among the 1,403 states with a correct Head route but an incorrect hybrid action. Of these, 935 have mismatched parameters for the same tool, 359 have a tool/final-output kind mismatch, 18 differ in final JSON fields, and 91 are rejected by route or target consistency checks. These are reference-matching outcomes, distinct from environment success. The paired test reuses one saved native generation per state for both policies, including states whose hybrid action bypasses generation. Its execution time covers this shared-generation offline procedure.

Validation selects epoch 1 for Head + LoRA and epoch 22 for Head-only. At the same completed epoch 13, their joint accuracies are 86.9% and 85.5%, respectively. This provides an equal-epoch descriptive comparison, alongside the selected checkpoints in Table 20. All results come from one training seed, and states can share a task. Earlier text-only experiments use a different input protocol and are excluded from this multimodal comparison.

E.7.4 Online Task Completion and Timing

The online comparison adapts the long-horizon-SFT NorthCUA-Nano checkpoint before joint RL and retains the original WebAgent runner. Each inference replica occupies one GPU. Full generation sends every decision through the native generation path. Both Ultrafast variants and the case in Figure 33 use the generation-preferred profile with a finite-choice confidence threshold of 0.8. A direct action requires both its operation and target probabilities to meet this threshold. Lower-confidence selections, missing or stale current references, and unsupported observation surfaces defer to native generation. When generation is used, its complete valid action is accepted without replacing its arguments with the classifier's selection. The gateway records any disagreement. This differs from the strict agreement checks used in the offline comparison above.

Task outcomes and matched execution records. The evaluator assigns success to 521, 481, and 492 of the 812 tasks for Full generation, Head-only, and Head + LoRA, respectively. Failures, timeouts, and evaluation errors remain in the denominator. The latest Head-only and LoRA scores come from rescoring existing trajectories. Their efficiency statistics retain the same execution records.

Task time and trajectory length use a fixed matched subset of 320 tasks, selected from 343 tasks jointly successful in the timing snapshot. The remaining 23 lack complete timing or action-dispatch evidence for at least one policy. This fixed set supports comparisons on the same tasks and defines the population for the conditional means. Rescoring updates the success totals while leaving this timing population unchanged.

Step and action boundaries. A step is one decision round, including a final response. Classification followed by generation or a length retry remains one step. An action counts a tool call dispatched to its handler, including calls that subsequently fail during execution. A step may issue multiple actions or none, and a final response is not a tool action. Only the final selected attempt is counted. The 320-task set contains 3,025, 3,260, and 3,386 steps, respectively, yielding

the means in Table 14. Its dispatched-action totals are 2,683, 2,940, and 3,066, with means of 8.4, 9.2, and 9.6 actions.

Decoding bypass counts recorded dispatched actions produced directly by the choice head without invoking native generation. The fractions are 6,182/15,385 for Head-only and 4,248/13,652 for Head + LoRA, or 40.2% and 31.1%. These denominators include available actions from failed tasks and are not restricted to the 320-task set. Incomplete action records affect 12 and eight tasks, respectively. Missing actions are excluded from both counts. Full generation has zero bypass by definition.

Measured time and deployment. Task time spans the agent execution loop through terminal observation and result persistence. It includes decisions, browser operations, and observations, while excluding environment provisioning, browser teardown, and evaluator scoring. The measured mean for Full generation is 616.8 s on the matched task set.

Full generation uses temperature 0.7, top-p 0.95, top-k 40, high reasoning effort, and up to 16,384 output tokens. The hybrid runs combine temperature-zero attempts with temperature-one, top-p-one length retries. Each replica occupies one GPU, with run-specific replica counts, concurrency, and browser placement. The reported task times characterize these evaluation deployments.

F Web Repair and Testing Protocol

F.1 Qualifying Browser Evidence

The main-text case in Figure 18 records browser testing and code repair during local-page development. To decide whether an observation justifies a product change, the proposed evaluation protocol first checks that the relevant test actions completed. A mismatch may otherwise originate in the test specification, the agent's actions, or the environment (Yang et al., 2025, Wan et al., 2026).

Figure 35 uses a title-persistence requirement to make this decision concrete. Record R17 should retain *Delivery plan v2* after reload. The Web Agent completes the edit and save, observes the new title, then reloads the record and finds the old title. Together, these observations support a candidate persistence defect. The report pairs the expected and observed values with the completed actions, tested build, reset state, and supporting artifacts, giving the Coding Agent a specific discrepancy to investigate.

The same old title has a different interpretation if Save could not be activated or reload did not complete. In that case, the test has not established that the product lost a saved value. The proposed adjudication rule therefore sends unresolved execution failures to diagnosis or bounded retesting. Test steps may be repaired and exploratory checks added while the issue's acceptance goals and regression requirements stay fixed.

Repair feedback and final acceptance also serve different purposes. Feedback helps the agents revise a candidate. Final acceptance evaluates the submitted patch from a protected clean state, using checks the repair agents cannot change. Its verdict is withheld from further revision, so acceptance tests whether the fix survives beyond the session used to develop it.

Appendix I.4 details the common assignments, resets, and denominators needed to compare feedback conditions.

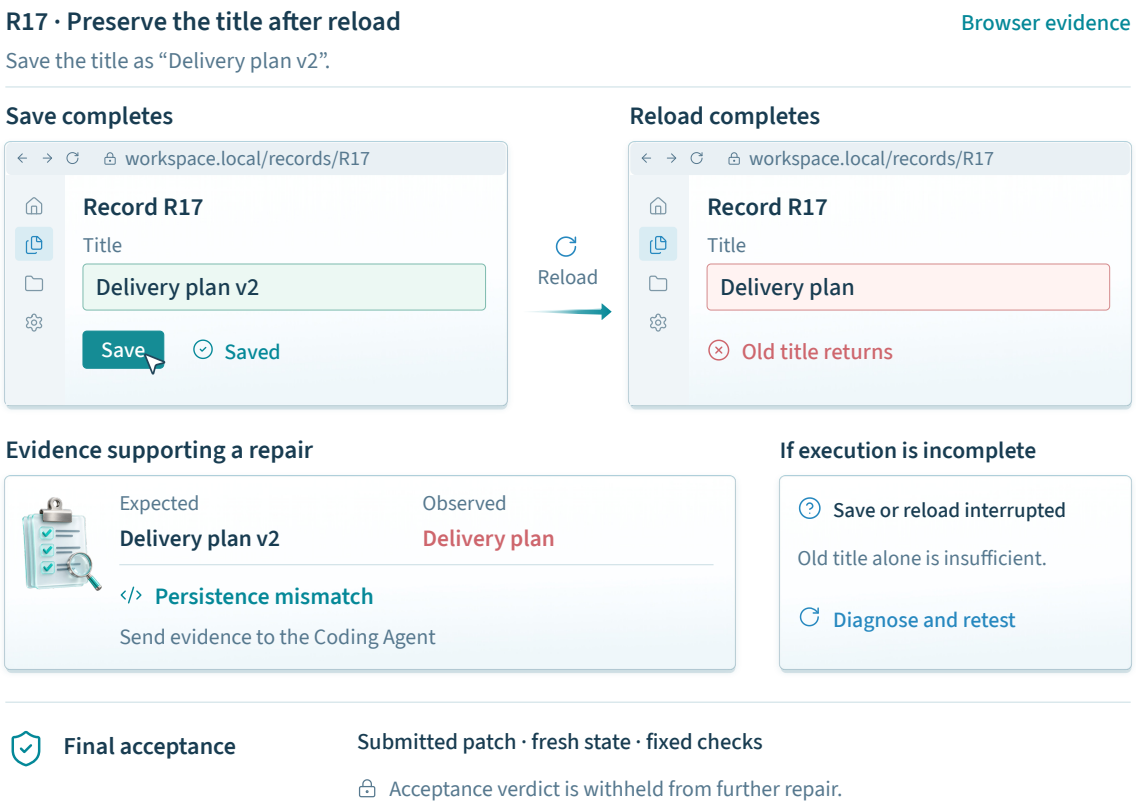


Figure 35. A persistence example for evidence qualification. The new title appears after saving but reverts after a completed reload. The report preserves this mismatch for repair. An interrupted save or reload requires diagnosis and retesting before assigning a product defect. Final acceptance uses fixed checks in a fresh state outside the repair loop. GUI states and inspection materials illustrate the proposed protocol.

These controls support the proposed comparison of browser-feedback conditions (Wang et al., 2026b). Inclusion of an episode in model training follows the separate source and consumption accounting in Appendix D.

G NorthCUA Case Studies

The following cases trace complex software tasks from an initial workspace or source records to editable deliverables and native task outcomes. They illustrate sustained GUI interaction across geometry, data analysis, visual composition, circuit simulation, and gameplay. The GeoGebra, office, and TIDAL workflows use NorthCUA's desktop MCP tools. The Virtuoso workflow uses native CUA Driver MCP tools in a prepared engineering environment. All use an external agent controller. The Zuma case extends this setting to screenshot-and-mouse gameplay. These demonstrations exercise the tool interface and are separate from the NorthCUA checkpoint evaluations in Section 6. The recorded step counts include GUI actions, observations, preparation, checks, and retries. The release packages preserve the full traces, infrastructure repairs, native outputs, and acceptance records.

G.1 Ruby Heart

The task is to turn supplied equations into a GeoGebra heart that can pulse, change depth and color, and reveal a moving cross-section. The difficulty is keeping the cutting plane and section linked to the changing surface. Figure 36 begins with an empty workspace. The agent enters native expressions to build a first construction, then arranges the geometry and four controls into an interactive exhibit. Construction view reveals the auxiliary geometry linked to the surface for inspection.

The agent then exercises these dependencies through playback and direct manipulation. Holding pulse phase and slice position fixed, it checks two depth settings that visibly change the surface and its section. It also reopens the saved file and later drags the slice to another position. The trace covers 18 combinations of depth, slice position, and pulse phase, connecting the visible shape to its underlying parameters. The 440-step record ends with the final saved `.ggb` and three separate view images. The editable construction supports continued interaction with the geometry established during modeling.

G.2 Silicon Atlas

This task asks for five editable plots from synthetic chip-test records. The challenge is preserving measurement meaning across spreadsheet analysis and plotting. Recorded failures take precedence. Otherwise, incomplete tests remain UNKNOWN, while missing current measurements stay missing. In Figure 37, the agent imports 2,142 sweep rows into Calc and fills formulas that classify 1,071 operating groups. A passing group requires two clean expected repeats, and energy is computed only for passing points. These rules govern the subsequent plots.

The classified table first feeds a Shmoo map of voltage and frequency in LabPlot. Matching axes on a second temperature map make the operating regions directly comparable. The agent next plots the maximum tested passing frequencies and passing-point energy. A separate 32-device sample supplies the final count plot, completing the atlas. The agent then reopens the workbook to check counts and units, repairs crowded labels, and exports the revised figure. The audit retains 573 passing, 489 failing, and nine unknown groups. After 928 recorded steps, the workbook and plotting project preserve the calculations and editable plots alongside the PDF and PNG outputs.

G.3 NOVA Presentation

The NOVA task asks for a six-slide presentation from a product brief, supplied images, and numerical data. The challenge is preserving accurate, editable scientific content while building a coherent visual story. Figure 38 starts with an empty presentation and a first cover. To support the performance slide, the agent computes mean signal-to-noise ratios from five repeats in Calc. The resulting table and supplied spectra feed native charts, which it embeds in Impress. Editable shapes and connectors then form the five-stage signal-path diagram.

The agent reads back the embedded chart data to check that application transfer preserved the calculation. The figure shows the same mean of 17.6268 at 5 ms in both views, making this check directly inspectable. A presentation-mode inspection covers all six slides before the 691-step record ends with an editable `.odp`, the calculation workbook, and a matching PDF. The task thus preserves a concrete numerical result through calculation, chart transfer, and final presentation. NOVA is a fictional product concept with illustrative imagery and numerical data.

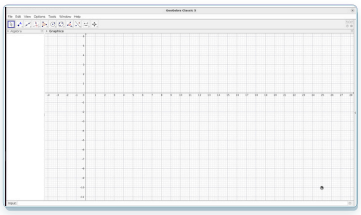
From equations to a live 3D exhibit

GeoGebra

440
recorded steps

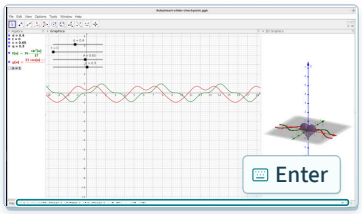
Build a live 3D heart with pulse, depth, color and mathematically linked slice controls.

1 Blank workspace 1



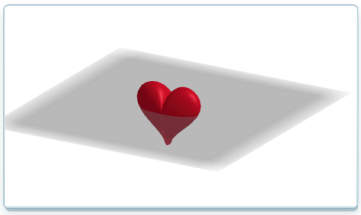
Start with an empty construction

2 Define the surface 44



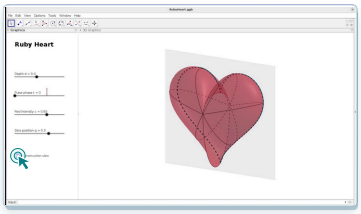
Type H = Surface(...)

3 First construction 79



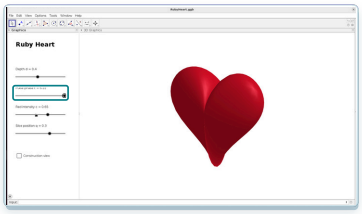
Inspect the first surface and plane

4 Reveal geometry 172



Click Construction view

5 Test the pulse 218

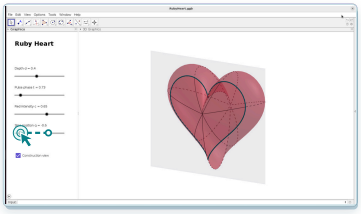


Play phase t from 0 to 6.22

6 Test depth 249 / 270

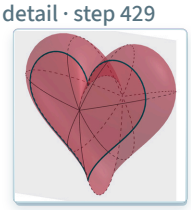
Hold pulse and slice position fixed

7 Move the slice 429

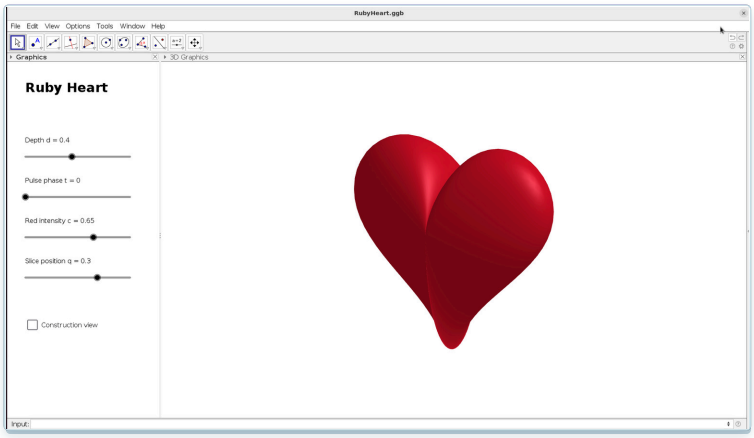


Drag q to inspect a new section

Section detail · step 429



8 Deliver the interactive exhibit 440



Native .ggb, four controls, three saved views

State IDs shown. Overlays summarize nearby actions.

18 parameter states checked

Figure 36. NorthCUA case study in GeoGebra. The trajectory connects expression entry to geometry, playback, and parameter checks. The first construction is enlarged from state 79. Panel 6 compares depths 0.20 and 0.65 at fixed pulse phase 0 and slice position -0.8. The section detail enlarges state 429. Numbers identify captured states in row order. Editorial overlays summarize nearby actions, with intervening actions omitted.

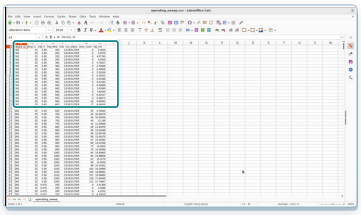
From raw logs to a five-panel atlas

Calc + LabPlot

928
recorded steps

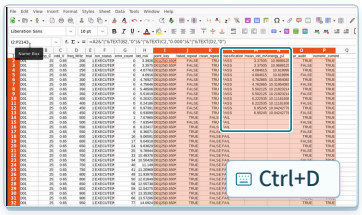
Build five editable chip-test plots, preserving failed and unknown operating points.

1 Raw test records 15



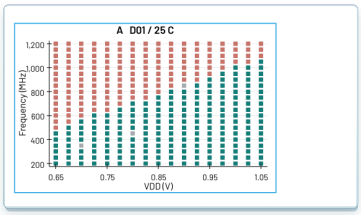
Import the operating-test CSV

2 Classify repeats 54



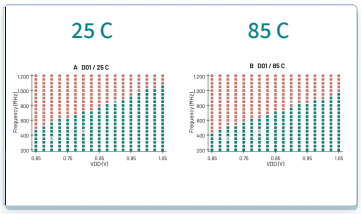
Fill PASS / FAIL / UNKNOWN rules

3 First Shmoo map 395



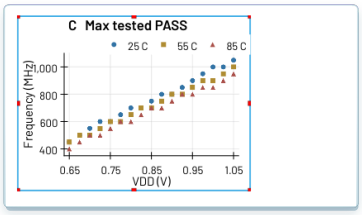
Plot PASS / FAIL / UNKNOWN

4 Match temperatures 482



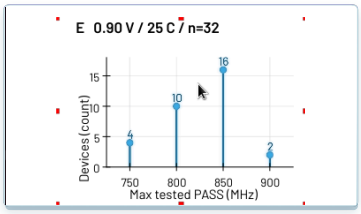
Align the 25 C and 85 C maps

5 Passing maxima 588



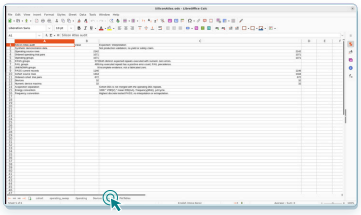
Compare three temperatures

6 Device variation 732



Count the 32-device sample

7 Audit & reopen 877

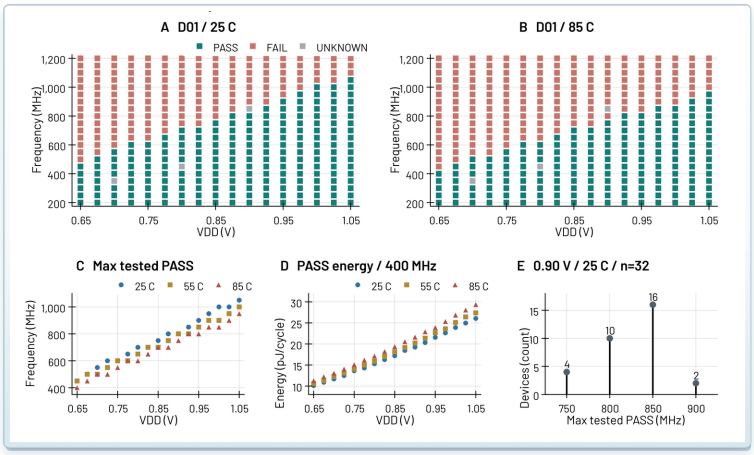


Check the workbook audit sheet

Operating-point counts

PASS	573
FAIL	489
UNKNOWN	9

8 Deliver five editable plots 928



Saved workbook, native plots, PDF and PNG

2,142 sweep rows. 1,071 operating groups.

Synthetic demonstration data

Figure 37. NorthCUA case study in Calc and LabPlot. Repeat-aware classification supplies the temperature maps and derived plots. Focused crops distinguish temperature comparison, passing-frequency maxima, and 32-device counts before the complete atlas. The count inset transcribes the audit totals at state 877. Numbers identify captured states, and editorial overlays summarize nearby actions. Intervening actions are omitted. The data describe a synthetic demonstration.

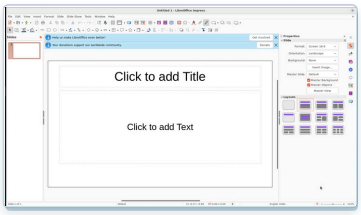
From source material to a six-slide story

Calc + Impress

691 recorded steps

Turn a brief, photos and data into six slides with editable charts and a native diagram.

1 Blank presentation 1

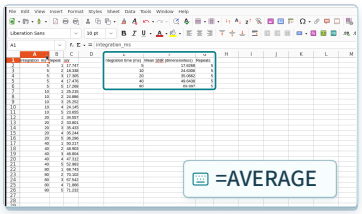


Start from the empty slide

2 Lay out the cover 91

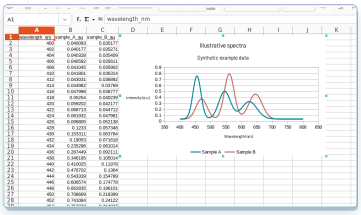
Place the supplied product image

3 Compute means 136



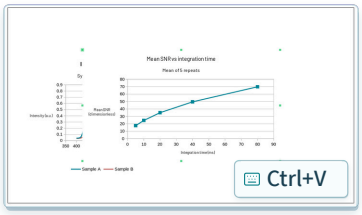
5 ms gives mean SNR 17.6268

4 Build native charts 247



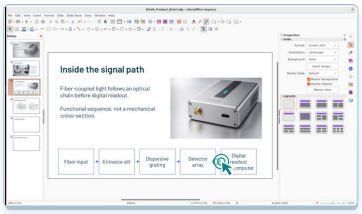
Bind spectra and SNR data

5 Embed the chart 282



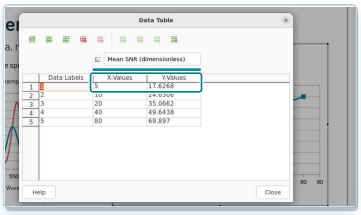
Paste the editable chart into Impress

6 Draw the signal path 444



Connect the five native shapes

7 Read back the data 649



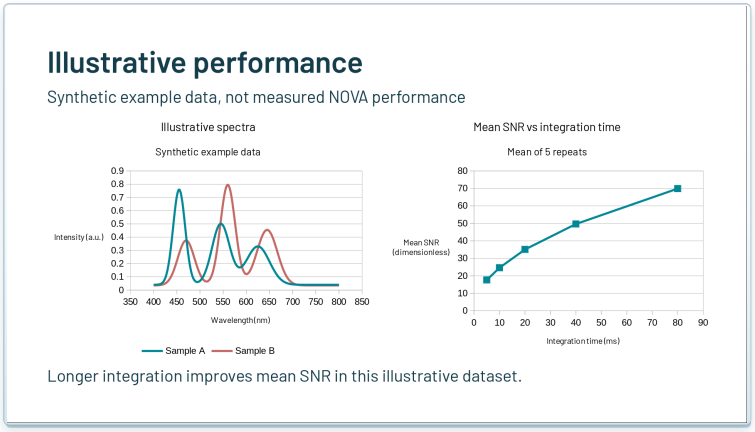
5 ms still gives mean SNR 17.6268

Native signal-path diagram

Six editable slides

Two embedded charts

8 Present the finished deck 685–691



Editable .odp + calculation workbook + PDF

State IDs shown. Overlays summarize nearby actions.

Concept product. Illustrative data.

Figure 38. NorthCUA case study in Calc and Impress. Selected states follow calculation, chart transfer, diagram construction, and data readback. The chart at state 282 has just been inserted and is later resized and positioned. The lower-left inset enlarges the native diagram at state 687, while the large view shows the finished evidence slide at state 688. Enlarged chart and data-table views connect the mean at state 136 to its embedded value at state 649. Numbers identify captured states. Editorial overlays summarize nearby actions.

G.4 CMOS Inverter Analysis

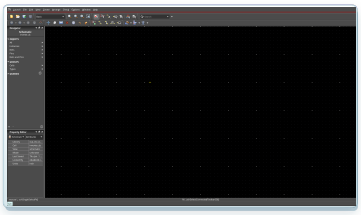
From a blank schematic to circuit insight

Virtuoso + Spectre

367 recorded steps

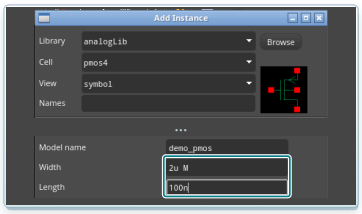
Build a 1 V CMOS inverter, sweep its input, and inspect switching and device states.

1 Blank schematic 49



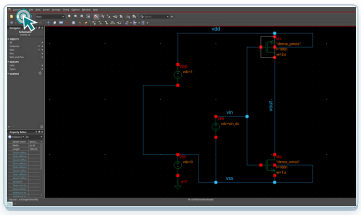
Start with an empty circuit

2 Size the PMOS 78



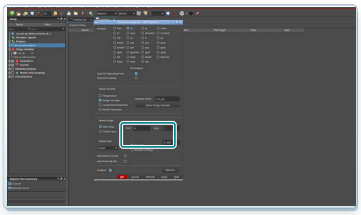
Set width 2 µm and length 100 nm

3 Wire and check 190



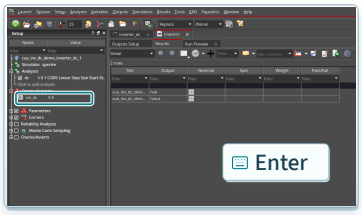
Connect, label, Check and Save

4 Set the sweep 232



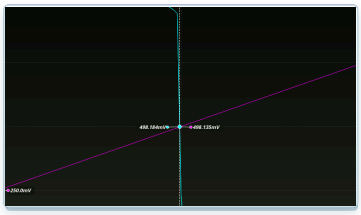
0–1 V sweep · 5 mV step

5 Commit the bias 268



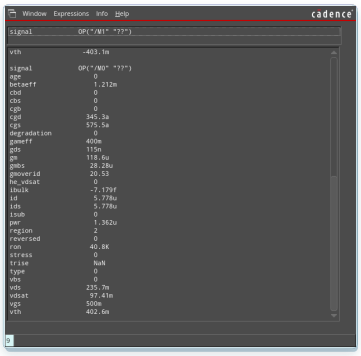
Commit 0 → 0.5 V and rerun

6 Locate the crossing 318



Find where vin ≈ vout

7 Check device states 340

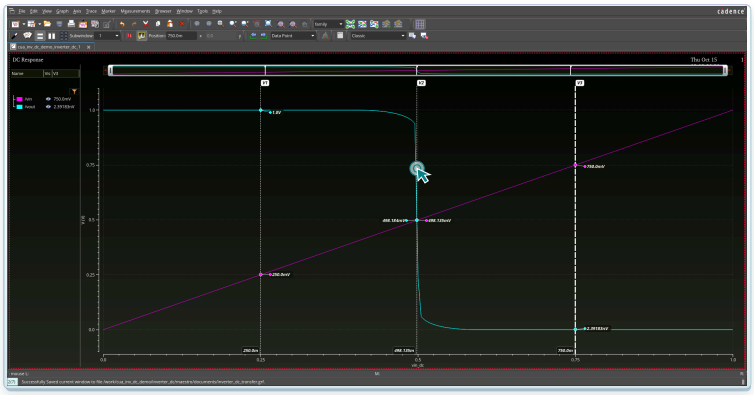


Nominal input = 0.5 V

NMOS +5.778 µA

PMOS −5.778 µA

8 Retain the native result 367



Switching estimate ≈ 0.498 V

5 mV sweep · interpolated crossing

State IDs shown. Intervening actions omitted.

Constructed MOS1 teaching models

Figure 39. NorthCUA case study in Virtuoso and Spectre. Panel 2 pairs device identity and size fields from the same native dialog at state 78. Panel 6 enlarges the crossing at state 318, estimated by interpolation between the 0.495 and 0.500 V sweep samples. Current annotations transcribe the PMOS and NMOS reports at states 335 and 340, respectively, both at nominal input 0.5 V. Numbers identify captured states. Editorial overlays summarize nearby actions, with intervening actions omitted.

The task asks for a 1 V CMOS inverter, its DC transfer curve, and transistor operating-point checks in Virtuoso and Spectre. The difficulty is carrying the intended device and analysis settings from the schematic into the simulator, then connecting numerical readouts to the condi-

tions that produced them. Figure 39 begins with an empty schematic. The agent sizes the two transistors, with the PMOS width and length fields shown in the figure. It then wires and labels the circuit and checks the completed design before configuring a 0–1 V input sweep with a 5 mV step. An uncommitted input edit then exposes a subtle GUI-state mismatch. The first sweep completes successfully, but the saved nominal input remains at 0 V. The agent returns to ADE, enters 0.5 V, commits it with Enter, and reruns the analysis so that the operating-point report uses the intended bias.

With the corrected configuration, the agent inspects the transfer curve in ViVA and locates the crossing of input and output near 0.498 V. This is an interpolated estimate from the 5 mV sweep. It separately reads both device reports at the nominal 0.5 V input, where the drain currents are $+5.778\,\mu$A and $-5.778\,\mu$A under the simulator's sign convention. The 367-request record ends with the native schematic, simulator setup, waveform state, and operating-point evidence retained for inspection. The case shows how a sustained GUI workflow can carry a circuit through construction, configuration repair, and numerical interpretation while preserving the engineering artifacts behind the result. The supplied MOS1 models are constructed teaching models.

G.5 TIDAL-01 Modeling and Animation

TIDAL-01 asks for a silver-and-glass craft with a thin hull, swept inlays, and an opening iris beneath the body. The challenge is carrying this visual design from a two-dimensional drawing into editable geometry whose separate parts can be animated and inspected from different angles. Figure 40 begins with the top and side outlines in Inkscape, then shows the completed three-view sketch. The agent places this drawing beside a Blender blockout to match the thin side profile. It then shapes a swept curve on the hull and fits three inlays to its surface. To animate the underside opening, it assembles the iris from separately pivoted leaves and keys their opening poses. The close view of the eight-leaf assembly makes this change from static shape to articulated geometry visible.

An earlier render still shows pale, opaque inlays. The agent sets glass transmission to 1.0, then inspects a later render to assess transparency and contact with the hull. To refine the panel seats, it constructs three matching cutter volumes by extruding and rotating the inlay outlines. After further geometry work, an underside render exposes the optical ring and open iris. With these parts established, it keys a tracked camera arc that moves from the upper hull to the underside and returns to a rear-quarter view. The final strip combines whole-object views with glass and iris close-ups from the same editable scene, connecting the sketch's silhouette and underside layout to the rendered object. Across 1,805 recorded steps, including observations, checks, and retries, the workflow produces an original SVG, a packed `.blend`, and a 12-second native animation with 288 frames at 24 fps. Reopening the packed scene reports no missing files. The case demonstrates a sustained 3D-authoring workflow in which geometric construction, material revision, and camera control contribute to one reusable animated asset.

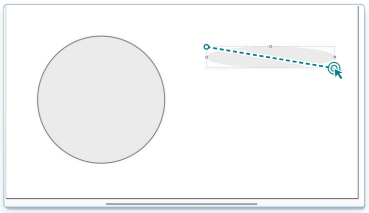
From a sketch to an articulated 3D model

TIDAL-01 · Inkscape + Blender

1,805 recorded steps

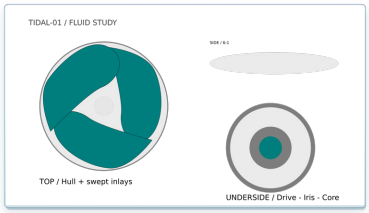
Sketch and model a silver-and-glass craft with an opening iris and an animated camera orbit.

1 Draw the silhouette 18



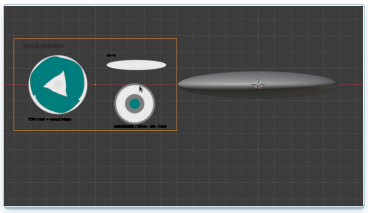
Set the top and side proportions

2 Three-view sketch 86



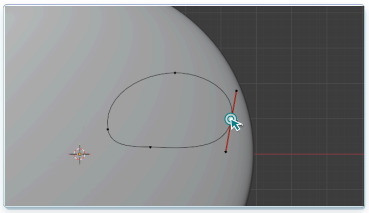
Save top, side and underside

3 Block out the hull 186



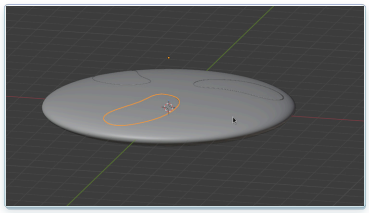
Match the sketch's thin profile

4 Shape an inlay 201



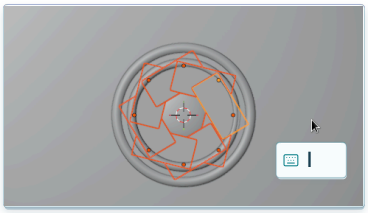
Refine the curve on the hull

5 Fit three inlays 247



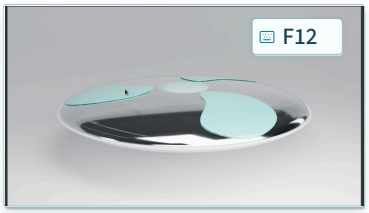
Conform each panel to the shell

6 Key the iris 393



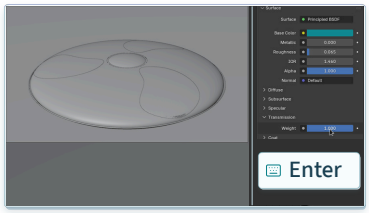
Animate eight independent leaves

7 Test the materials 661



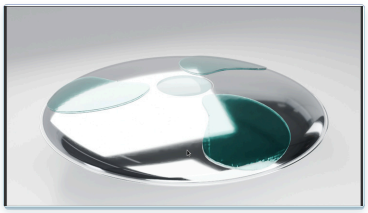
Inspect the shaded surface

8 Tune transmission 1133



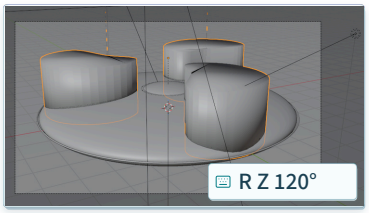
Set Transmission to 1.0

9 Review the glass 1168



Inspect transparency and contact

10 Build seat cutters 1373

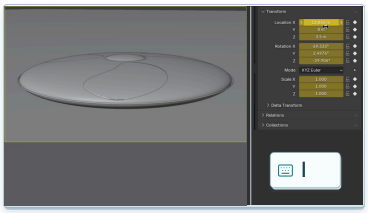


Extrude and rotate three volumes

11 Inspect the opening 1529

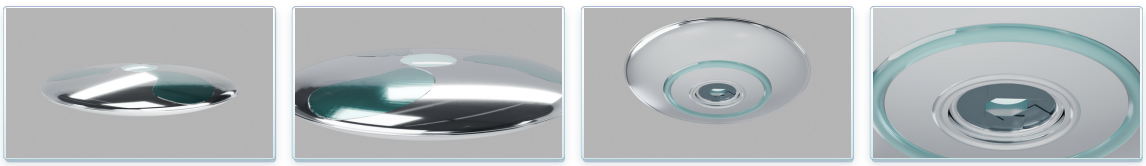
Check the underside in render

12 Key the camera 1577



Set the rear-quarter orbit key

Completed model · complementary views Same editable Blender scene



Hero view | Glass close-up | Underside | Iris close-up

Recorded states. Intermediate actions omitted. Editable .blend · 12 s native film

Figure 40. NorthCUA case study in Inkscape and Blender. Twelve selected states trace drawing, modeling, material revision, and camera keying. Panel 2 is the exported sketch. Panels 8 and 12 pair viewport and settings crops from their respective states. Editorial cues summarize nearby actions. The compact final row combines native hero, glass-detail, underside, and iris-detail renders from the same scene. Intermediate actions are omitted.

Figure 41. Screenshot-driven Zuma gameplay. Panels follow row order. # denotes zero-based after-action IDs. Editorial aim guides precede actions 0, 191, and 197. Paired ammunition crops straddle action 144. L/R denotes mouse buttons. Intervening actions are omitted. The game pauses between actions.

G.6 Zuma Gameplay

Turns and tunnels change the accessible targets in this six-color Jaguar-level task. Figure 41 begins with a purple match and progresses to a six-shot chain bonus. At 5,020 points, the progress bar turns green with balls still in play. The agent swaps blue ammunition for green, clears the exposed pair, and works through the bend to a short bottom-lane tail. It eliminates the last purple and red groups, waits for the completion bonus, and reaches native STATS with 9,070 points after 199 GUI actions.

H Runtime and Admission Protocols

This appendix defines the runtime choices needed to interpret an interaction record, including the observations presented to the policy, the tools available to each role, environment ownership, and demonstration selection. The learning-side segment and update rules are consolidated in Appendix E.5.

H.1 Runtime Configuration

A context profile determines which retained interactions enter each model call. The sliding visual profile counts the latest clean frame within its image budget and keeps textual action/result records after image eviction. Other profiles replay visual history within a compaction epoch. Text windows, image budgets, and semantic compaction are separate settings. The generic context manager defaults to off. Shadow mode records decisions without applying them, and enforce mode applies its configured policy.

Historical overlays mark issued actions on pre-action images, while the latest clean frame shows the currently observed state. Text-only tools, including notes, do not add visual frames. A later application change can require waiting or a fresh observation. After a timeout or failed refresh, state inspection precedes retry. Web element references belong to the observed page state, and failed refreshes invalidate affected aliases.

Compaction retains a structured account of the goal, constraints, supported facts, completed work, failures, current state, open items, and artifact references. Summaries and notes carry their temporal provenance and remain distinguishable from observations. The summary prompt, protected recent interactions, trigger/target budgets, and failure policy determine the resulting input. Note categories are fact, correction, precaution, constraint, decision, and checkpoint. Compatibility names map `mistake` to correction and `plan/other` to checkpoint. The exposed schema, replacement behavior, and prompt budget control which notes remain visible.

Execution scopes and tool interfaces. A session binds the agent to its environment and resources. A round tracks one request and execution budget, and a step records one policy-decision and action-execution iteration. A step can contain several tool calls, so model decisions, tool calls, and GUI interactions are different counts. The dispatcher validates declared arguments before invoking local handlers or connected MCP tools and returns results to the requesting step. Appendix H.2 specifies the model-facing interfaces. Runtime-only probes are not additional actor permissions.

Evaluator profiles and attribution. Evaluation binds a task and rubric to an evidence source, inspection permissions, budget, and report schema. The rubric-generation template requests three to seven criteria. The Agent Kit desktop entry path can use a request-level fallback, while the web profile supplies eight fixed criteria. The selected profile determines which criteria, weights, applicability rules, and reward conversion apply. Report-format validation checks the report's structure.

Artifact tools inspect retained records. Live tools additionally require authorization and a dispatcher bound to the evaluated workspace. Artifact-only replay has no live device access. Read-only probes are preferred. Shell inspection uses an allowlist, and state-changing GUI or browser

inspection requires explicit permission. The workspace remains reserved through required live checks.

Actor credit depends on the state before evaluator intervention. Terminal evidence and evaluator actions are therefore recorded separately. Inspection may reveal an artifact or page result, but cannot complete or repair the submitted work on the actor's behalf. A potentially mutating check needs a procedure that preserves the original state, such as a prepared evaluation copy. When that basis cannot be preserved, assessment uses retained evidence or leaves the condition unresolved. Probe failure, conflicting evidence, and an exhausted inspection budget are distinct from a supported task-failure verdict. When evaluator protection is enabled, recognized evaluation failures remain unresolved without replacing the actor's terminal status.

H.2 Action Spaces and Parameter Schemas

The desktop and web instances in Section 3.1.3 expose different tools through the same dispatcher. The tables below summarize their model-facing contracts and the evaluator interfaces in Section 3.3. Fields without defaults are required. A question mark marks an optional field with no supplied value, and `T[]` denotes an array. Empty arguments are written as `{}`. These are tool signatures, independent of provider-specific call envelopes. Runtime-only probes and environment-setup commands are outside the acting agents' action spaces.

H.2.1 Desktop Agent

Desktop actions address the current screen and focused application. Coordinates follow the run's declared coordinate policy, which maps the model's image coordinates to device coordinates. A drag point has the schema `{x: number, y: number}`. The `button` argument accepts `left`, `right`, or `middle`.

Table 21. Desktop agent action space. Defaults and argument constraints are shown explicitly.

Tool	Parameter schema	Operation or constraint
`click`	`x, y: number` `button: string = left`	Click at the specified point.
`double_click`	`x, y: number` `button: string = left`	Double click at the specified point.
`move`	`x, y: number`	Move the pointer without clicking.
`drag`	`path: Point[]` `button: string = left`	Follow an ordered path containing at least two points.

`scroll`	`x?, y?: number`
	`delta_x: number = 0`
	`delta_y: number = 0`
		Scroll at the pointer or a supplied location. Positive vertical deltas scroll up, negative deltas down.

`type`	`text: string` `target_kind: enum = auto`	Enter literal text into the focused control. Target kind is `auto`, `editable`, or `terminal`.

Continued on next page

Table 21 (continued)

Tool	Parameter schema	Operation or constraint
keypress	key?: string keys?: string[]	Supply exactly one field. Array entries are ordered key-press units, with + joining a chord.
wait	ms: integer = 1000	Pause for the given milliseconds, then obtain a fresh screenshot.
screenshot	all_screens: boolean = false	Capture the primary display, or the virtual desktop when enabled.
note	text: string reason: string = "" kind: enum = fact	Preserve working memory without GUI input. Text is at most 8,000 characters and reason at most 2,000.

The note categories are `fact`, `mistake`, `precaution`, `plan`, `constraint`, `decision`, and `other`. These exposed names map to the working-memory categories described in Appendix H.1. The same note fields apply to `web_note` below.

H.2.2 Web Agent

Web actions can use element references from the latest page observation or viewport-relative CSS coordinates. In Table 22, `Ref` is a string or integer element label. References are refreshed with page state, while `tab_ref` identifies a browser tab. Point actions use nonnegative CSS coordinates measured from the current viewport's top-left corner. These differ from full-page or resized-image coordinates.

Table 22. Web agent action space. Each row gives one tool's arguments and operation.

Tool	Parameter schema	Operation or constraint
web_open	url: string	Navigate after URL-policy validation.
web_snapshot	{}	Read the current page structure.
web_get_url	{}	Read the active tab's URL.
web_list_tabs	{}	List open browser tabs.
web_click_ref	ref: Ref	Click the referenced element.
web_hover_ref	ref: Ref	Hover over the referenced element.
web_check_ref	ref: Ref	Set the referenced checkbox to checked.
web_uncheck_ref	ref: Ref	Set the referenced checkbox to unchecked.
web_get_text	ref: Ref	Read text from the referenced element.

Continued on next page

Table 22 (continued)

Tool	Parameter schema	Operation or constraint
`web_fill_ref`	`ref: Ref` `text: string`	Replace the content of an editable element.
`web_select_option_ref`	`ref: Ref` `values: string[]`	Select one or more nonempty HTML option values.
`web_drag_ref`	`from_ref: Ref` `to_ref: Ref`	Drag between two referenced elements.
`web_click_point`	`x, y: number` `button: enum = left`	Click at a viewport point. Button is `left`, `right`, or `middle`.
`web_double_click_point`	`x, y: number` `button: enum = left`	Double click at a viewport point. Button is `left`, `right`, or `middle`.
`web_drag_point`	`from_x, from_y: number` `to_x, to_y: number`	Drag between viewport points using the left button.
`web_type_text`	`text: string`	Type into the focused element.
`web_press`	`key: string`	Send a browser key or supported key expression to the focused element.
`web_scroll`	`direction: enum` `amount: integer = 600`	Direction is `up`, `down`, `left`, or `right`. Amount is a positive CSS-pixel distance.

`web_wait`	`duration_ms?: integer`
	`load_state?: enum`
	`text?: string`
	`url_pattern?: string`
	`selector?: string`
		Choose exactly one target. Duration is nonnegative. Load state is `load`, `domcontentloaded`, or `networkidle`.

`web_screenshot`	`annotate: boolean = false`	Capture the viewport. Annotation adds current Set-of-Mark labels.
`web_new_tab`	`url?: string` `label?: string`	Open a tab, optionally with a URL and stable label.
`web_close_tab`	`tab_ref?: string`	Close the specified tab, or the active tab when omitted.
`web_focus_tab`	`tab_ref: string`	Focus the specified browser tab.
`web_note`	Same fields as `note` in Table 21.	Preserve nonempty text without interacting with the page.

H.2.3 Evaluator

The evaluator retrieves recorded evidence and, when authorized, inspects the evaluated workspace through live tools. Table 23 separates these interfaces. The active desktop or web profile selects the applicable tools. Evidence references identify saved artifacts, independently of the web agent's element references. Retrieval tools with `detail_level` use `standard` by default and accept `debug` for expanded records.

Table 23. Evaluator inspection interfaces. Live tools require access to the evaluated workspace and the corresponding permissions.

Tool	Parameter schema	Purpose or scope
Recorded evidence		
`read_round_bundle`	`section?: string` `detail_level: string = standard`	Read a compact bundle or a selected section, such as trajectory or final state.
`read_step`	`session_step: integer` `detail_level: string = standard`	Retrieve one recorded step.
`read_trace`	`session_step: integer` `detail_level: string = standard`	Alias of `read_step`.
`query_artifacts`	`kind?: string`	List evidence, optionally by kind.
`query_runtime_events`	`query?: string`	Search runtime event summaries.

`query_tool_results`	`tool_name?: string`
	`session_step?: integer`
	`detail_level: string = standard`
		Retrieve results by tool or step.

`query_a11y_snapshot`	`evidence_ref?: string`
	`session_step?: integer`
	`detail_level: string = standard`
		Read saved desktop accessibility evidence.

`query_dom_snapshot`	`evidence_ref?: string`
	`session_step?: integer`
	`detail_level: string = standard`
		Read saved web DOM evidence.

`query_web_page_state`	`session_step: integer` `detail_level: string = standard`	Read a recorded URL, title, and page-state summary.

`query_som_annotation`	`evidence_ref?: string`
	`session_step?: integer`
	`include_geometry: boolean = false`
	`ref_filter?: string[]`
	`detail_level: string = standard`
		Read saved Set-of-Mark metadata. In standard detail, geometry requires the flag and a reference filter.

`query_web_terminal_verification`	`session_step?: integer` `detail_level: string = standard`	Retrieve terminal browser-verification evidence.
`compare_images`	`left_ref: string` `right_ref: string`	Compare two evidence images using deterministic similarity metrics.
`extract_image_text_summary`	`evidence_ref: string`	Extract an OCR summary when the profile includes this tool.
Authorized live inspection		
`client_screenshot`	`all_screens: boolean = false`	Capture the evaluated desktop.
`gui_inspect`	`remove_empty: boolean = true` `refresh_app_windows: boolean = true`	Inspect desktop window targets. The refresh field is retained for compatibility.

Continued on next page

Table 23 (continued)

Tool	Parameter schema	Purpose or scope
client_bash_readonly	cmd: string	Run a nonempty command under the read-only command allowlist.
client_browser_snapshot	{}	Read the evaluated browser's page structure.
client_browser_get_url	{}	Read the evaluated browser's current URL.
client_browser_tabs	{}	List tabs in the evaluated browser session.
client_browser_get_text	ref: Ref	Read a current browser element.
client_browser_screenshot	annotate: boolean = false	Capture the evaluated browser viewport.

Authorized desktop GUI inspection reuses the schemas in Table 21 through client_gui_screenshot, client_gui_click, client_gui_type, client_gui_keypress, and client_gui_wait. Additional browser operations use the matching client_browser_ wrappers and schemas from Table 22, excluding web_note. State-changing inspection requires explicit permission and follows the attribution protocol in Appendix H.1.

Evaluation ends with final_report, whose required fields appear in Table 24. Optional structured fields carry criterion scores, diagnoses, side effects, safety and efficiency assessments, and evidence-linked divergence information. The web report additionally records the final URL and title, verification flags, a blocker category, and critical evidence references.

Table 24. Required fields of the evaluator's final_report argument object. Platform-specific and diagnostic fields extend this shared core.

Field	Type and allowed values
overall_verdict	Enum success, partial, failure, or uncertain.
outcome_label	Enum success, failure, or uncertain.
process_rating	Integer from 1 to 5 rating execution quality.
completion_rating	Integer from 1 to 5 rating task completion.
confidence_rating	Integer from 1 to 5 rating confidence in the assessment.
controllability	Enum controllable, uncontrollable, mixed, or unknown.
reason	String giving the evidence-grounded rationale.

H.3 Batch Collection and Resource Ownership

Task preparation binds CaseSpec to its inputs, initialization procedure, application/tool profile, budgets, and optional post-run checks. Placement considers configured evaluator requirements as well as actor requirements. Resumed work uses the original task and configuration identity.

Supported context-managed settings are retained in batch-local snapshots, while external dependencies keep their own version records.

Fresh single-use sandboxes isolate new collection attempts. Task-specific fixtures also cover external accounts and service state. An optional prepared queue, disabled by library default, performs initialization ahead of execution. Reservations are task/attempt-specific and assigned once. On-demand acquisition is a separate fallback. Prepared queues and warm pools have different allocation and reuse rules. An externally owned environment retains its own termination authority.

Creation, preparation, execution, submission, and status-query limits govern different resources. Assignment checks the remaining provider lifetime against the task timeout and release margin. Attempt identifiers link scheduler state to controller state, and reconciliation checks an existing attempt before starting another. A scheduler claim and a device execution lease represent distinct ownership decisions.

Live evaluation and environment-resident output capture finish before owned resources are released. Once a sandbox is destroyed, replay is limited to retained artifacts. Unconfirmed cleanup remains unresolved and continues to incur resource cost until destruction is confirmed. This lifecycle distinguishes transport disconnection from environment termination.

Table 25. Shared execution interfaces with different collection and training lifecycle owners.

Responsibility	Supervised collection	Online RL
Acting policy	Configured demonstration policy	Current policy being optimized
Scheduling	Standalone batch service	Training-side group orchestration
Environment lifetime	Owned sandboxes released after required checks and capture	External orchestrator retains release authority
Learning output	Selected trajectories and role-specific demonstrations	Attempt outcomes and captured calls for task groups

H.4 Cohort Admission

Outcome-verified completion demonstrations require actor-only extraction, valid action/context alignment, required artifacts, full-task acceptance, and the cohort's quality checks. Broad judge-filtered mid-training uses its separately declared selector, and evaluator demonstrations are separate judgment-task samples (Appendix E.1). Online RL retains valid negative outcomes within task groups.

Table 26. Selection distinguishes task outcomes from missing or invalid evaluation and defines completion-demonstration eligibility.

Attempt evidence	Completion-demonstration disposition
Verified complete with quality checks	Eligible; retain source and outcome evidence
Supported infeasibility or uncontrollable blocker	Excluded; retain separately if used
Partial or valid failed outcome	Excluded; retain as failure/diagnostic experience if declared
Uncertain verdict or conflicting evidence	Unresolved; requires assessment
Required evaluation error or missing outcome evidence	Not admitted as a verified completion

I Benchmark Evaluation Details

Section 6.1 introduces the six benchmarks. This appendix collects the metric definitions, task sets, and implementation settings used to interpret the results, together with the sources of published baseline scores.

Comparable tasks and interfaces. Interactive evaluation uses the benchmark's scorer independently of the harness evaluator used for collection. OSWorld-Verified uses the supported 361-task setting without the eight Google Drive tasks. Our OSWorld 2.0 binary-completion and partial scores both use the full 108-task set. Grounding records identify the OSWorld-G instruction variant and absent-target convention, and map crop/resize predictions back to annotation coordinates.

WebArena-Verified retains the responses and network traces required by its scorer. Our WebTestBench evaluation supplies the gold checklist directly, following the benchmark's oracle setting. Agents return Pass/Fail judgments, with Fail as the positive class. Precision, recall, and F1 measure defect detection. Checklist generation is outside this setting, so coverage is not reported. We first compute precision, recall, and F1 from the checklist judgments within each task, then average each metric over all evaluated tasks. For task-level scores $P_t, R_t, F_{1,t} \in [0, 1]$ across N tasks, the reported percentages are

$$\overline{P} = \frac{100}{N}\sum_{t=1}^{N} P_t, \qquad \overline{R} = \frac{100}{N}\sum_{t=1}^{N} R_t, \qquad \overline{F}_1 = \frac{100}{N}\sum_{t=1}^{N} F_{1,t}. \tag{6}$$

This task-level macro average gives equal weight to tasks with different checklist lengths. It differs from pooling all checklist judgments into one confusion matrix. In particular, $\overline{F}_1$ averages the per-task F1 scores and generally differs from the harmonic mean of $\overline{P}$ and $\overline{R}$. Table 11 reports all three metrics for each included model and omits models without measured results. The NorthCUA Nano, Mini, and Pro results all use final checkpoints after joint desktop–web RL.

The source notes below record the task scope, tool permissions, budgets, and scoring conventions of published baselines. Parameter columns distinguish total from activated language-model counts. Within each NorthCUA tier, stage comparisons use the declared benchmark tasks, tools, observations, and execution budgets.

Repeated predictions. Table 29 summarizes repeated predictions on the same grounding examples, including run-to-run variation and pass@3 (Dror et al., 2018). The accompanying task-level

Table 27. SSP accuracy (%) from the supplied per-example records. The application groups form one partition, and target type forms a second.

SSP group	Examples	NorthCUA-Nano	NorthCUA-Mini	NorthCUA-Pro
Office	230	82.6	93.0	93.0
Creative	341	71.3	90.6	91.5
Operating systems	196	68.9	93.9	93.9
Development	299	72.9	93.7	92.6
Scientific	254	71.7	92.1	94.5
CAD	261	70.5	92.0	95.0
Text targets	977	83.9	94.8	95.2
Icon targets	604	55.0	88.6	90.2
Overall	1,581	72.9	92.4	93.3

Table 28. SSP accuracy (%) by annotated target extent and type. All six rows are disjoint and together cover the full SSP evaluation.

Target size	Target type	Items	NorthCUA-Nano	NorthCUA-Mini	NorthCUA-Pro
< 16 px	Text	31	58.1	87.1	90.3
	Icon	57	50.9	84.2	93.0
16–< 32 px	Text	661	84.6	94.1	94.6
	Icon	418	53.1	89.0	89.0
≥ 32 px	Text	285	85.3	97.2	97.2
	Icon	129	62.8	89.2	93.0

breakdowns explain coverage and execution behavior. Showcase demonstrations are described separately from model checkpoint evaluation.

I.1 Grounding Scores and Repeated Sampling

The direct SSP evaluation produces one coordinate prediction from a full screenshot. Table 27 breaks down the available final-model records by application group and target type. Every domain uses the same sample set for Nano, Mini, and Pro. This makes the coverage differences visible without changing the task distribution between models.

We further stratify SSP by target size and type (Table 28). Target size is the smaller of the annotated box's width and height, $\min(x_2 - x_1, y_2 - y_1)$ in the original image coordinates. Fixed boundaries at 16 and 32 pixels define the three exploratory size bins. Predictions are paired across all three models by the exact image filename and query. All 1,581 identities are unique and matched, with identical reference boxes and target-type labels. The bins measure geometric extent in the original screenshot before model preprocessing.

The paired Nano–Mini predictions show 1,086 items correct for both models, 375 correct only for Mini, 66 correct only for Nano, and 54 missed by both. Mini locates 375 of Nano's 429 misses, but each model also has unique successes. For Mini and Pro, 1,373 items are correct for both, 102 only for Pro, 88 only for Mini, and 18 for neither. Pro's net gain of 14 correct items therefore coexists with distinct successes from each model. The paired counts distinguish shared errors from checkpoint-specific successes.

Table 29. Grounding summaries (%). OSWorld-G Refined SD is the sample standard deviation across three complete runs. SSP accuracies and subgroup results are recomputed from one-prediction records for all three models. SSP pass@3 values use author-reported summaries.

Model	SSP		OSWorld-G		
	Accuracy	Pass@3	Mean ± SD	Best run	Pass@3
NorthCUA-Nano	72.9	74.6	82.5 ± 0.77	83.0	86.0
NorthCUA-Mini	92.4	93.6	87.2 ± 0.71	87.9	90.6
NorthCUA-Pro	93.3	95.6	90.2 ± 0.80	91.0	94.0

Pro's remaining 106 errors all produce valid coordinates outside the target box. Icons account for 59 errors, including 46 in the 16–$<$ 32 pixel band. That band contains 418 icon targets, of which 11.0% are missed, and contributes 43.4% of Pro's errors overall. Its smallest icons perform better at 93.0%, so target size alone does not explain the remaining errors. Across application groups, Pro gains most over Mini in CAD (3.1 points) and scientific software (2.4 points), while development software is 1.0 point lower.

OSWorld-G Refined uses temperature 0.7, top-p 0.95, top-k 40, and a 1,024-token output limit. Table 29 reports the mean and sample standard deviation over three complete runs, alongside the best complete-run accuracy used in the main comparison. Pass@3 counts an item as correct when at least one of its three predictions succeeds.

On SSP, Pro raises single-run accuracy by 0.9 percentage points over Mini and reaches 95.6% pass@3, 2.3 points above its single-run result.

The Refined records contain 564 unique item IDs in each run, with matched annotations across all three models and all three draws. Of these, 510 have visible targets, comprising 470 rectangular and 40 polygonal reference regions. The other 54 require an absent-target refusal. Visible-target accuracy uses the corresponding region test, and refusal accuracy uses the benchmark's designated no-target output.

All three tiers share the grounding system prompt, sampling settings, and coordinate scoring. Nano and Mini use a grounding-specific click schema that describes normalized coordinates and refusal, requires `x`, `y`, and `button`, and enumerates button choices. Pro uses the frozen NorthCUA desktop click schema, whose descriptions refer to desktop interaction and whose button argument is optional. The reported tier differences therefore describe these evaluated model and interface configurations.

Table 30 reports how often each item succeeds. Nano's three complete runs contain 467, 460, and 468 correct predictions, while Mini's contain 488, 496, and 492, and Pro's contain 504, 509, and 513. All runs have zero request errors and unparsed predictions. Pro's second run includes one separately completed missing item, yielding the same 564 unique IDs as the other runs. This completion contributes one prediction for that item to run 2 and adds no extra draw to pass@3.

Pro's 34 persistently missed items contribute 102 of its 166 incorrect predictions (61.5%). Its remaining 41 items vary across draws, while 489 succeed in all three. Mini and Pro share 17 persistent misses, and 16 items remain persistently incorrect across all three model tiers. These counts show that Pro reduces the recurring error set while leaving a smaller common subset unresolved. Repeated predictions share 564 items and 251 screenshots, so they are not treated as 1,692 independent tasks.

Table 30. Per-item success counts across three OSWorld-G Refined draws. Each row sums to 564 items, including absent-target cases.

Model	0/3 correct	1/3 correct	2/3 correct	3/3 correct
NorthCUA-Nano	79	20	20	445
NorthCUA-Mini	53	15	27	469
NorthCUA-Pro	34	23	18	489

Table 31. Mean OSWorld-G accuracy (%) across three draws by reference geometry. These disjoint groups cover all 564 items.

Reference region	Items	NorthCUA-Nano	NorthCUA-Mini	NorthCUA-Pro
Rectangle	470	83.1	87.9	91.1
Polygon	40	95.0	97.5	98.3
Absent target	54	67.9	74.1	75.9

The improvement also spans reference geometries (Table 31). On rectangular targets, Pro averages 91.1%, compared with Mini's 87.9%. Polygonal regions are already high-performing for Mini at 97.5%, and Pro reaches 98.3%. The geometry labels identify the benchmark's scoring regions, not separate semantic task categories. Together with the absent-target analysis in the main text, they identify refusal decisions as a larger remaining gap than polygonal targeting.

I.2 Interactive Results and Trajectory Breakdowns

The main WebArena-Verified table reports final RL checkpoints at 67.2% for Nano, 70.4% for Mini, and 73.9% for Pro. Pro's score comes from one complete evaluation of all 812 tasks. The trajectory breakdowns below cover Nano and Mini after SFT and Pro after RL, with 521, 552, and 600 successful tasks, respectively. Mini uses the second evaluation round, matching Figure 27.

The earlier SFT evaluations additionally report pass@3 of 69.3% for Nano and 73.9% for Mini. Figure 27 includes Mini's SFT website-level breakdown alongside Nano SFT and Pro RL.

Table 32 groups the 812 tasks by website scope and requested outcome. Website groups use the task's site set, pooling multi-site combinations into one group. Outcome types follow the evaluator's expected category. Each grouping therefore partitions the same task set and reveals a different aspect of task completion.

Table 32. WebArena-Verified task counts for Nano SFT, Mini SFT, and Pro RL. Website scope and expected task type each partition the same 812 tasks. Forum denotes Reddit, and state change denotes the evaluator's `MUTATE` category.

Task group	Tasks	NorthCUA-Nano passes	NorthCUA-Mini passes	NorthCUA-Pro passes
Website scope				
GitLab only	180	115	126	131
Map only	109	67	71	77
Forum only	106	87	85	89
Shopping only	187	123	123	135
Store administration only	182	122	135	151
Cross-site	48	7	12	17
Expected task type				
State change	374	240	253	274
Retrieval	325	204	223	245
Navigation	113	77	76	81
Overall	812	521	552	600

Mini's additional completions over Nano concentrate in state-changing and retrieval tasks, with 13 and 19 more successes, respectively, while their navigation counts are nearly identical. Pro reaches 75.4% on retrieval, 73.3% on state change, and 71.7% on navigation. Cross-site work is more demanding for all three checkpoints, with 7, 12, and 17 successes out of 48 tasks. In comparison, Pro completes 76.3% of the 764 single-site tasks.

Mini's records show how this difference relates to execution length within one checkpoint. It completes 540 of 764 single-site tasks (70.7%) and 12 of 48 cross-site tasks (25.0%). Their median lengths are 10 and 27.5 decisions, respectively. Even among successful tasks, the cross-site median is 17 decisions, compared with nine for single-site work. These tasks combine lower completion with longer interactions, making them a useful focus for experience that trains the agent to carry constraints and intermediate results between websites.

Table 33 summarizes decision counts by evaluator outcome. Each record contributes its current assistant response once, including the final response. Nano has 811 traced tasks and one untraced failure, while Mini and Pro each have 812. All 812 tasks enter each model's success-rate denominator. Quartiles use linear interpolation at $(n-1)p$ for percentile fraction p.

Successful tasks have similar median lengths of 9, 10, and 10 decisions for Nano, Mini, and Pro. Failed tasks have longer medians of 19, 18, and 15.5 decisions and contribute 58.5%, 50.6%, and 38.4% of their recorded decisions, respectively. For Mini, this is 7,698 of 15,224 decisions across 260 failures. Separating outcomes thus exposes the disproportionate execution devoted to unsuccessful tasks. At the same time, Pro completes 24 tasks with at least 50 decisions, including a 93-decision success, showing that extended execution can also support successful completion.

Table 33. Decision records by task outcome for Nano SFT, Mini SFT, and Pro RL. Counts include terminal responses. Nano uses the archived record boundary, which includes interrupted execution, while Mini and Pro use one retained attempt per task.

Model	Outcome	Traces	Mean	Median	Q25–Q75
NorthCUA-Nano	Passed	521	13.4	9	7–16
	Failed	290	33.9	19	10–44
NorthCUA-Mini	Passed	552	13.6	10	7–16
	Failed	260	29.6	18	10–36.3
NorthCUA-Pro	Passed	600	15.0	10	7–17.3
	Failed	212	26.5	15.5	10–33.3

Mini likewise completes 13 tasks with at least 50 decisions, with its longest success taking 75. Across all three checkpoints, successful-task means exceed their medians, reflecting a tail of extended but productive execution. This profile favors training examples that retain successful long trajectories alongside concise workflows. In our learning pipeline, outcome-based selection retains successful long workflows for supervision, while RL also uses unsuccessful attempts to provide negative feedback.

The browser case in Figure 31 is task 1 in the Pro export. Its 13 decisions include an unavailable-option error, inspection of the actual dropdown choices, and successful completion after revising the report settings. The screenshots show how inspecting the available choices resolves the error and lets the agent finish the report query.

The desktop checkpoint summaries cover 361 OSWorld-Verified tasks with a 100-step limit and the full 108-task OSWorld 2.0 set with a 500-step limit. The OSWorld-Verified main table uses the RL checkpoints at 78.9%, 85.3%, and 88.6% for Nano, Mini, and Pro. Tables 12 and 13 retain the earlier checkpoints for stage comparisons. On OSWorld 2.0, the final RL checkpoints have Binary/Partial scores of 13.0/45.6 for Nano, 18.5/50.1 for Mini, and 30.6/66.8 for Pro. The Mid pairs are 0.9/24.2, 2.8/25.6, and 9.3/30.6, respectively. The long-horizon SFT checkpoints score 12.0/40.9, 18.5/46.3, and 27.3/63.6 in the same tier order. These SFT checkpoints serve as the Before-RL baselines in Table 13. The binary and partial scores describe full completion and intermediate progress on the same tasks.

I.3 External Comparison Settings

Table 34 summarizes the evaluation settings of published specialist and general-purpose models. Scores retain their reported task scopes, metrics, and aggregation rules.

Additional OSWorld-Verified references. Table 8 extends the comparison across model generations and specialist families. OpenCUA-72B's 45.0% is the 100-step, three-run mean in its final conference paper (Wang et al., 2025c). GUI-Owl-1.5 uses the 8B and 32B Instruct checkpoints (Xu et al., 2026b), while UI-Venus-2 contributes its 9B and 27B computer-use models (Venus Team, 2026b). These are distinct from UI-Venus-1.5 in the grounding comparison. MiniMax M3's 75.2% and Seed 2.1 Pro's 78.8% follow the Qwen-UI-Agent comparison. Qwen3.8-Max's 86.1% and GPT-5.6 Sol's 83.2% follow the Qwen3.8 release (Zhou et al., 2026b, Qwen Team, 2026c). The Nex-N2.5 release adds 76.7% for DeepSeek-V4-Flash-Vision on OSWorld-Verified.

It also reports WebArena-Verified success of 71.6% for Kimi-K3 and 62.3% for DeepSeek-V4-Flash-Vision (Nex-AGI, 2026b). We retain the release's DeepSeek model label.

The benchmark authors separately report 86.0% for Claude Fable 5 [1m] on 358 scored tasks and 83.4% for Claude Opus 5 [1m] on 360, both with 100 steps (XLANG Lab, 2026b). Google's Gemini 3.6 Flash card reports 83.0%, using a five-run mean with 100 steps (Google DeepMind, 2026b).

GLM-5.3-Flash's 84.7% is an independent public evaluation by hojuna (2026), with released code and trajectories. It reports 305.0 earned credits over 360 tasks after excluding one grader failure from the non-Google-Drive set. The run uses screenshot observations, PyAutoGUI, a 100-step budget, and no task retries. The main-table citation identifies the model, while the source above identifies this independent evaluation.

Our DeepSeek-V4.1-Flash evaluation yields 71.7% on the 361-task OSWorld-Verified setting with a 100-step limit. Its starred entry in Table 8 is separate from the published DeepSeek-V4-Flash-Vision result.

OSWorld 2.0 task scopes. Our GLM-5.3-Flash evaluation uses the full 108-task set and a 500-step budget, yielding 14.8% binary completion and 48.7% partial credit. The star in Table 9 identifies this measurement as our evaluation, with the model citation identifying the tested system. For GPT-5.6 Sol and Claude Opus 5, Table 9 uses the official full 108-task v2026.08.08 results, evaluated with batched actions, maximum effort, and 500 steps (XLANG Lab, 2026a). GPT-5.6 Sol averages 27.3% binary and 62.7% partial over two runs. Claude Opus 5 averages 31.4% and 68.3% over seven runs. These references match our full-set task scope, while their execution settings and aggregation remain source-specific. OpenAI's 65.7% GPT-5.6 Sol result uses the 82-task offline subset (OpenAI, 2026d). MiniMax's release comparison reports 70.6% for Claude Opus 5 without identifying its task subset (MiniMax, 2026a). Both alternative scores are omitted from the main table in favor of the full-set pairs. The earlier GPT-5.5 and Claude Opus 4.8 pairs use the official June full-set release, also at 500 steps with batched actions. UI-Venus-2 uses 108 tasks with a shorter 150-step budget (Venus Team, 2026b).

Claude Fable 5 and Fable 5.1 report binary/partial pairs of 36.1%/72.9% and 41.7%/77.9%, respectively, on 108 tasks (Anthropic, 2026a). Binary corresponds to the source's strict pass rate. Both are five-run means at maximum effort with a 500-step budget and 1080p observations. They use the August 2026 task release with subsequent task, setup, and grading fixes, so their scores differ in protocol from earlier releases. Safeguard interventions count as failures.

MiniMax's comparison also reports partial scores of 61.7% for MiniMax M3.1 Flash, 54.3% for DeepSeek-V4.1-Flash, and 58.3% for Kimi-K3 (MiniMax, 2026a). These remain additional published references, with task versions and subsets unspecified. Corresponding binary results are unavailable for these reported settings and remain dashes. We retain DeepSeek-V4.1-Flash without transferring its score to the later 1017 variant. DeepSeek's size entry uses 552B backbone parameters and 16B activated during decoding. Its model card separately reports 8B activated during prefill and 196B of additional Engram memory (DeepSeek-AI, 2026a).

Other reports provide both metrics without an explicit task count. Qwen3.8-Max reports 19.4% binary and 46.7% partial, while Qwen-CUA, Qwen-CUA-Max, and Qwen-UI-Agent retain their published pairs (Qwen Team, 2026c, Lu et al., 2026, Zhou et al., 2026b). Gemini 3.8 Flash's 59.0% partial score is the maximum of three 500-step benchmark runs on the pre-August-patch setting (Google DeepMind, 2026c). It is not task-wise pass@3.

The later Opus 5.5 system card reports 48.7% binary and 81.8% partial for Opus 5.5, alongside 42.8%/80.7% for Fable 5.1 (Anthropic, 2026e). These five-run means use 108 tasks and a 500-step budget, but update the task assets to September 10, 2026 and retain screenshots until server-side context compaction. The release announcement labels this revision OSWorld 2.1 (Anthropic, 2026d). We keep these later-revision results separate from the OSWorld 2.0 main table because both the task files and harness configuration changed.

Grounding and browser references. For grounding, the dagger-marked general-purpose rows in Table 6 are measurements by the Qwen-UI-Agent authors. Its SSP table separates direct inference from zoom-assisted results (Zhou et al., 2026b). GUI-Owl-1.5 uses the Instruct checkpoints, and UI-Venus-1.5 uses the final unified models. We take their refined OSWorld-G results from the respective model papers (Xu et al., 2026b, Venus Team, 2026a). MAI-UI's refined score is its 73.9% no-zoom result, with the 75.0% zoom-assisted result excluded (Zhou et al., 2025). Claude Opus 4.8's 82.3% SSP result is a five-run mean using adaptive thinking at maximum effort without Python tools (Anthropic, 2026c). GPT-5.6 Sol's 76.9% SSP accuracy is OpenAI's reported no-tools result (OpenAI, 2026d). Its 77.7% OSWorld-G score comes from the Nex-N2.5 comparison (Nex-AGI, 2026b). We confirmed that this result uses Refined instructions. Qwen3.8-Max's 84.5% SSP score follows its August release report (Qwen Team, 2026c), while its 71.6% OSWorld-G score comes from our Refined evaluation. Its star in Table 6 therefore applies only to OSWorld-G. The published references retain each source's inference settings and aggregation. Nex-N2.5-mini and Pro contribute reported Refined OSWorld-G accuracies of 82.9% and 87.4%, respectively (Nex-AGI, 2026a,b).

The other starred rows in Table 6 contain our single-run Refined evaluations of Claude Opus 5.5, Claude Fable 5.1, Kimi-K3, GPT-6 Astra, GPT-6.1 Sol, DeepSeek-V4.1-Flash, GLM-5.3-Flash, and Gemini 4 Argon. Their model citations identify the evaluated systems, while the stars identify the measurement source. The stars also identify our SSP evaluations of Claude Opus 5.5, Claude Fable 5.1, Kimi-K3, GPT-6 Astra, GPT-6.1 Sol, DeepSeek-V4.1-Flash, GLM-5.3-Flash, and Gemini 4 Argon.

For browser specialists, Qwen-UI-Agent reports original-WebArena success of 73.6% for 27B and 69.2% for 35B-A3B, with author corrections to some reference answers and evaluators (Zhou et al., 2026b). Qwen-CUA reports 64.2% on WebArena (Lu et al., 2026). These are useful neighboring results but are not relabeled as WebArena-Verified. Likewise, OpenGCUA evaluates an 82-task offline OSWorld 2.0 subset and varies the harness around existing models (Zhou et al., 2026a). It is a system comparison, not an additional independently trained model row in our table.

In Table 10, the Qwen3.7-Plus, Qwen3.8-Max, Claude Opus 4.8, Claude Fable 5, Gemini 3.1 Pro, and GPT-5.6 Sol scores come from the August Qwen3.8-Max release report (Qwen Team, 2026c). It uses WebArena's official grader within the OSWorld scaffold. We preserve its reported model variants and scores, including Claude Fable 5 and GPT-5.6 Sol, without assigning later snapshot results or an unreported reasoning-effort setting to those rows. Qwen3.8-Max's 2.4T total and 95B active parameters describe its language-model backbone (Qwen Team, 2026d). The evaluated hosted model supports vision, while the downloadable Qwen3.8-2.4T-A95B checkpoint is text-only.

The starred GPT-6 Astra, GPT-6.1 Sol, Claude Opus 5.5, Gemini 4 Argon, GLM-5.3-Flash, and DeepSeek-V4.1-Flash entries are our WebArena-Verified evaluations. Their citations identify the evaluated models, while the stars distinguish these measurements from the published reference

Table 34. Evaluation settings of published comparison models.

Reference	Comparison setting
Qwen-UI-Agent (Zhou et al., 2026b)	SSP uses the no-zoom row. OSWorld-G uses refined instructions. Desktop evaluation allows CLI and batched actions, with 361 OSWorld-Verified tasks. The source labels 79.5 as success rate in its table but describes partial progress in the text, so a strict binary-completion interpretation remains unestablished.
Qwen-CUA / Qwen-CUA-Max (Lu et al., 2026)	The reported 86.2 is earned credit on 360 tasks with a 100-step limit. Max reports 87.6 under the same protocol. Their native interface uses screenshots and keyboard/mouse actions. Max discloses over 1T total parameters without an activated count.
MAI-Gym-9B (Tong et al., 2026)	GUI and CLI actions. The paper lists 361 OSWorld-Verified tasks with 100 steps and 108 OSWorld 2.0 tasks with 500 steps. It evaluates three trials per task, without a complete aggregation rule in the main table.
ScaleCUA-Qwen3.5-9B (Lv et al., 2026)	Four-run mean with a 50-step budget. The official reproduction setup uses the 361-task non-Google-Drive list. The original paper calls the benchmark OSWorld.
Nex-N2.5-mini / Pro (Nex-AGI, 2026a,b)	These general-purpose models have 35B and 397B total parameters, respectively. Provider-reported computer/browser scores use the NorthCUA harness family. The release reports OSWorld 2.0 partial credit of 30.5/56.4 and WebArena-Verified success of 63.4/67.6 for mini/Pro. Corresponding binary scores are unavailable.
MiniMax M3 (MiniMax, 2026b)	Benchmark-author OSWorld 2.0 evaluation on 108 tasks with 500 steps, single-action calls, and thinking enabled (Yuan et al., 2026).

scores. The main table reports task success, with stage-specific trace accounting above. The Mid-to-SFT comparison appears in Section 6.5.

I.4 Web Repair Evaluation Boundaries

The recorded case in Figure 18 illustrates browser feedback and code repair. A controlled repair comparison additionally fixes the issue set, starting patch, Coding Agent, test generator, adjudicator, budgets, and protected acceptance conditions. Role-specific information remains separate even when roles share model weights. This comparison design is specified separately from that case's observed outcomes.

Each assigned issue begins with the nominated defect reproduced. Before a new condition or patch is tested, browser storage, account/backend data, service-worker state, and external fixtures are restored. Intentional stateful test dependencies remain explicit. Reports bind the build, reset state, test item, and attempt. Only evidence-supported product discrepancies justify a repair.

Acceptance uses the common assigned issue set as its denominator, retaining tester noncompletion. Infrastructure-invalid attempts follow the same bounded rerun rule, and unresolved issues remain visible in the accounting. Final acceptance runs the submitted patch from a protected clean state outside the repair loop, with its outputs withheld from further revision. Changing a whole feedback configuration and replacing only the adaptive browser policy answer different attribution questions.

J OSWorld 2.0 Execution Cases

These cases follow NorthCUA through selected decisions in complete desktop tasks. Counts refer to model decisions, each of which can issue multiple tools. Agent callouts summarize recorded decisions for readability.

The pair in Figure 28 has identical requests but different dataset revisions. The earlier run records unavailable OCR, while the later run uses it. This comparison describes behavior and outcomes without isolating training from initialization or software availability.

J.1 Preserving a Delivery Constraint Across Long Contexts

The photo-delivery task in Figure 42 requires the agent to reconcile invoices against bank payments, identify two clients' photos, and send different deliveries. Sarah has paid her $800 invoice in full and should receive her complete gallery. Emily has paid $250 of a $500 invoice and should receive a watermarked solo-indoor preview with a balance reminder. The difficulty is carrying this distinction through image preparation and email composition after the original banking interaction leaves recent context.

After exact-name payment matching, the agent records Emily's outstanding $250 and preview-only rule at decision 46. The compacted input at decision 150 retains both and the unfinished delivery checks, carrying the payment constraint beyond the earlier interaction history.

At decision 233, two action-marked historical screenshots identify the attachment click and selected PNG, while the clean current frame shows the file attached to Emily's draft. The agent sends the message and checks its sent copy at decision 236, confirming the preview and $250 reminder. A final image inspection confirms the watermark, and the 243-decision attempt receives full credit. Notes and compaction preserve the constraint, while the bounded image window exposes the current attachment state.

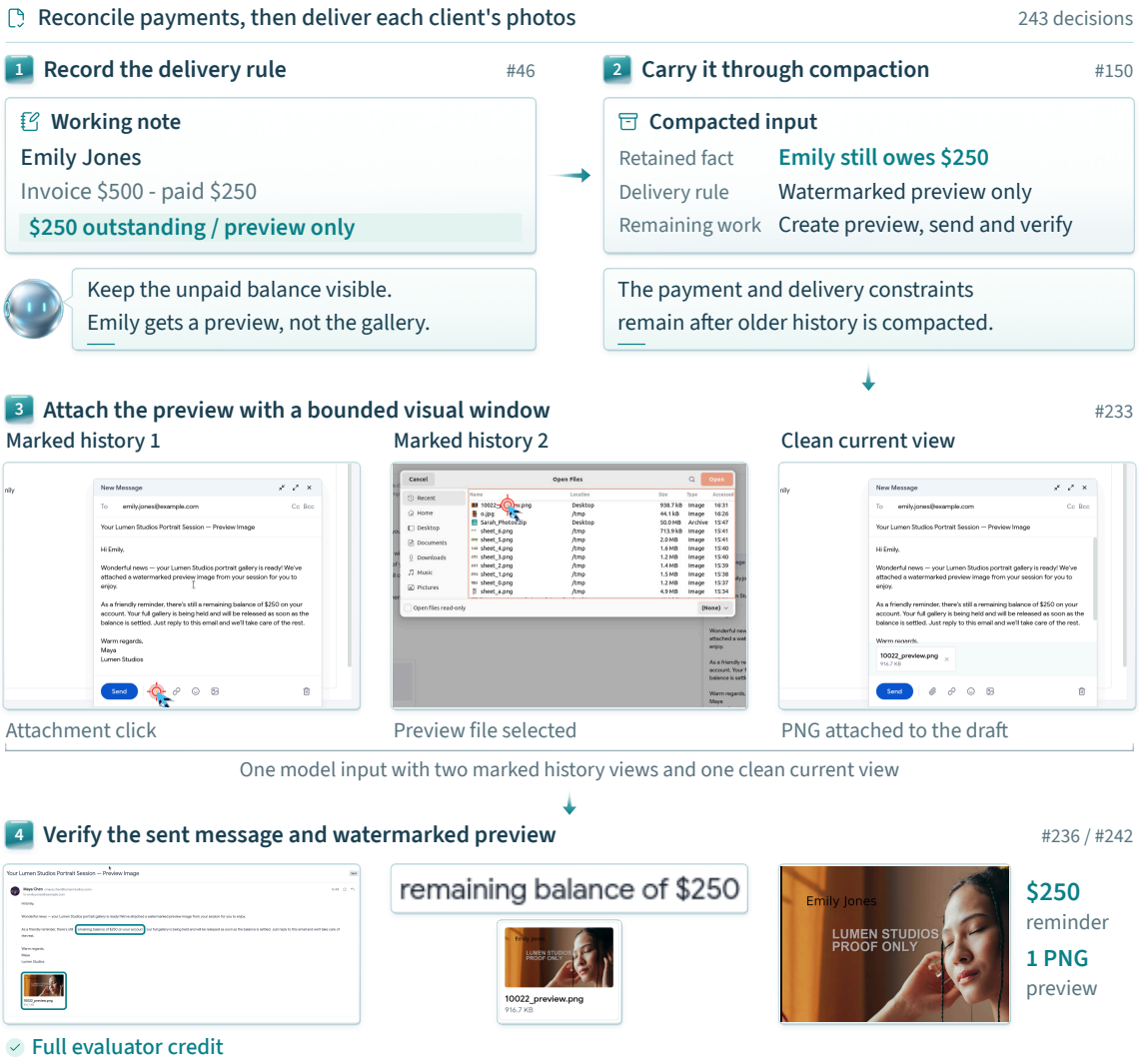


Figure 42. A payment constraint retained through a long-context task. Stages follow note 46, compacted input 150, the three image inputs at decision 233, and delivery checks. Historical action marks are preserved. Text artifacts are abridged and the agent callout summarizes a recorded decision. Screenshot crops precede the indexed decisions, with enlarged details.

J.2 Using a Terminal Through the GUI

Figure 43 follows a 3D course-logo task requiring Bauhaus 93 text, a 120-frame rotation, and a 960×540 WebM video at 24 fps inside Impress. To control scene settings and automate repeated rendering, the agent opens a terminal through GUI keypresses and types commands. Its exposed tools are desktop actions and notes, and the task permits terminal and application-console use.

After opening a terminal at decision 3 to identify fonts, the agent builds the scene in Blender's scripting workspace. When direct video export rejects `WEBM` and `FFMPEG` settings, it renders 120 PNG frames and encodes them with terminal-based `ffmpeg` using VP9 at 24 fps, preserving the 960×540 resolution. Separating rendering from encoding lets it complete the repeated frame processing programmatically.

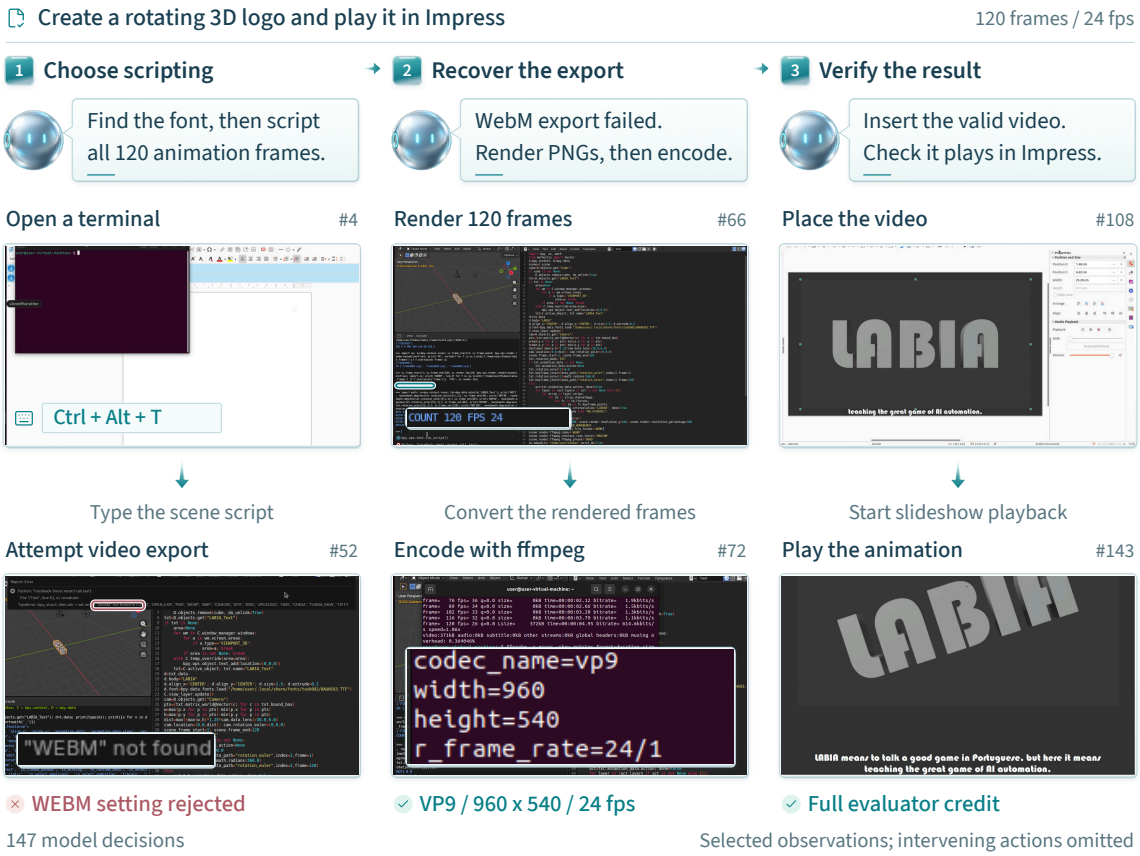


Figure 43. GUI-mediated scripting and recovery in an animation task. Each numbered phase reads top to bottom. Callouts paraphrase recorded decisions. Screenshots show cropped pre-decision observations with enlarged diagnostic details. Intermediate interactions are omitted.

After checking video properties, the agent positions the video in Impress and starts the slideshow. Observations 143 and 144 show changing logo orientations. The task receives full credit after 147 decisions and 253 tool calls, including four notes. GUI-accessible scripting handles the repetitive work, and GUI playback verifies the result. A GUI-only timing comparison was not performed.